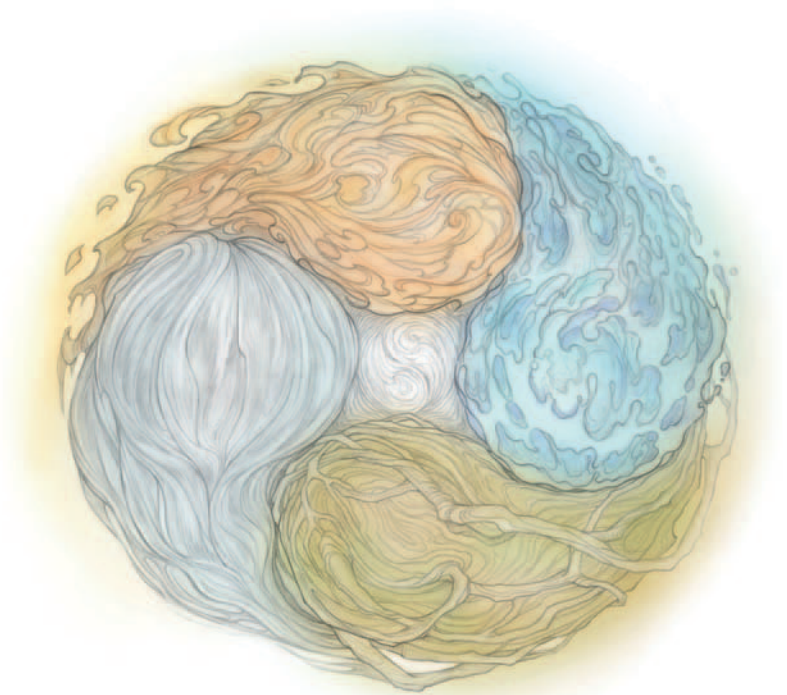
Elemental Magic

The Art of Special Effects Animation

Joseph Gilland



With a Foreword by Michel Gagne

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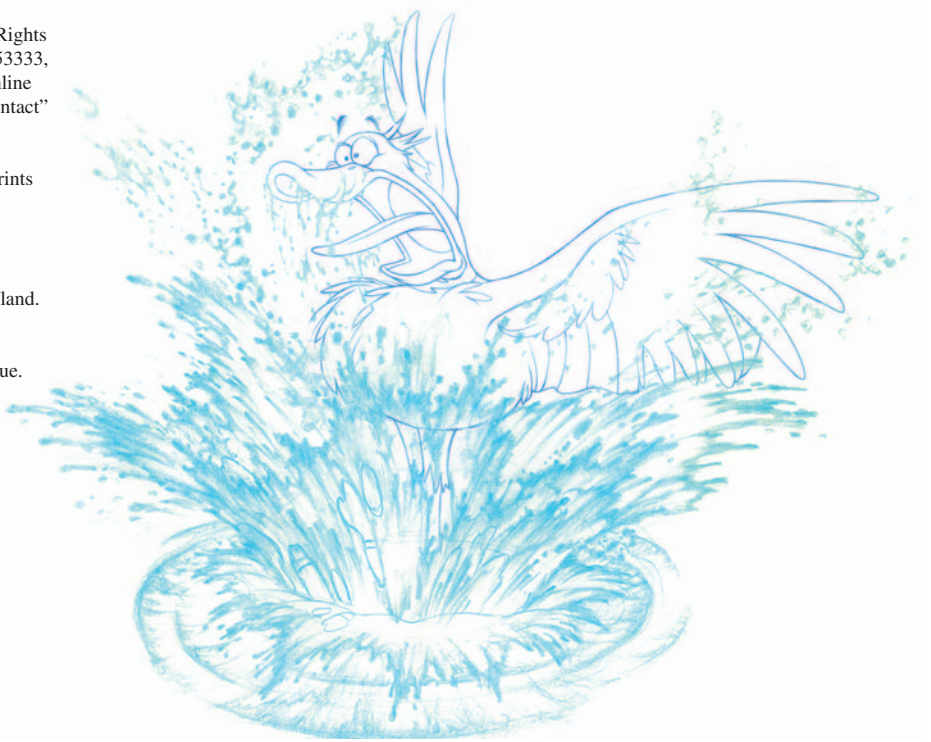

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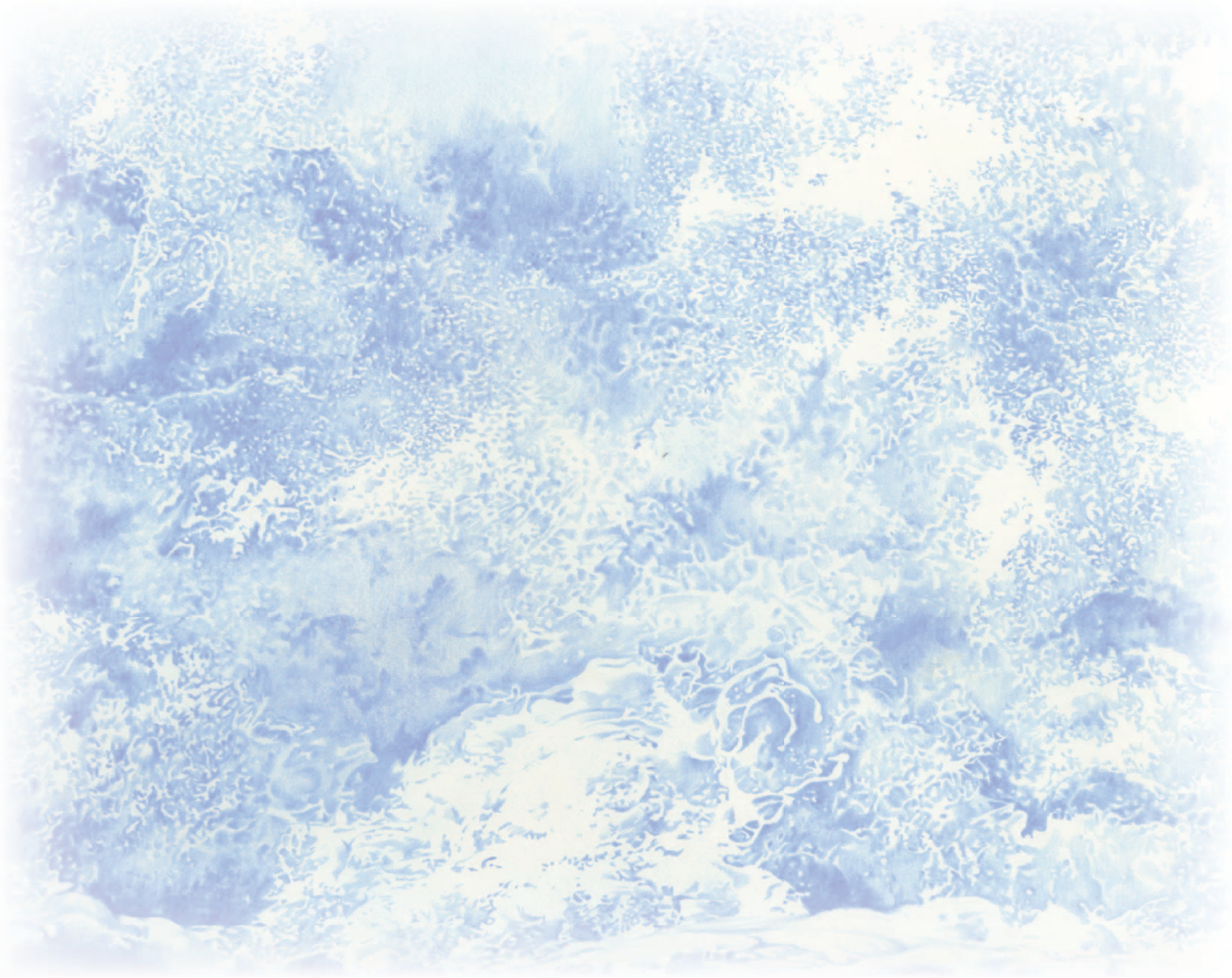
Dedicated to my parents, Frank and Tommy



Contents

Foreword by Michel Gagne XI
Preface XV
Acknowledgments XXV

Chapter 1 - A Brief History of Classical Animated Special Effects 1

Chapter 2 - The Art of Drawing and Animating Special Effects 15

Chapter 3 - The Specific Effects Categories 73

Chapter 4 - Liquids 87

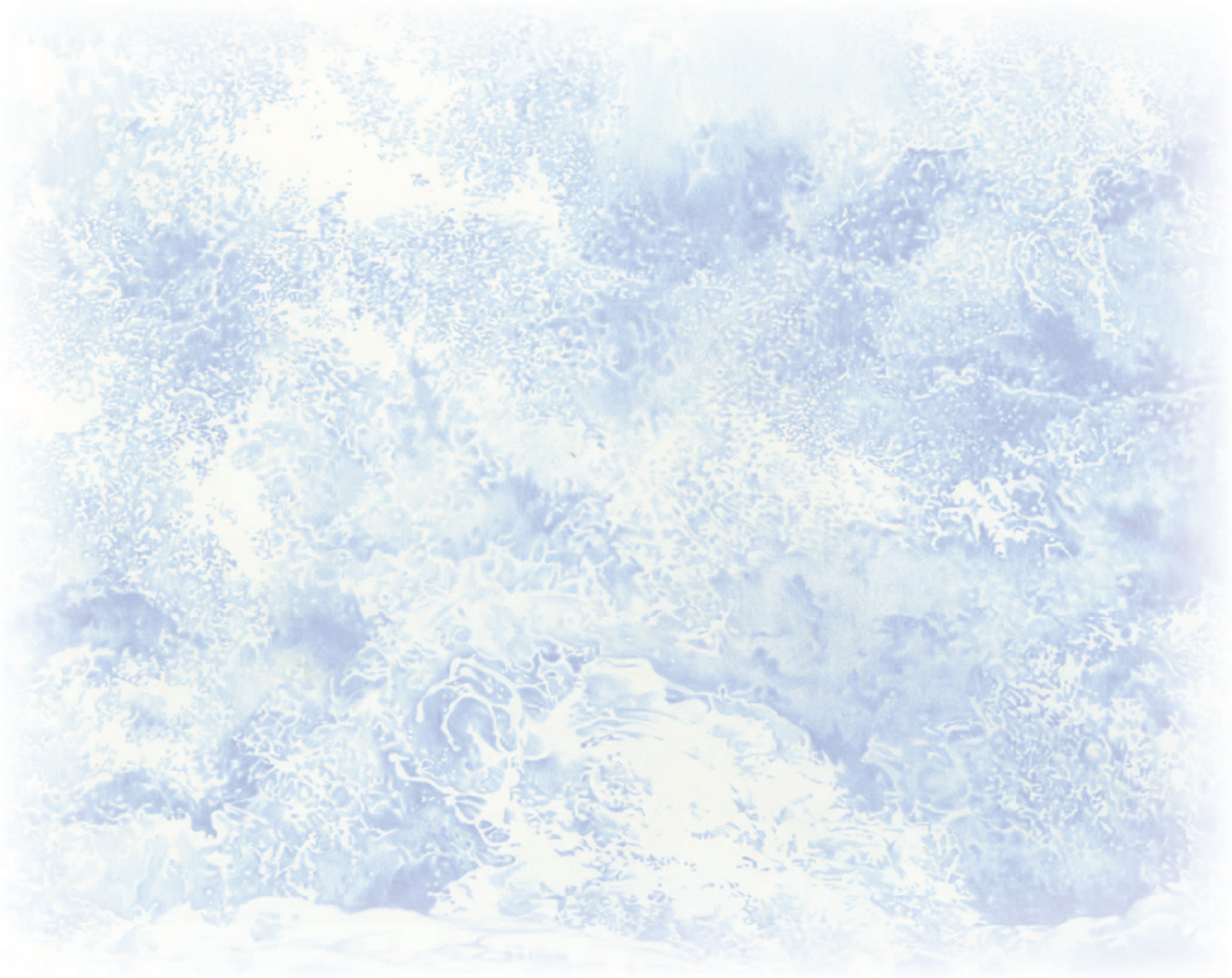
Chapter 5 - Fire, Smoke, and Explosions 161

Chapter 6 - Magic 225

Chapter 7 - Props and Everything 249

Chapter 8 - The Last Element 293

Index 297



Foreword

I was the luckiest man alive when I landed the job of my dreams and started working for Don Bluth as a character animator in 1986. Somehow though, right from the start, I felt miscast. I watched my finished work move on to the effects department, where all the cool stuff was being added: fire, water, mud, lava … you name it! So, after three difficult years in character animation, I asked to be transferred to the effects department. By far, one of the best decisions I have ever made!

Effects animation was like my newfound freedom. While staying within the stylistic boundaries of the movie, I could design my drawings without having to stick to the strict character model sheets. I could stylize the timing and motion without having to adhere to the confines of a dialogue track. I felt like a scientific magician, moving water molecules, creating lightning, and conjuring fiery whirlpools, all from the tip of my pencil.

I remember the first time I saw a scene animated by Joe Gilland. I was watching a pencil test of a duck splashing around in a pond. I was mesmerized by the intricacy of the water animation and the attention to every detail.

My first exposure to Joe's work left an indelible impression on me. Through the years, while I've drifted onto other mediums, Joe has remained dedicated to the craft of effects animation, creating wonderful work for many feature films and teaching new generations of creative professionals.

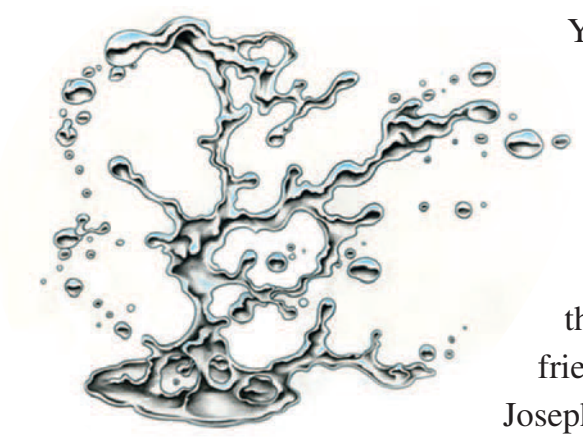
Many animators and students have asked me to write a volume such as the one you now hold. Although I saw the necessity for such a project, I never felt the calling to create it. I love doing effects animation, but I have never really tried to understand it beyond what comes to me instinctively. Joe, on the other hand, is a true scholar on the subject. When he told me of his intention to write this book, I breathed a sigh of relief and felt great enthusiasm for Joe. I knew the subject matter was in the most capable hands.

As I've said, I truly love the art of effects animation. To me, it is poetry in motion, inspired by the most primal forces in creation. I hope that the wealth of knowledge and gorgeous artwork contained within these pages will ignite the same passion in the hearts of young animators, and will encourage a whole new generation to carry on the swirling, smoking and sparkling torch of classical effects animation.

Michel Gagne July 2008

Preface

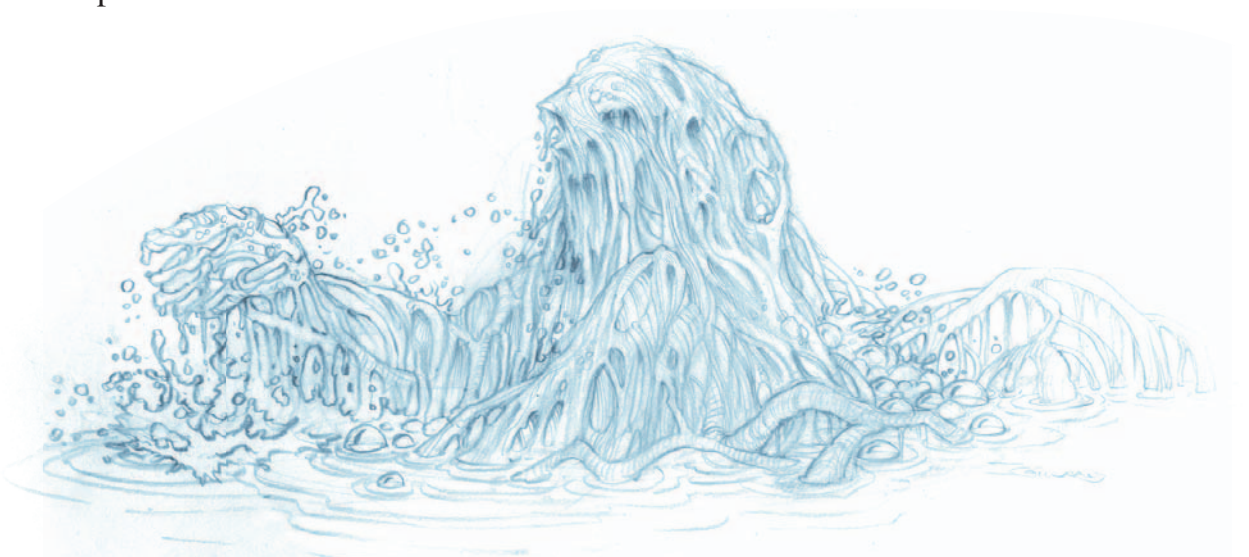
In the spring of 2004, at the Annecy International Animation Festival in France, a dear friend and I had a conversation about animation books. Interesting, we noted, that there is not a single book dedicated to classical hand-drawn special effects animation, even though it is an enormously important aspect of the animation business, and a very specialized and specific art form in its own right. While there is a proliferation of books dealing with various kinds of visual effects work, most of these are focused purely on the kinds of polished photorealistic visual effects that we are used to seeing in Hollywood action and science fiction films, using optical live-action film techniques, and more recently, computer-generated techniques.



Yet the fine art of animating special effects—like water, fire, smoke, pixie dust, and every imaginable kind of magic—by hand with a pencil and paper, has never really been documented thoroughly (much less broken down and analyzed, explained and discussed in depth). On that fateful day, in the picturesque alpine village of Annecy, my dear friend uttered the words, "YOU should write that book Joseph!" After that, I realized that I really didn't have any choice: This book simply had to happen.

So here I sit, late at night, staring into my laptop and pondering the infinite world of hand-drawn animated special effects, and the almost thirty years of my life that I have devoted to animating them. What kind of book should this be? A handbook? A textbook? A “how to” book? A pretty coffee table book? A historical document? Maybe it will be a bit of each of those things. I’ll leave that for you to decide.

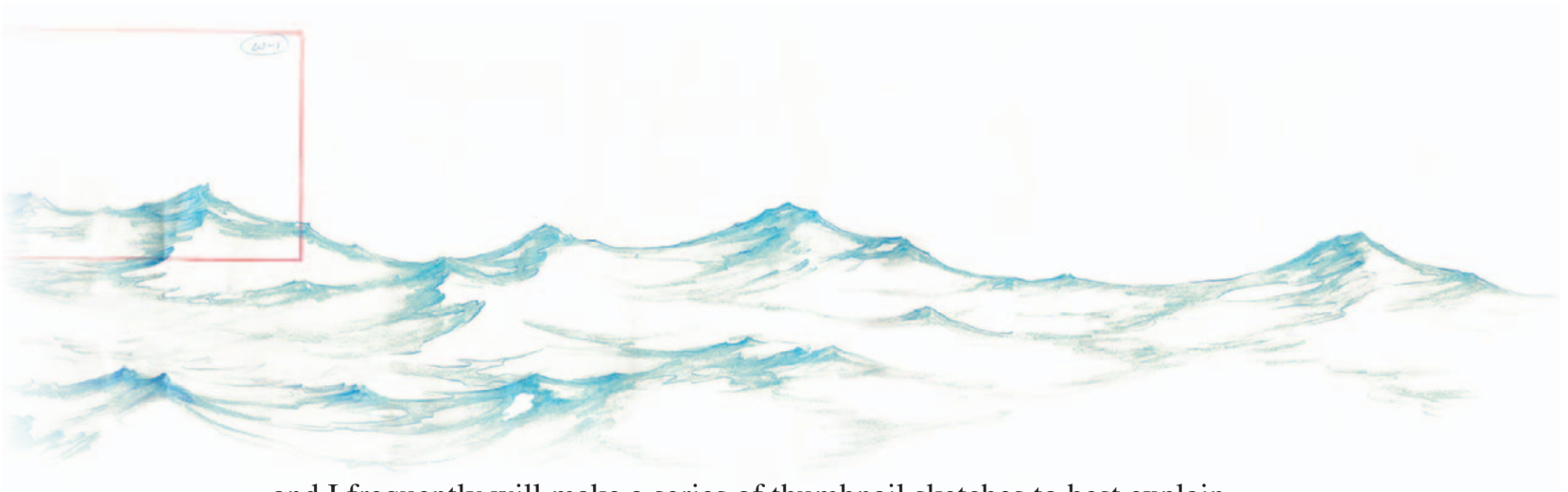
First and foremost, I hope this will be a book which pays homage to the men and women who have created special effects for animated films spanning the last 70 years. Most of them have worked thanklessly in the trenches, often in complete artistic obscurity, fueled only by their intense dedication to their craft. Considering the insane pencil mileage and the technical difficulties required by such work, one must have true passion to draw effects.



Keep in mind, too, that this has never been a path to great recognition or filthy lucre. For all of the countless animated films that we have all watched over the past seven decades, there is a universe of gorgeous artwork that is largely unknown to the average viewer. It is my goal to honor both the artisans and the craft with this book, to bring to the fore what has often been treated as a secondary, less vital part of the animation process.

Our planet is an infinite source of design perfection!

Regardless of your interest, whether you are a student of 2D or 3D animation or a casual moviegoer, I hope you will find this volume informative, entertaining, and somehow imbued with the magic that initially drew me into this mysterious world. Since September of 2003, I have been involved in teaching animation to both classical animation students and CGI animation students for the first time in my long animation career. Students often bring their special effects questions to me,



and I frequently will make a series of thumbnail sketches to best explain a particular technique or idea. I have accumulated an interesting collection of research material and lecture notes through the years, and I often make copies of these for the students, but again and again it becomes apparent that a special effects handbook would be an extremely valuable companion to other animation books such as “The Illusion of Life” and the “The Animation Survival Kit.” For my students, and anyone out there starting a career in animation, I hope that this book will delight you and inspire you to keep on drawing, and pushing the effects envelope. For those of you moving into the wonderful world of 3D visual effects, I hope this book will pique your interest and get you to try drawing these effects, and help you understand them more deeply, before using the fantastic digital tools that you have at your disposal.

An enormous aspect of learning to be an effects animator has to do with learning how to see, how to observe and soak in the phenomenon of pure energy and life all around us. As with “art” in general, and every form of it—be it acting, or writing, or dance, or painting or sculpting—we need to learn to be observers of life, to be the antennae of society, the first ones out there, inquisitive, probing, observing, questioning, examining and interpreting life.

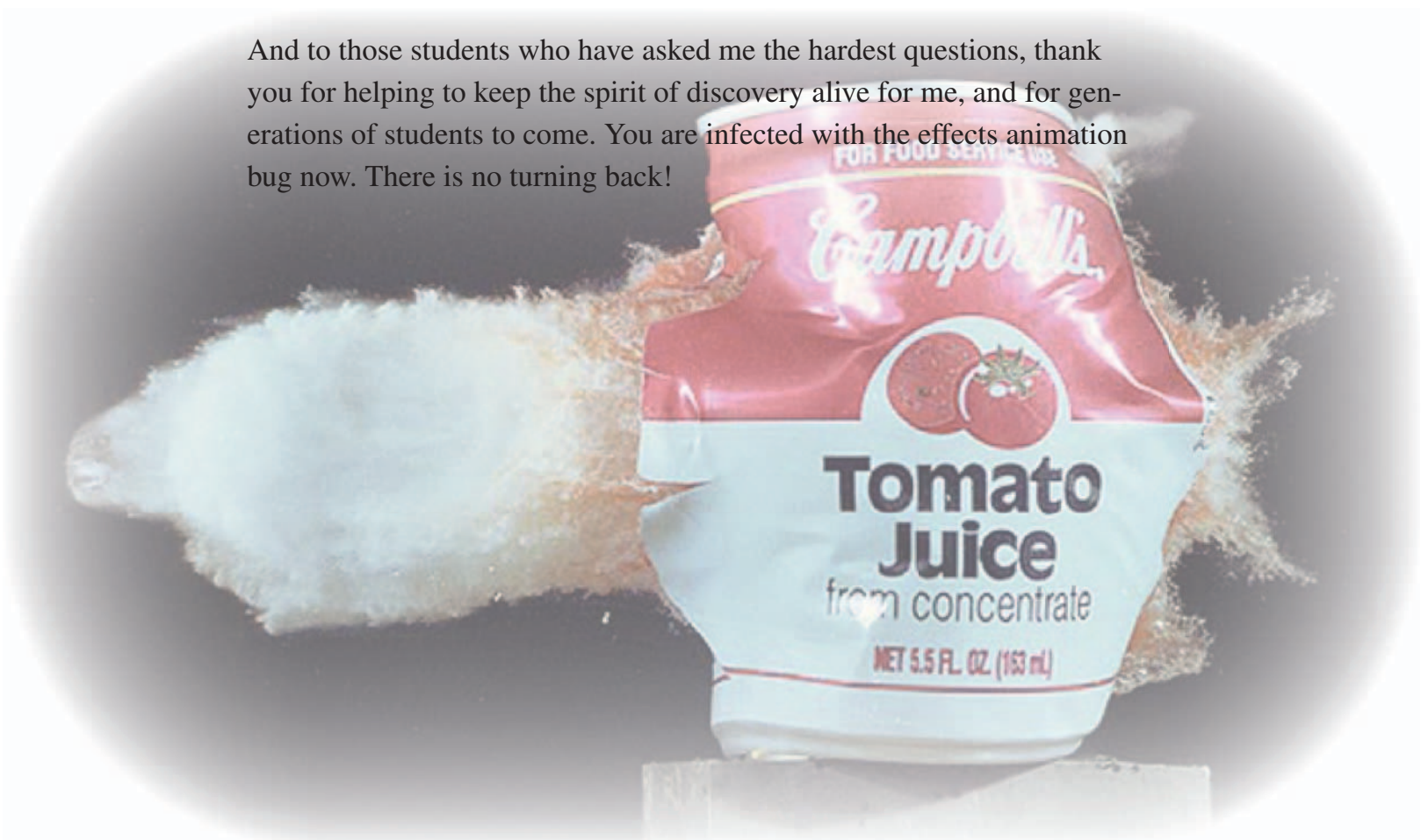
Never a dull moment conducting effects research!

It is that inquisitive spirit that the best teachers are able to imbue in their students. When we are successful as art teachers, our students are infected for life, and incapable of ever seeing this world as anything but magical, mysterious and infinite. The dedicated effects artist will never know boredom. One never needs to stop learning on this journey.

If I can somehow express that spirit of discovery and life itself here in this book, I will have succeeded far beyond my wildest dreams. It certainly has been a continuing magical journey for me, even after 30 years of animating special effects, there remains an infinity of new things to see, to draw, and to bring to life. Not to say that life is without its hardships and low points, of course. Life is certainly always a handful. But the spirit of creative discovery keeps me feeling vibrant and alive, excited, and ultimately very pleased to be writing this preface to this long overdue effects animation book.

I will bring a lot of this accumulated information to bear on this book, and I want to thank everyone I have ever worked with for adding to this wealth of information so that I may share it in this book. I cannot thank you all here, there are too many of you, and I can never thank you enough. But if you've worked with me in the past, you know who you are, and I thank you all. It has been an honor working with you.

And to those students who have asked me the hardest questions, thank you for helping to keep the spirit of discovery alive for me, and for generations of students to come. You are infected with the effects animation bug now. There is no turning back!



It is important as well, for me to note here, what this book is not. It will touch on some of the history of special effects animation, but it is not a history book. I will touch on some of the technical aspects of special effects, but this is not a book about the "camera trick" aspects of special effects animation, or the deep technical side of special effects animation, the putting together of all the artwork, the compositing, the art direction, the picking and choosing of exposures, densities, or colors. Volumes could and should be written about all of these fascinating topics, but I want to keep this one straightforward and to the point. How to *draw* and *understand* special effects animation.

Acknowledgments

It has been a long journey to finally get this book out of my system and out into the world, and there are countless people to whom I would like to express my gratitude for helping me along the way. Not only those who actually helped me complete the book, but the many who helped me earlier on in my animation career, and my life, long before this book was even an idea in the back of my head. Surely I will leave someone out who deserves mention here, but it means a lot to me, so I will do my best! For inspiration in the earlier stages of my animation journey, I would like to thank Keith Ingham, Paul Curassi, Ishu Patel, David Tidgwell, Anne Denman, Marv Newland, Barry Ward, Delna Besania, Roseann Tisserand, Dianne Landau, Darlene Hadrika, Jason Wolbert, Jennifer Claire Little, Victoria Goldner, Milt Grey, Tom Hush, Mauro Maressa, Pascal Blaise, Chris Sanders, Dean Deblois, Aaron Blaise, Chad Van de Keere, Thomas Haegle, Onni Pohl, Angela Steffen, all the Classical Animation students at VFS from classes 46 to 64, and many more. Extra special thanks to the late Ryan Larkin. And to my dear friend and kindred spirit, Mary Anne Shelton, who left this life far too soon, but left me with the greatest gift of all, you will live on always in everything that I do.

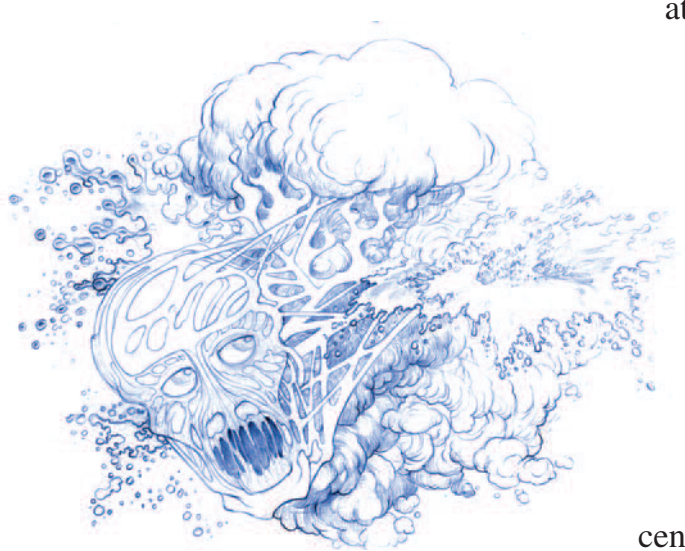
And then there are those who have contributed very much directly to this book, each in their own special way. First of all, to Rita Street, for imploring me to write this book in the first place. Thank you Rita, for seeing it inside me before I could! To Jazno Francoeur, for helping me so much with the writing along the way, you know you should probably have coauthor credits, Jazno, and I really may not have ever done it without you! To Ed Hooks, for his words of encouragement and his indomitable spirit. John Canemaker for his generous guidance. Phil Roberts, and Rick Doyle, for their inspiring artwork, photos, and encouraging words along the way. To Myriam Casper, for opening my eyes to so much, and sharing her life, her vision and her incredible photography with me. To Jim and Claire for helping me to initially get a handle on Adobe InDesign, and Peter Bromley for helping me get started with the bizarre anomaly of Quark. Special huge thanks to Michel Gagne and his wonderful wife Nancy, for believing in me and getting me started, and continuing to help all the way through to the end. To Carole McClendon at Waterfront for her invaluable patience, guidance and experience. To Chris Simpson, Mónica Mendoza and everyone at Focal Press who believed in my book. And to Marilyn, I could not have made it here without you ...

Chapter 1

A Brief History of Classical Animated Special Effects

Although this is not intended to be a history book at all, I thought it would make sense to run through a very brief history of how hand-drawn special effects animation evolved to become a well-developed art form all its own. It has been, through the decades ill defined at best, but nevertheless, special effects animation can stand on its own as an important, exciting, and vital facet of the art of animation. I'd like to take it from cave paintings to computers, and see where we end up!

While the history of animation as a whole took place not only in North America, but in Europe, Asia and pretty much all over the world, I am approaching the art of animating special effects for the most part from the Western or North American standpoint. Much as a book on the history of the fine arts might focus on a particular art period or place of origin, I cannot help but turn to the American Disney classics such as Pinocchio and Fantasia when discussing the art of hand-drawn



effects animation which, during the 1930s and ’40s at the Disney animation studios, was brought to a level of mastery that has not been matched since.

For starters, let’s go way, way back, to the dawn of man’s creative impulses, and the earliest recorded sculptures and cave paintings. What we have decided to call “art” has often reflected man’s fascination with the elements that shape the world around him. Throughout the early centuries of recorded history, we see the sun, the moon, mysterious constellations of stars in the night sky, volcanoes, floods and fires represented in man’s art. Elements and occurrences that must have frightened, confused and mystified early man. Events that we labeled “Acts of God” that awoke within us not only fear and misunderstanding, but also a profound curiosity about how the world around us might work. Artists throughout the ages have also continually depicted spiritual and magical energies. Angels bathed in light, wizards, faeries and cupids casting mystical spells. And as science, geology, astronomy, mathematics and physics evolved, there were always men and women who probed a little further into the unknown. The “mad scientists”, the inventors, the artists, the Leonardo Da Vinci’s

of the world. Often misunderstood by their peers, but nonetheless, the real shakers and movers of mankind's evolutionary path. We have within these two groups, a strange mixture of practical, scientific minds, and fanciful, creative minds. This is the legacy of the creative men and women who have evolved into the modern day special effects animation artists. Scientist/artist/magicians playing with a new set of alchemical tools, playing with light and shadow, and playing God with our imaginations!

Shortly after the turn of the century, an eclectic Frenchman named George Melies began experimenting with special effects techniques in his ground-breaking live-action films, using innovative matte paintings, double exposures, split screens, miniature models, and stop-motion animation to conjure up magical looking scenarios, the likes of which had never been seen before. Although he did not work on animated films,

Melies' did create many special effects that could be considered to be *animated*, and his contribution to the future of special effects animation cannot be underestimated. With his genius for inventiveness and experimentation, Melies innovated on the fly as he made his films, and his work paved the way for generations of animation and special effects artists to follow.

When the history of the animated film began, back in the very early 1900s, special effects such as shadows, water, fire, smoke and motion effects were not yet considered as anything but an occasional addition to the main character animation of the day.

Some of the most influential and beautiful hand-drawn special effects of the day were seen in Winsor McCay's 1905 classic "Little Nemo in Slumberland" comic strip, and his 1912 comic strip "In the Land of the Wonderful Dream" Winsor McCay was masterful with his renditions of clouds, magic, water splashes, fire and smoke. His drawings were highly stylized yet somehow incredibly realistic and true to the real underlying physics of the effects being drawn, especially considering that in that day and age there was no way of filming these types of phenomena and examining them frame-by-frame as we can today.

In the earliest animated cartoons, the special effects were generally quite rudimentary. There were often exclamation-like designs emanating from character's heads to indicate when they were surprised or afraid before sound was added to cartoons, or maybe a character spitting into a metal spittoon with a loud dinging sound, in time with the music, in the earliest sound cartoons. A character might pick up an object, like Mickey in "Steamboat Willie", to bang out a tune with spoon on pots and pans and other various props scattered around his steamboat.

When in “Gertie the Dinosaur”, Winsor McCay’s ground-breaking animation classic, a sea serpent spat water, there was some simply fantastic water animation, but not a great deal of thought or research really went into it, as the focus was entirely on the magic of bringing a character to life for the first time.

In 1930, Walt Disney Studio’s special effects department was made up of only two artists. Ugo D’Orsi, who would go on to animate the incredible water effects in Fantasia’s “The Sorcerer’s Apprentice” and Cy Young, who animated the dancing flowers in Fantasia’s “Nutcracker Suite”. But that was soon to change drastically. By 1935 and throughout what is often referred to as the “Golden Years” of animation (1935-1941) the effects department at Walt Disney Animation continued to grow and develop into a very specialized department, and by the late ’30’s with the production of Fantasia, the effects department had blossomed into a diverse collection of well over 100 highly specialized artists. Led by the genius of artists like Carleton (Jack) Boyd, Josh Meador, and Dan McManus, this department elevated special effects animation to the level of a truly fine art.

"The men and women who solved these problems were skilled craftspeople of diverse background who had been thrown together in the catchall department, special effects."

—Frank Thomas and Ollie Johnson from "The Illusion of Life"

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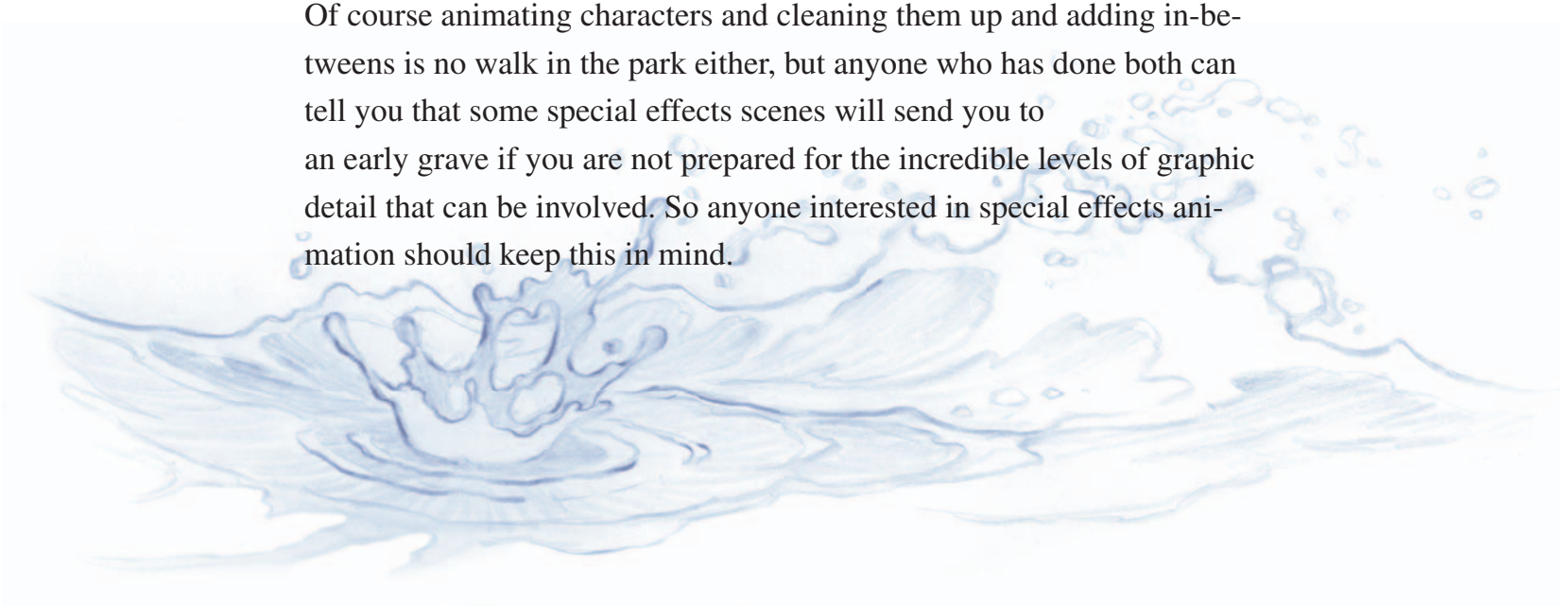
Through experimentation and observation and a great deal of very hard work, what was once a secondary aspect of animation became more and more vital to the art of animation. These were a new breed of animators, artists with a pronounced interest in nature and the elements, often with a deeply scientific approach, bringing artistic flair and originality to their work. Special effects became a passionate calling for these individuals (as it has for me) who went to incredible lengths to understand the physics behind fire and smoke, and water in all its forms, light and shadow and its interplay with backgrounds and characters, even the simplest falling leaves or flower petals would be studied

intensely to achieve a sense of magical realism. The mastery of the water animation in Pinocchio's "Monstro" the whale sequence, and the birthing of a planet, with volcanoes, earthquakes, and typhoons in Fantasia's "The Rite of Spring", were inspired achievements of animation which have yet to be equaled to this day. In Fantasia, virtually every type of effects animation imaginable was explored, researched, and to a great degree, mastered. In the film Fantasia alone, the effects animation department handled a staggering array of phenomena.

Earthquakes, volcanoes, lightning, rain, bubbling and spewing lava, bubbling mud, magical fairy dust, dandelion seed pods, glistening water drops on a spider web, ocean waves, steam, snowflakes, leaves and flower petals, magic spells, meteors and swirling galaxies, smoke, spirits, frost, clouds, and countless forms of water splashes, waves, and ripples. Even this is only a short list! Special effects, although still a very loosely defined term covering an incredibly broad range of animation styles, techniques and technologies, had truly come into its own,

and other studios soon followed suit, turning to artists with a particular penchant for understanding and animating the natural elements necessary to make their cartoons truly come to life. While it is all good and well to romanticize this amazing discipline of bringing the elements to life in animated drawings, there is another side to special effects animation that is not romantic or exciting at all. The sheer volume of work and endless pencil miles required to animate something like a heaving ocean surface or an enormous avalanche rushing down the side of a mountain, is tortuous at the best of times.

Of course animating characters and cleaning them up and adding in-betweens is no walk in the park either, but anyone who has done both can tell you that some special effects scenes will send you to an early grave if you are not prepared for the incredible levels of graphic detail that can be involved. So anyone interested in special effects animation should keep this in mind.

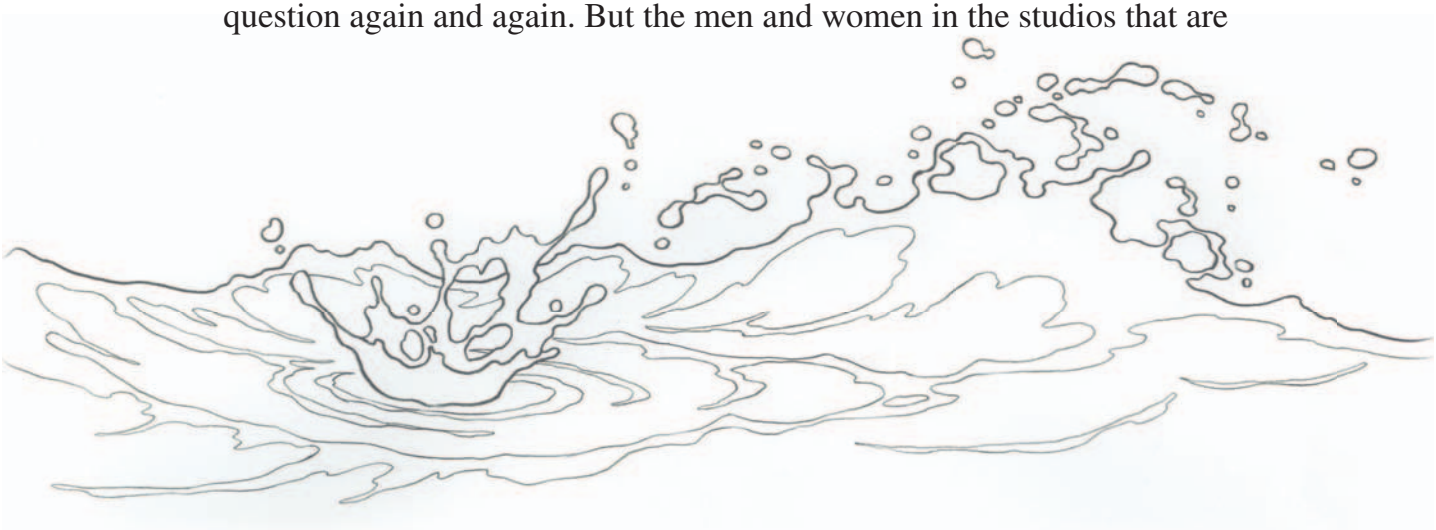


The men and women who pioneered this work were not only ingenious, inventive and brilliant, they were also absolutely crazy! And there were those in the business of making animated films who took full advantage of this newly born department full of inventors and creative outcasts. Thus was born the new hybrid effects animation department, the department that does all the stuff that nobody else in the studio wants to

do. So if it were discovered that no one in the studio had ever drawn or animated an automobile, it would get sent to the special effects department. If a character had to handle a difficult prop of some sort that the character animators didn't want to deal with, it got sent to the special effects department. Of course making animated films is always a highly

collaborative venture, there is generally a lot of back and forth and sharing of difficult tasks, and it would be an oversimplification to say that all the difficult, gnarly animation jobs got sent to the effects department. However, I can say without a moment's hesitation that the effects departments in most animation studios I ever worked in, evolved into being catchall departments for all the oddball jobs that come up when you are making an animated film.

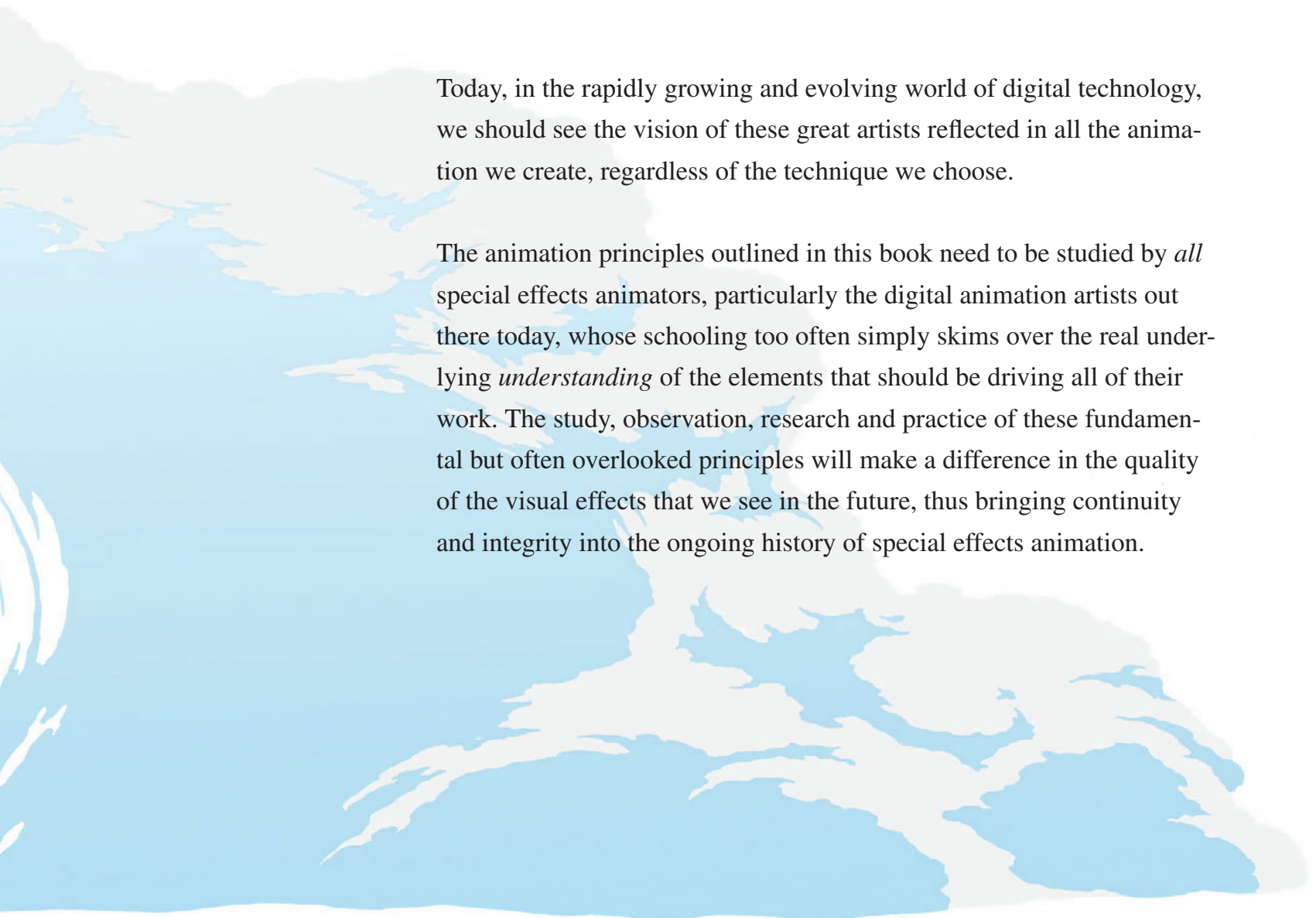
Today, the modern art of animation filmmaking relies more and more heavily on computers. Animation artists have adapted quickly, realizing that computer technology removes a great deal of the difficulty and drudgery involved in producing animated films. Recently, the most successful animated films have been almost entirely computer generated (with the notable exception of the Japanese animé market) and the future of hand-drawn animation in feature films has been brought into question again and again. But the men and women in the studios that are



making the most successful films out there today know better. The success of these 3D films relies heavily on the experience and expertise of classically trained artists who have successfully learned how to work with computers.

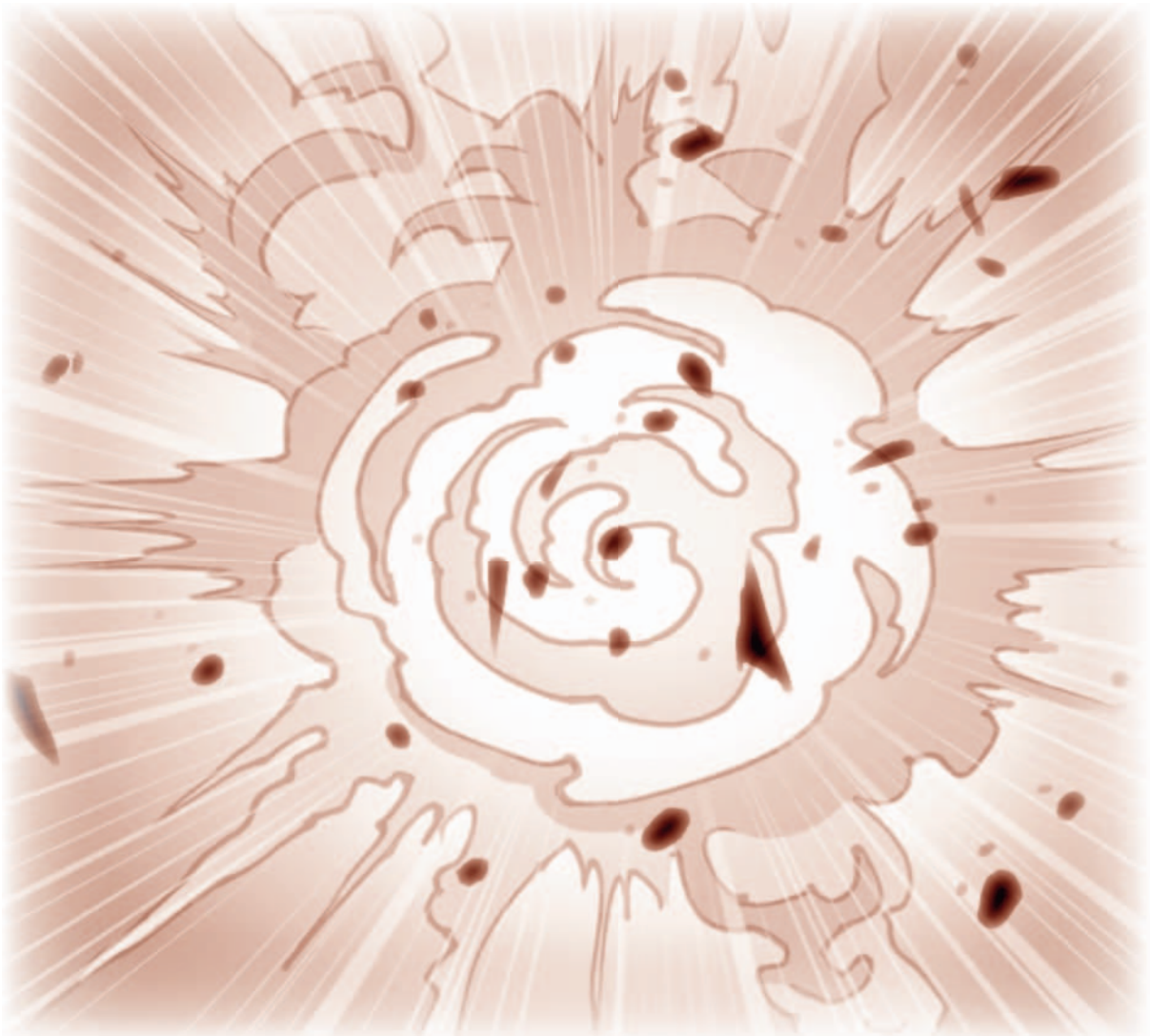
The background knowledge of how to bring our artwork to life has not changed. Only our paintbrush has changed. With every new advantage that computers have given us, new challenges have arisen, and the most serious challenge today, is to ensure that the classical art form of learning how things really work is preserved.

Let this book be a testament to the men and women who mastered the art form back in the 1930s and ’40s, and left behind for us a body of work that is staggering in its depth of creativity and in its understanding of the elements and physics in general. Anyone who attempts to learn these old school special effects techniques can’t help but greatly admire these phenomenal animation artists. And the skill and understanding that this study will bring to any animation artist’s work cannot be underestimated.



Today, in the rapidly growing and evolving world of digital technology, we should see the vision of these great artists reflected in all the animation we create, regardless of the technique we choose.

The animation principles outlined in this book need to be studied by *all* special effects animators, particularly the digital animation artists out there today, whose schooling too often simply skims over the real underlying *understanding* of the elements that should be driving all of their work. The study, observation, research and practice of these fundamental but often overlooked principles will make a difference in the quality of the visual effects that we see in the future, thus bringing continuity and integrity into the ongoing history of special effects animation.

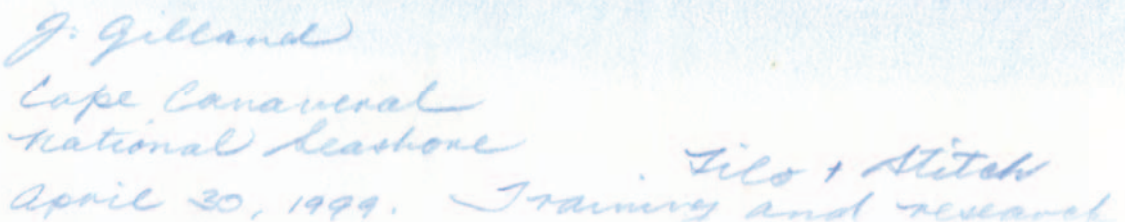


Chapter 2

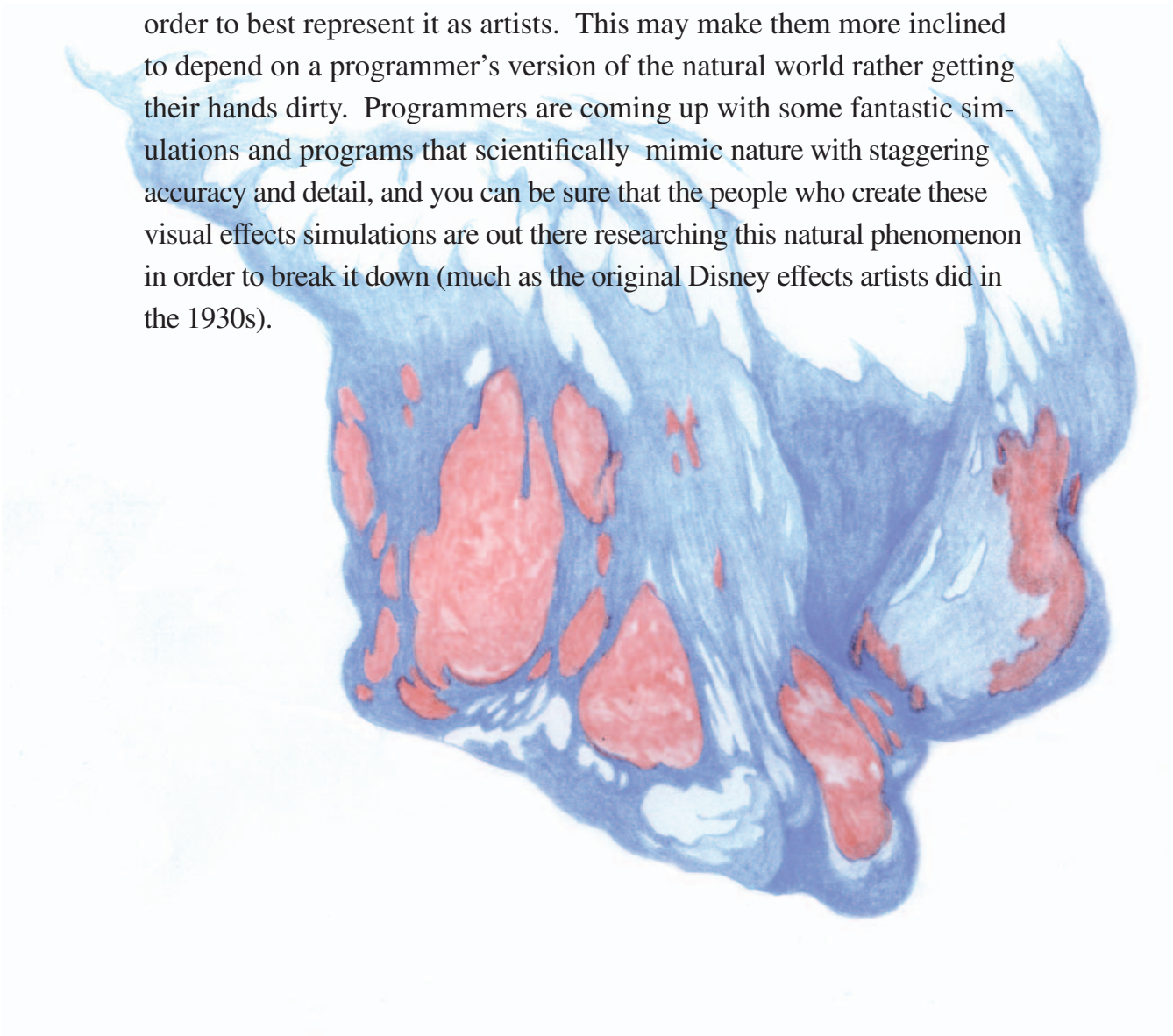
The Art of Drawing and Animating Special Effects

Given the amount of visual effects that bombard us on movie screens around the world today, you would think there would be at least one book dedicated to the process of effects design and visual physics from a more “classical” approach. My greatest incentive for creating this book is the complete lack of such information currently available. Neither animation professionals nor students have adequate references describing in detail the classical approach to drawing and thoroughly understanding special effects animation.

With the incredible digital design and animation tools we now have at our disposal, countless books are being written on this topic, with their primary focus being on understanding how to use these relatively

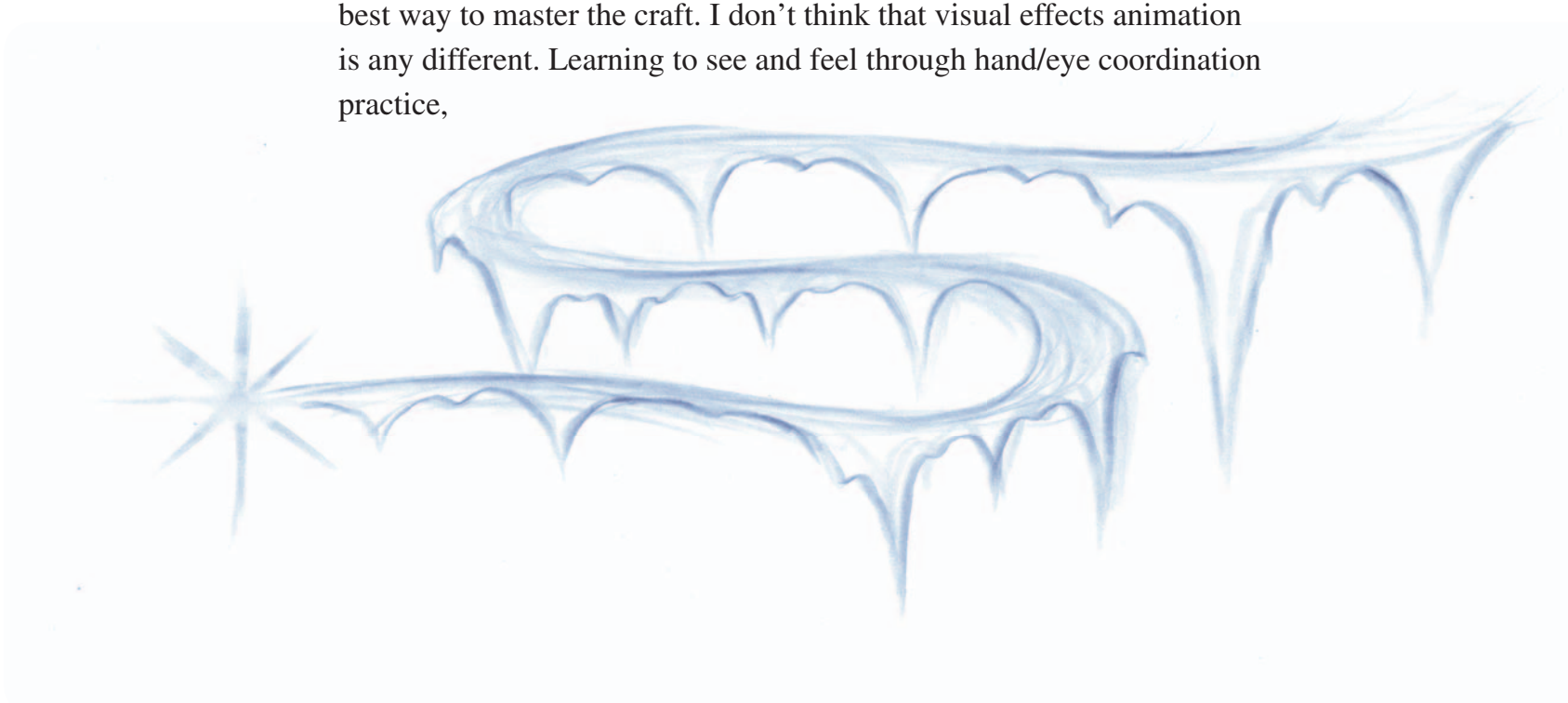


new digital tools, while less emphasis is being put on understanding the actual phenomena which we are attempting to recreate. More time is being spent teaching young artists how to manipulate complex computer graphics software, and less time is spent teaching them how to observe, how to see and intuit the incredible subtleties of nature's splendor in order to best represent it as artists. This may make them more inclined to depend on a programmer's version of the natural world rather getting their hands dirty. Programmers are coming up with some fantastic simulations and programs that scientifically mimic nature with staggering accuracy and detail, and you can be sure that the people who create these visual effects simulations are out there researching this natural phenomenon in order to break it down (much as the original Disney effects artists did in the 1930s).

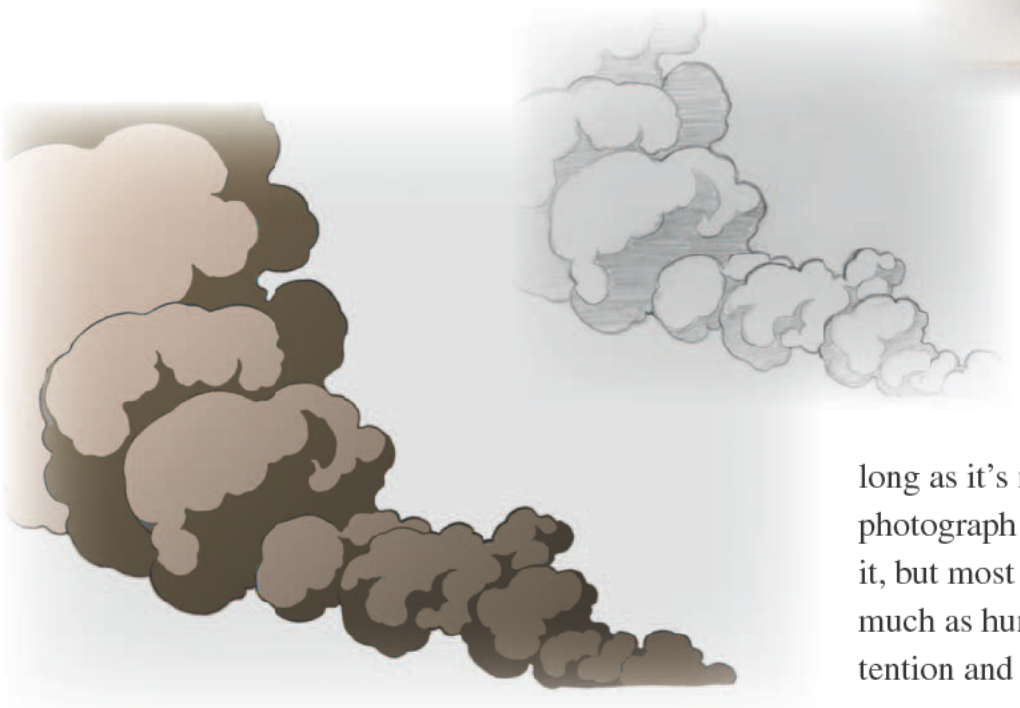


In my experience supervising digital artists over the last ten years, I have consistently come across individuals who have developed a high level of design skills as well as a high degree of proficiency with their digital tools, but who haven't developed the sense of feeling the physics of what they were trying to animate. During the course of their schooling, the steps of understanding natural phenomena through the practice of research, touch, feel, personal interaction, and drawing had been all but left out.

This is true of both the digital visual effects animators and digital character animators entering the animation business today. The top CGI studios and directors are well aware that the study of character animation through traditional observation and drawing techniques is the best way to master the craft. I don't think that visual effects animation is any different. Learning to see and feel through hand/eye coordination practice,



observing, drawing, sculpting, animating by hand–all of these are invaluable if an effects artist wants to get to that mystical place where time, space, physics and motion become fully natural and intuitive. Get out there and research your subject, get to know it inside and out, like a method actor who throws himself into



a demanding role: if you want to animate something well, become intimate with how it works and feels. Touch it, taste it (so long as it's not toxic), draw it, photograph it, sculpt it, analyze it, but most of all observe it as much as humanly possible. Attention and observation is key!

I have spent endless hours at the beach watching the ocean, letting it fill in and around my senses. If you are attentive, and receptive, you can let the visual information inform your subconscious simply by being there with your eyes open. Sketching and taking photographic reference is beneficial, but I cannot stress enough the value of simply letting the information in.

Balance this relatively inactive practice with lots and lots of drawing. The hand/eye coordination is of the utmost importance, whether you end up drawing or using a mouse or tablet to draw digitally. Remember that no matter how well you master your drawing technique, if you can't feel the effects you are trying to animate, they will not sing with life. An imagination that is full of life, must inform the well-trained drawing hand!

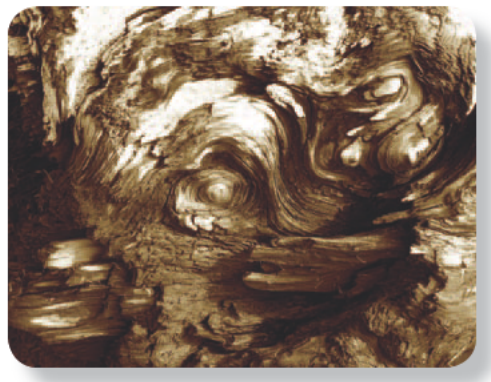
The incredible studies of reef structures and wave behaviors of master surf artist Phil Roberts, are the result of decades of experience and research, coupled with his incredible drawing and painting skills. Wherever your life takes you, keep your eyes opened, as the magical energy of the elements dances all around us, at every moment. Whether we are in a tropical paradise, waiting for the next ocean swell on our surfboards, or being splashed by a taxi in downtown rush-hour traffic, the information we seek is right under our noses!

On the following pages, I have illustrated the relationship between forms commonly found in nature, and the energy patterns that inform all elemental special effects animation, like smoke, fire and water. Whether it is a gnarled up old piece of driftwood, the root system of a tree, a puddle in the sand, a random pile of beach debris, or a perfect curling wave, the same energy patterns are inherent in all forms of nature.

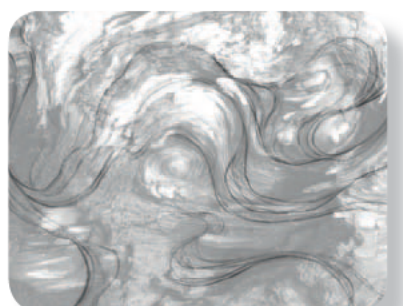
These building blocks of life, connect the very tissue of reality as we know it. It is everywhere around us, if only we open up our eyes to it. From the macrocosm, to the microcosm, the same energy holds it all together, and this magical energy, which science can't yet even begin to fully understand, must inform every stroke of our pencil, when we design and animate elemental magic.

This magical energy must inform every stroke of our pencil.....

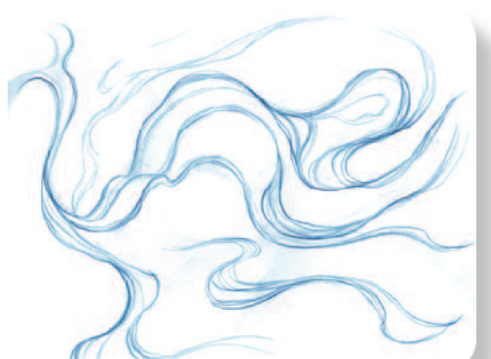
As the head of the visual effects department at Walt Disney Feature Animation in Florida, I took my effects crew on several very interesting field trips. On one such outing we spent an afternoon drawing and cavorting at the beach, on another occasion we went to Cape Canaveral to watch the space shuttle lift off.

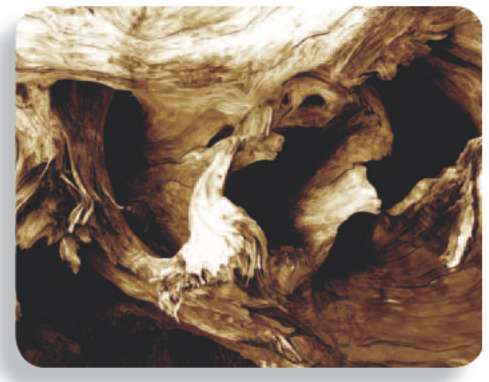


I also encouraged experimentation whenever possible. The practice of conducting experiments to see how things really work is one of the most fascinating aspects of understanding special effects animation.

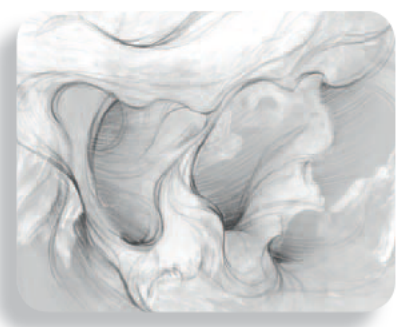


Sometimes the experiments and demonstrations were surreal (as evidenced by a fire chief who seemed to enjoy lighting things on fire a bit too much), and other times they were downright scary (as when the Disney theme park folks incinerated a giant cement bunker filled with expired fireworks).

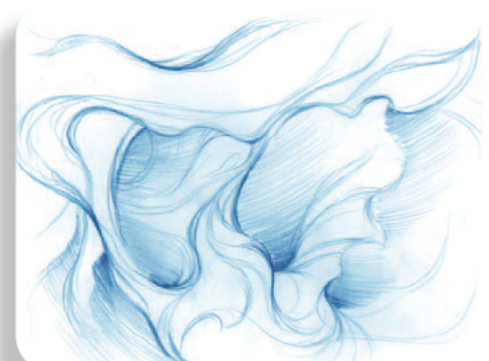




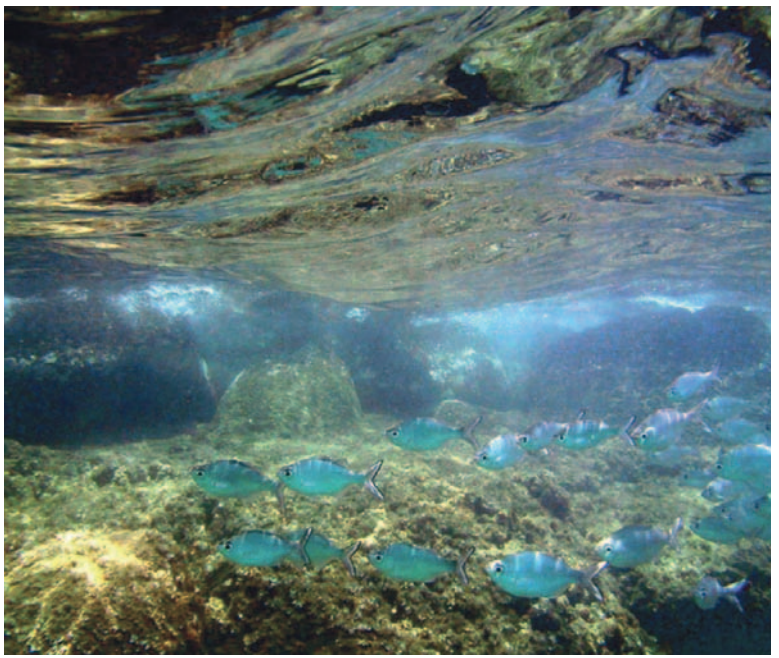
By far the most hilarious experiment was our department's annual "Destructo Day," where all manner of objects were smashed, blown up, or otherwise obliterated. Some of us drew pictures, some took still photography or video, and the rest simply watched.



Not only were these entertaining team-building experiences, but I believe every one of us learned more than we ever could have by sitting in front of a drafting table or a computer.

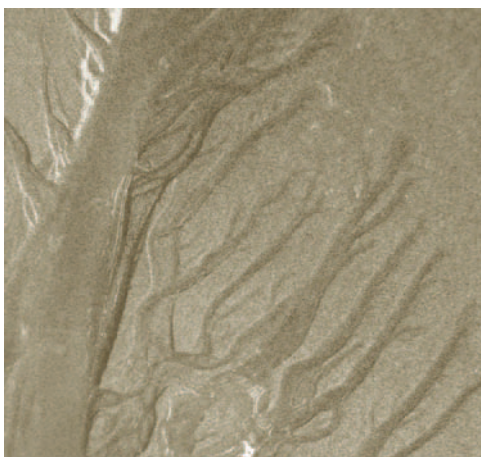


While on assignment in Hawaii to do visual development research for the film "Lilo & Stitch," it was my job to get out there and study the waves of the beaches, rocks and cliffs of the Hawaiian shoreline. I watched surfers for hours, and shot hours of videotape. I got in the water and felt the swells moving me,



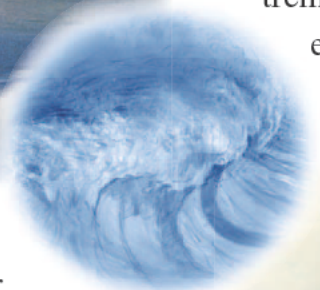
I got under the water and looked up to examine the way the sky, clouds, mountains and trees distorted through the surface. While I was playing in the water, our directors, art director, production designer and animators were observing the Hawaiian people, the villages, the mountains, and the forests through their respective media, i.e., drawings, paintings, photography and video.

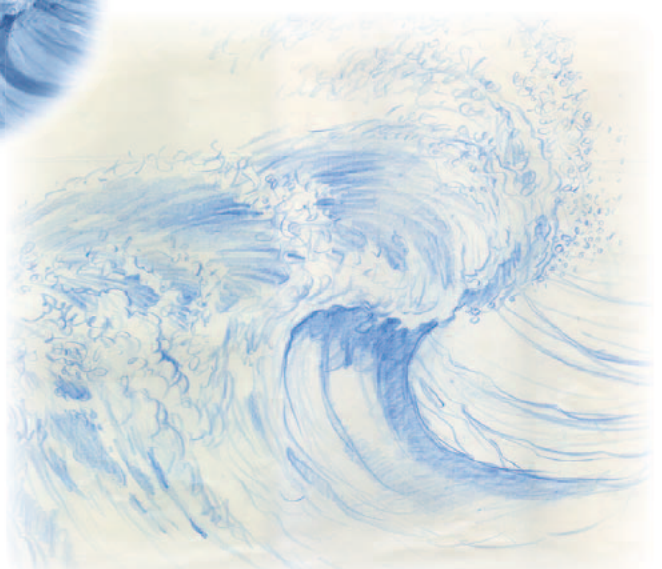
This was not frivolous spending on the part of Walt Disney Animation Studios, or some crazy junket contrived by artists chained to their tables too long (and wishing they were in, say, Hawaii). This was because the company knew full well that observation of your subject matter is paramount if you want to imbue your art with soul and integrity.
In a nutshell: Know your subject matter. Get out there and experience it, even if it's just in your own back yard.



So, let's say we have done our research, our experimentation and our observation. How do we take all this information and get it onto paper?

What drawing techniques do we need to embrace? How does one capture the extremely complex shapes of elements, like fire and water, and invest them with "life"? The answer is simple: loosen up, and let your drawing hand flow. In the same way a good life-drawing teacher will encourage us to quickly draw a figure's "gesture," we need to spend a lot of time observing our model, which in our case is the natural phenomena we wish to recreate.



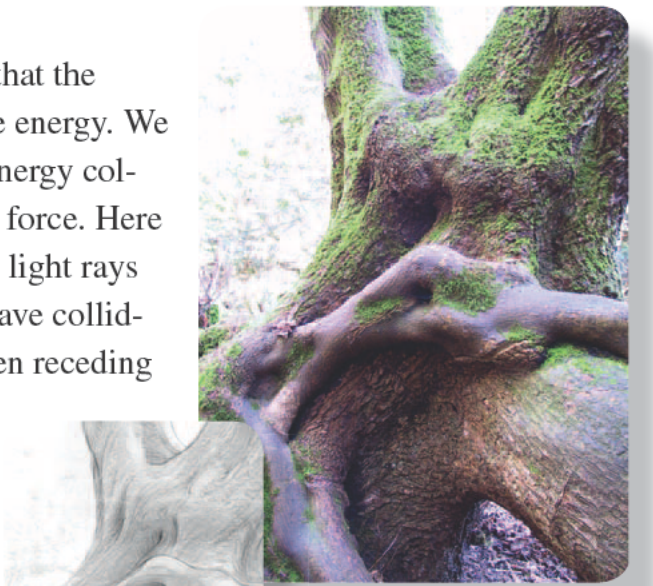


A pile of seaweed can contain all the energy of the universe!

Draw with as much abandon and freedom of movement as possible. Although the designs and patterns we wish to represent may be filled with intricate details, what we need to capture first on our blank sheet of paper is the energy and movement underlying the detail. Every wave, ripple, splash, lick of flame, or plume of smoke has beneath its details a natural energy form much like a human's underlying form of skeleton and muscle. Gravity, hot and cold air, and any number of outside forces may inform that energy pattern.

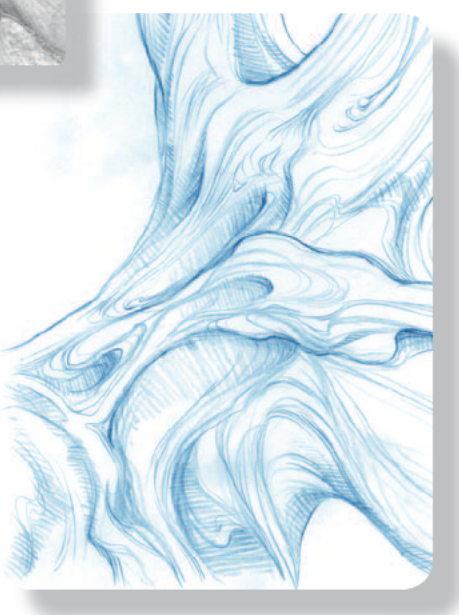
Once in motion, these energy patterns play out in a uniform force, moving in a beautifully logical way, until the underlying energy dissipates. Whether it is a house on fire, or a 40-foot wave rolling toward the coast of Hawaii, there is cause-and-effect playing out in the metamorphosis of energy patterns. That energy must inform our hand as we sketch our initial effects animation drawing. If we have studied our elements thoroughly, we will instinctively move our hand with arcs and paths of action that possess this very energy. In the chapters to come, we will discuss these patterns in

detail. The emphasis here is to understand that the underlying substance of every effect is pure energy. We will also discuss what happens when that energy collides with an object, or an opposing energy force. Here are just a few examples of this phenomena: light rays hitting an object and casting a shadow; a wave colliding with the shoreline creating splashes, then receding and colliding with the next approaching wave; smoke billowing upwards and spreading out across a ceiling; and raindrops landing on a sidewalk.

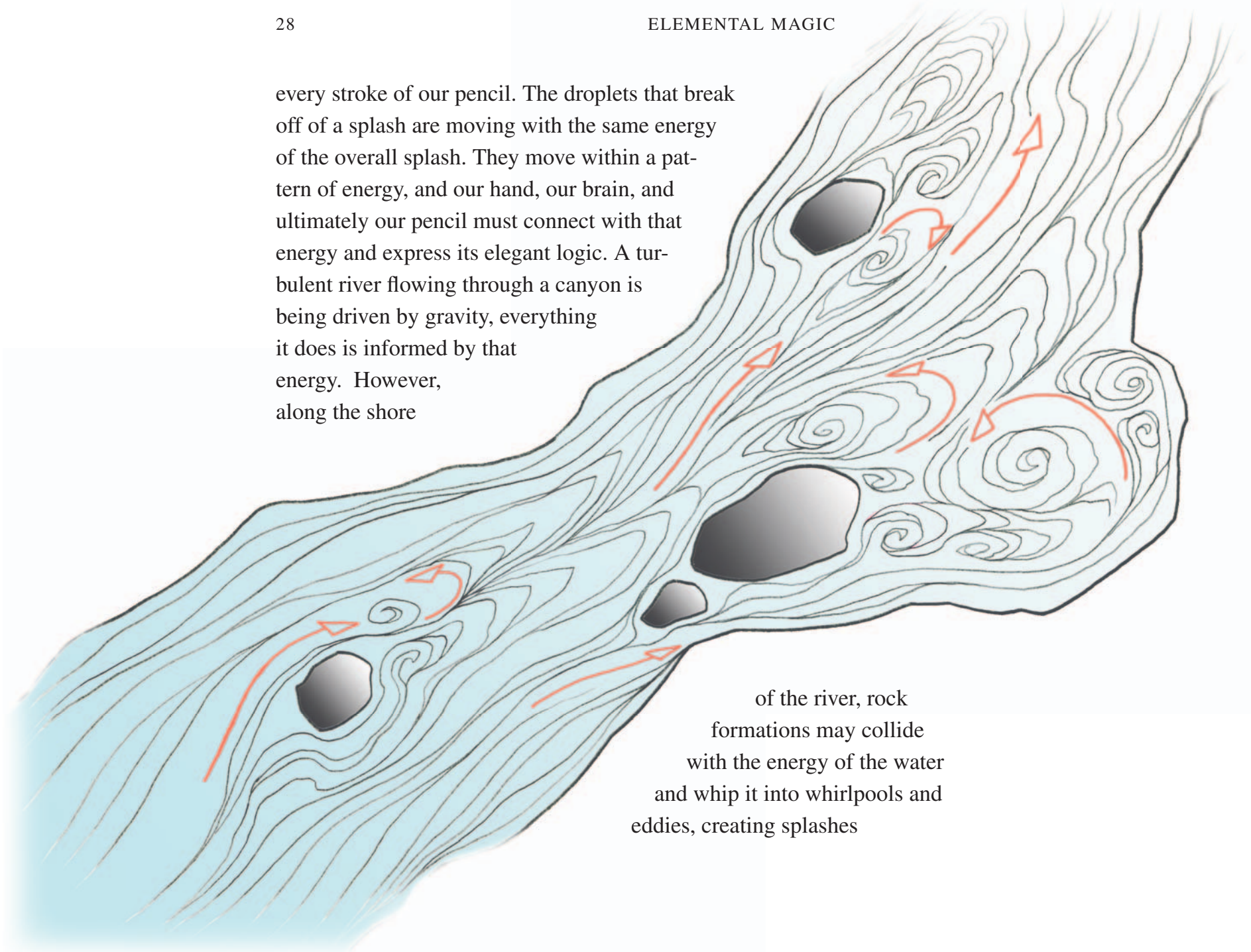


This tree trunk has an incredible amount of energy in its form!

So if the structure of our effects drawings contains this energy, this thrust of pattern and motion, then what of the details? Where and how do they fit into the picture? This is the key to every great effects drawing. Every tiny ripple on an ocean's surface, every droplet of a splash, every detail of a nuclear mushroom cloud is informed by a similar energy pattern. Sometimes it is the same force that drives the overall effect; sometimes it is a subset of that energy, or an opposing force. As we finesse our effects drawings and add details, these energy forces must inform

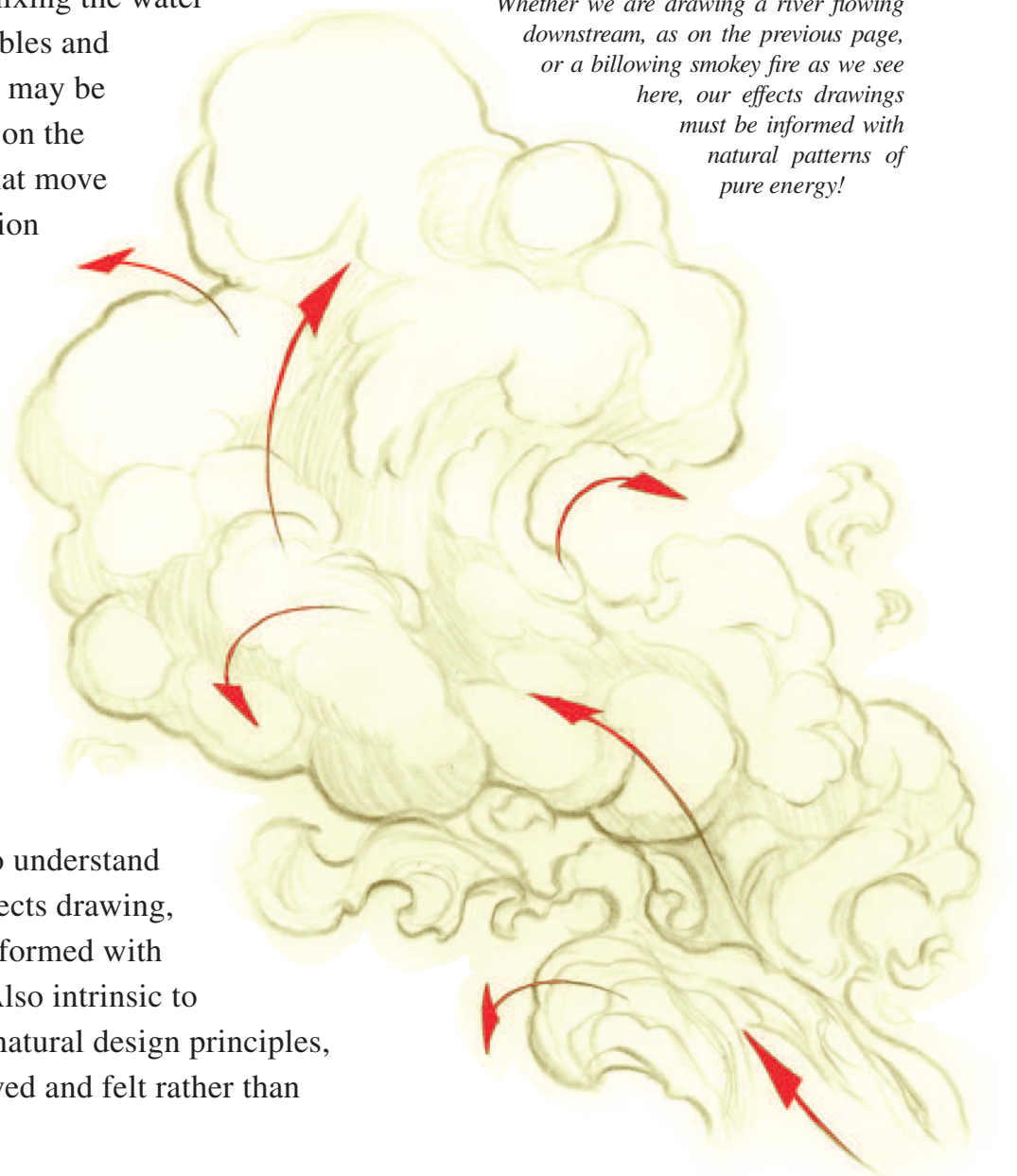


every stroke of our pencil. The droplets that break off of a splash are moving with the same energy of the overall splash. They move within a pattern of energy, and our hand, our brain, and ultimately our pencil must connect with that energy and express its elegant logic. A turbulent river flowing through a canyon is being driven by gravity, everything it does is informed by that energy. However, along the shore



of the river, rock formations may collide with the energy of the water and whip it into whirlpools and eddies, creating splashes

and spiral patterns, mixing the water with air to create bubbles and foam. A gust of wind may be creating small waves on the surface of the river that move in the opposite direction of the river's flow. All these forces, or overlapping actions, need to be taken into consideration when creating a special effects scene. While some highly detailed effects drawings may appear to be filled with random details, it is very important to understand that if it is a good effects drawing, every detail in it is informed with a pattern of energy. Also intrinsic to effects drawings are natural design principles, which are best observed and felt rather than



Whether we are drawing a river flowing downstream, as on the previous page, or a billowing smokey fire as we see here, our effects drawings must be informed with natural patterns of pure energy!

scientifically understood, though I will do my best to describe the more scientific aspects of these design principles in the following pages. By carefully studying patterns in nature, one begins to discover a cosmic similarity in all the natural elements on this planet, which are at once perfect and chaotic, yet wholly unique (for instance, every snowflake has its own discrete design).

The patterns in nature shift between the macrocosmic and the microcosmic, bridging the distance from the cosmic to the cellular. The spiral of a galaxy is reflected in the twist of our DNA, a budding fern, a snake's skeleton, and a snail's shell. The branching out of pure energy as it flows from place to place is evident in a bolt of lightning, in the spreading branches of an oak tree, in the veins that run through our bodies, in a crack on the sidewalk, or in the arterial channels of a river delta flowing into the ocean.

"The fractal beauty of a mountain range that takes our breath away can be seen in ocean waves heaving in a storm."

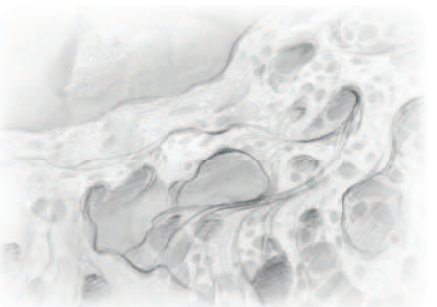
"The patterns in nature shift between the macrocosmic and the microcosmic, bridging the distance from the cosmic to the subatomic."

Study nature, observe and feel the patterns of energy inherent in the world we live in.

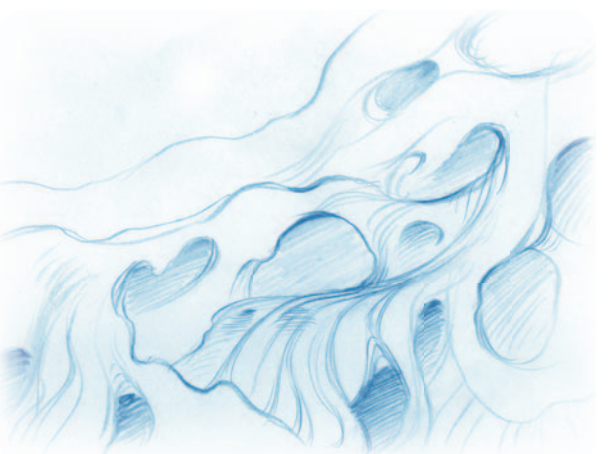
By carefully studying patterns in nature, one begins to discover a cosmic similarity in all the natural elements on this planet.

The fractal beauty of a mountain range that takes our breath away can be seen in ocean waves heaving in a storm, or in a single rock held in our hand. Even if we are animating supernatural phenomena, like pixie dust or flying spirit entities, there must be contained somewhere deep within our drawings these archetypal principles of natural design. They are everywhere around us, and yet escape the eye and the hand of a great many artists.

Study nature, observe and feel the patterns of energy inherent in the world we live in. To accurately illustrate these patterns is one of the greatest challenges in creating beautiful, yet accurate, special effects design drawings. For some reason, there is a human tendency to create repetitive shapes and to err on the side of bilateral design. Our left brain takes over and forces our drawing hand to repeat shapes



whether we want to or not. This is an aspect of drawing effects that confounds every practitioner of the craft. Even after all these years, I've noticed this annoying tendency can still grip my drawing and turn it into a boring, often symmetrical affair full of repeating shapes and parallel lines, which then need to be reworked again and again. Sometimes the intuitive feeling of natural phenomenon flows from every stroke of the pencil, other times I find it is necessary to seek out and destroy those pesky repetitious shapes, (or "parallelisms").

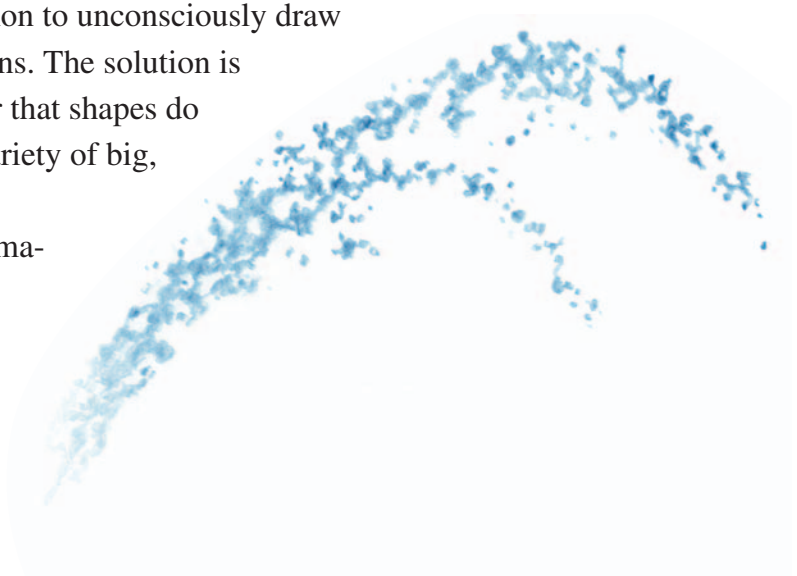


What is essential in most forms of architecture and technical drawing is the death of a successful organic effects drawing. (I say "organic" because effects that are "inorganic," such as props, moving machinery, crumbling buildings, etc. require a more linear, pragmatic mode of thinking and drawing.) Much of this tendency is a result of pattern recognition, and our equating certain phenomenon with clichéd images, like employing symmetrical repeating popcorn shapes to describe clouds. "Iconifying" an

If we walk by the same fountain every day, do we really "see" the intricate and wonderfully varied shapes of the water as it flies through the air?

object may work for hieroglyphics and corporate branding, such as the CBS eye logo, but it has nothing to do with the true mechanics of the object it describes. An eye is not an almond shape with a circle in the middle, just as water is not a series of repeating scallop shapes. If you were to take a single water splash and darken it in completely, we would only be left with its "silhouette," which is to say its primary shape. Within the primary shape, there are secondary shapes, where most of the intricate details occur; this also happens to be where much of the twinning occurs.

In hair, fur, and organic objects such as grass and leaves, it is common to unconsciously draw "buzz saw" patterns. The solution is to make sure the spacing is not too even or that shapes do not tangent directly, and that there are a variety of big, medium, and small shapes. This creates contrast, and in every aspect of effects animation one is confronted with the need for it, whether one is drawing splashing water, magic particles, powdery snow, bubbles, a cluster of rocks, or a shattered object. There is a tendency for the new effects artist to "noodle"



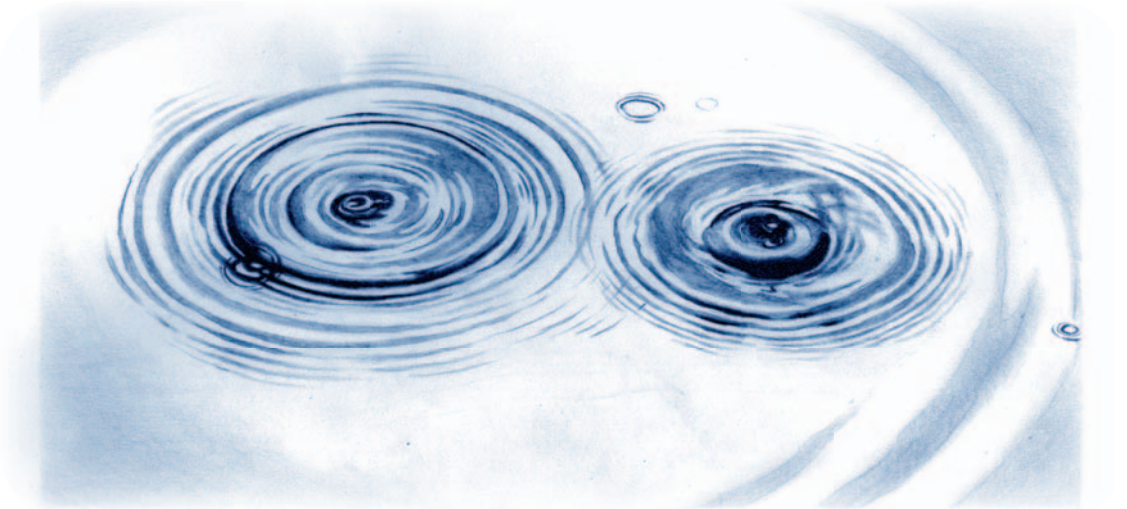
On the left we see an iconic symbol for water, with its scalloped, repetitive shapes, as we might see on a theatre set, or in a children's story book. Real water looks more like the random shapes on the right.

Here on the left we see the iconified "hot-rod" flames that we typically see on the gas tank of a motorcycle, or a pair of surf shorts. Real fire has far more chunky and unusual shapes!

The popcorn cloud on the left is very typical of a graphic cloud representation. Its shapes are repetitive and uninteresting. The cloud on the right is far more realistic, its shapes are more random.

the secondary shapes, but neglect the primary silhouette. This often results in highly detailed drawings which are confusing to look at and even more confusing to animate. Sometimes the most complex phenomena happen right under our noses, when we rinse our razor in the sink, when we pour cream into our morning coffee, or when we let out a large breath of air in the freezing cold as we reach for the paper. And yet we automatically default to a mode of non-observation, where these things have no particular significance. Every human being has an innate feeling of how the world works because of these rote, almost unconscious daily observations. That's why even a car mechanic or a civics

At first glance this beautiful photo of some ripples looks extremely simple!



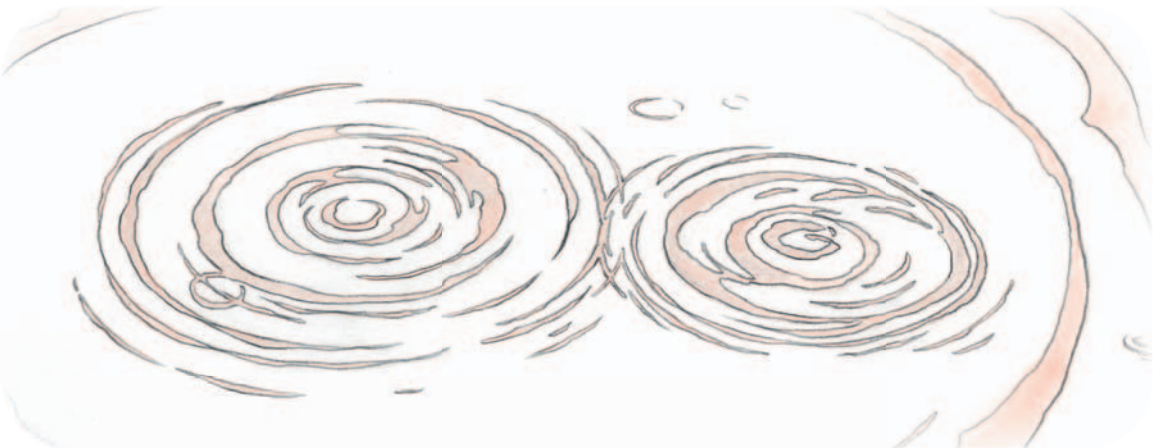
But if you try drawing it accurately, you'll soon find out that there are literally hundreds, if not thousands, of intricate little details which make it look like it does!

teacher could watch an animated film and actually sense whether or not the effects are well animated. Why? Because they know what water looks like … they drink it and shower every day! Therefore they have a built-in sense of what it should "look" like, even if they couldn't articulate why. But if you put a pencil in their hand and said, "Draw water," most laypersons would simply draw the iconic representations they have used since childhood.

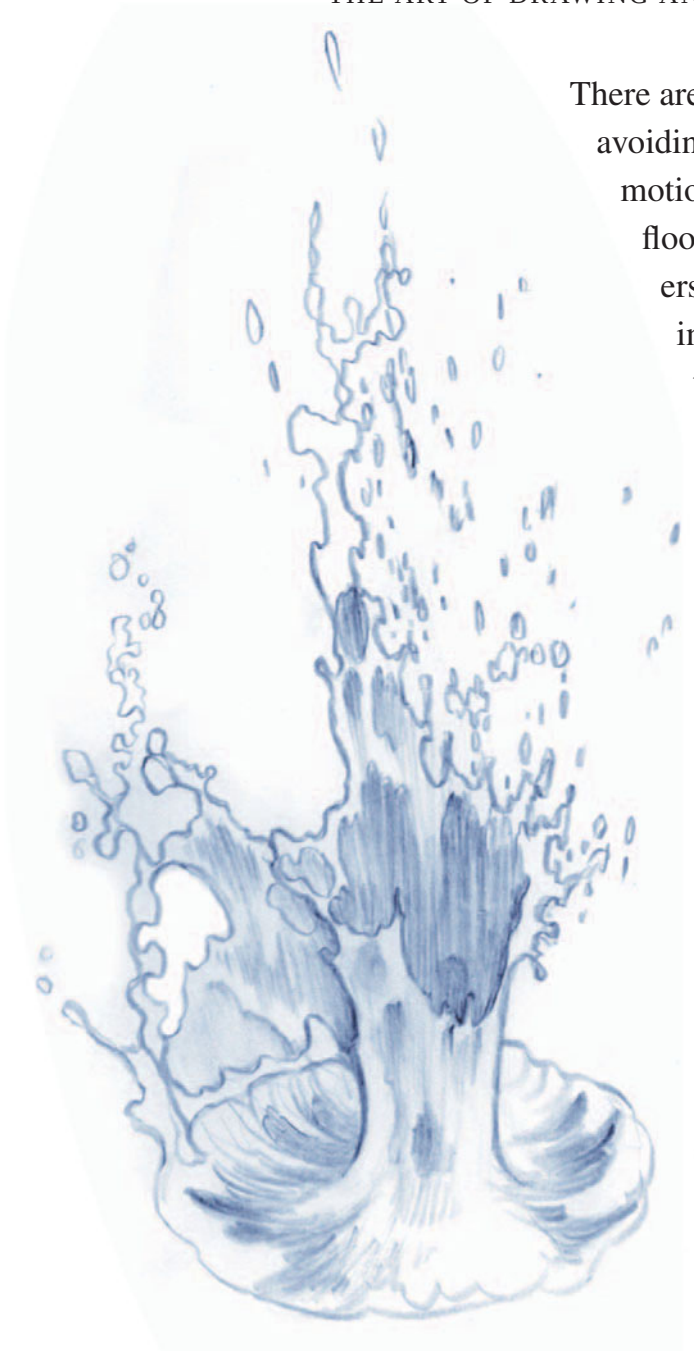
This brings me back to the importance of observing nature rather than observing modern art or the graphic arts advertising world, which is filled with simplified representations and clichés (which do little to honor nature's pure design).

If you want your special effects to feel natural, observe the forces at work in the natural world, and put those forces into play through your pencil! Remember, water doesn't think about what it's doing, neither does a ceramic bowl being dropped on the floor. All these phenomena are informed by the laws of physics and follow fundamental patterns of energy.

The same principle of avoiding patterns of repetition and symmetry can be applied to the timing of effects animation. Repeated, even timing quickly becomes apparent and unnatural; to make effects animation more dynamic, we need to overlap the timing of our elements and the directions of overall movement as well.



Reducing the amount of detail considerably, we get to a drawing that is actually manageable to animate. Even with 80% of the detail left out, it is still quite challenging to animate!

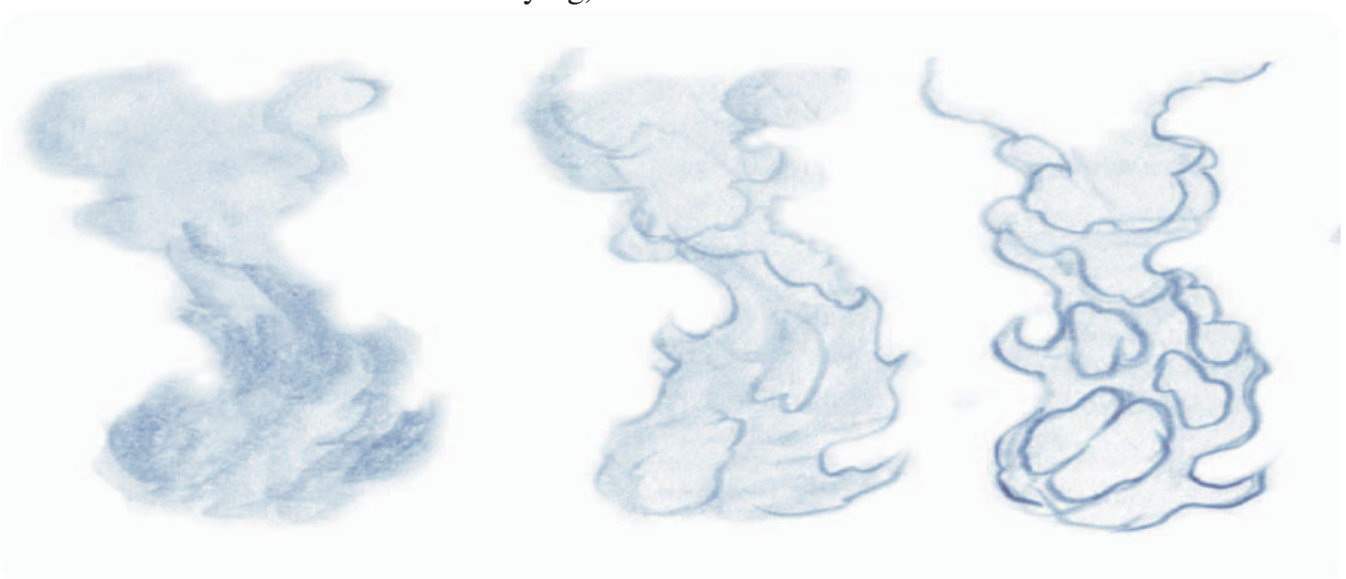


There are endless ways we can practice the art of avoiding repetition (or "twinning) in effects design, motion, and behavior. If a bottle smashes on the floor, some of its broken pieces will bounce, others will slide along the floor, some will fly off into the air, spinning. Yet if all the pieces do the same thing, if they all uniformly scatter and resolve at the same time, our breaking bottle will look very boring and unnatural. Animation appears stiff and mannered if there is no contrast in the shapes and their respective timing.

We call this variety of movement choreography. Think of it as a magic trick, or ad copy: We need to direct the viewer at all times where to look. Even in moments of supposed randomness and chaos, as in an exploding planet or a huge avalanche, there's always an underlying logic to what is happening, whether one is looking at a single drawing or at a series of drawings.

It is interesting to note that in every crew, there are always those more disposed to drawing certain types of effects. Some specialize in organic elements, and others thrive drawing inorganic things. Both are critical for completing a film. I have always had an affinity for animating water and magic elements, and found that both came naturally to me, whereas to others more comfortable drawing explosions or props, the learning curve might actually be steeper. There are nuances of fire or smoke some might understand more readily than others, whereas I sometimes struggle with a simple smoke scene.

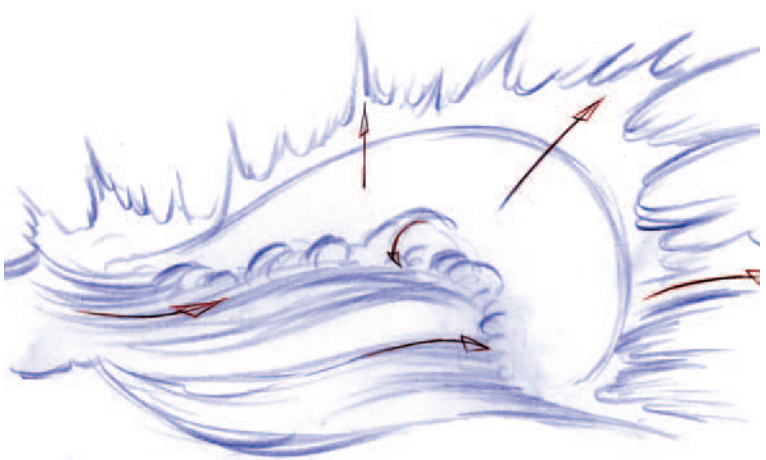
If you desire to become a truly proficient effects animator, you should work extra hard on those effects which do not come naturally. The single most critical yardstick for measuring success as an effects animator is how willingly we embrace the fine art of drawing rough (or "ruff," as veterans are fond of saying).



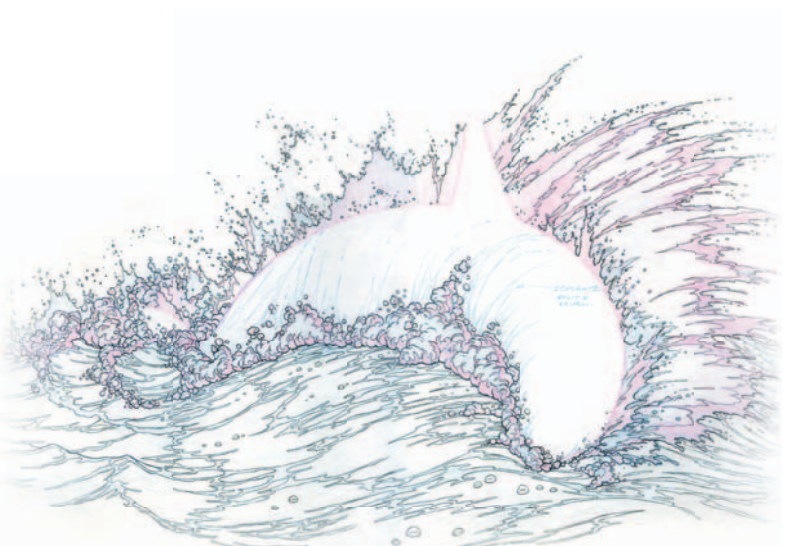
I mentioned earlier the importance of feeling the energy of a particular effect and allowing our hand and pencil to flow with the direction of that energy. If we are trying to draw with our pencil held tightly (that is, pinched in the classical writing style with the point of our pencil pointed directly down toward the drawing surface at almost a 90 degree angle), we cannot even begin to draw elegant, flowing effects animation. Heeding the sage advice from great life drawing artists and teachers, effects animators have learned to draw with the side of the pencil lead, holding the pencil lightly between the thumb and forefinger. This allows us a broader range of motion that originates with the shoulder, instead of the wrist. It also facilitates drawing with more flowing arcs and broader paths of action.

Some effects artists who have honed their skills like to use a very thick, dull, and soft pencil to animate their effects. Working with the side of a stick of charcoal could be a very good way to break an animator of holding a pencil rigidly. The trick is to stop trying to control the tip of the pencil in order to create "shapes" and to let the energy flow through our shoulder, arm, wrist, and hand to create "energy"!

Drawing in this unblemished, raw manner, allows us to create organic shapes and energy patterns that would otherwise elude us. From the sketchy-looking scribbles of a rough effects drawing, one can extract the most marvelous shapes and patterns!

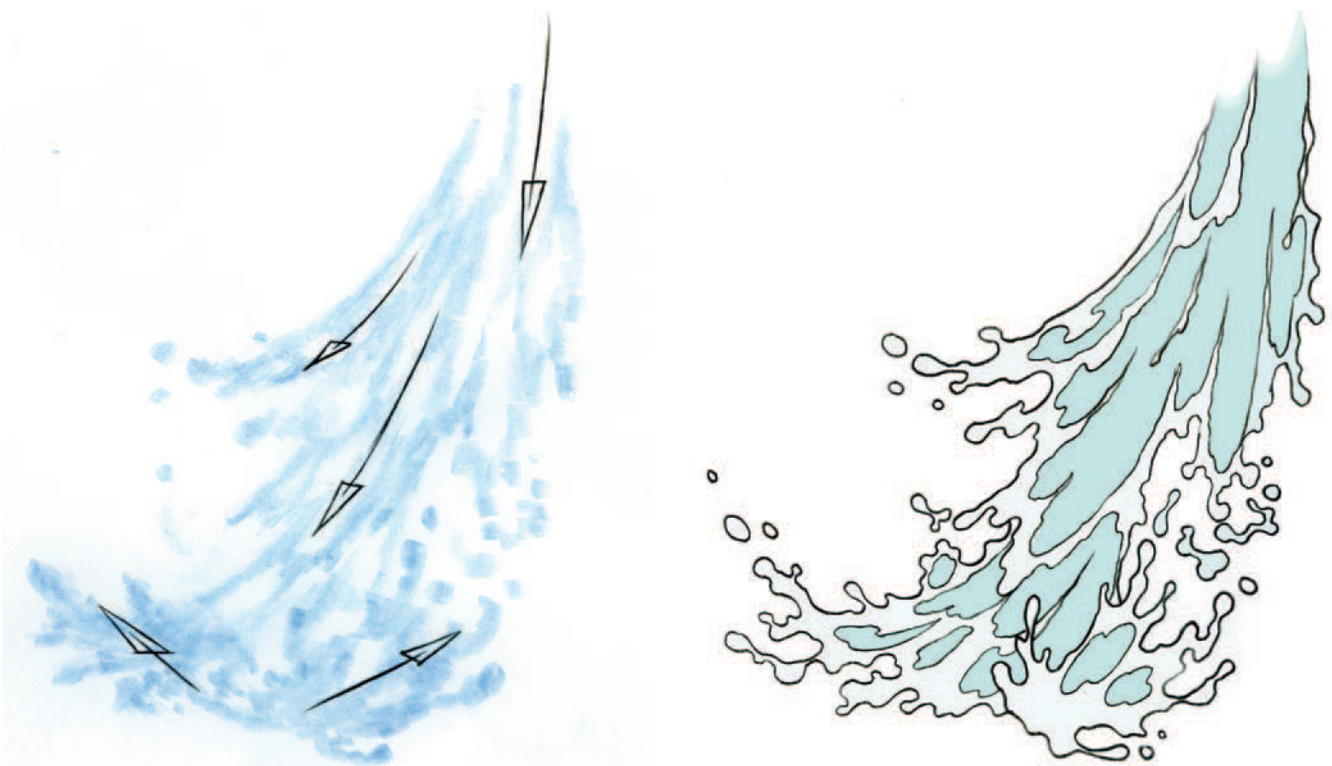


Keep the rough animation relatively simple and focus on powerful energy and dynamics rather than the intricate details.



Once the rough animation is working well, the details can be added to add scale and realism if necessary.

The sketchy, rough drawing on the left has a strong directional energy flow, and provides a strong basis for the elegant effects drawing design on the right. A quality effects drawing should always start this way!



Sensitivity to style is another crucial layer in the effects animation process, which determines how the look of the drawings fit in with the art direction of a film. It's also important to realize what purpose the effects are serving in any particular scene. In most cases, effects are used simply to add a subtle sense of realism to a scene, as with shadows, lighting, or reflection effects. A basic splash should never draw undue attention to itself if the audience's attention is intended to focus on a character in the scene. It's basic stagecraft: Always know your part, and never upstage another character or story element

These scenes from Disney's Mulan are an excellent example of special effects animation being used to drive the humor in a story. Having inadvertently set her matchmaker's rear end on fire, Mulan then gets herself in even more trouble by extinguishing it with a pot of tea! The effects were key in making this sequence work.

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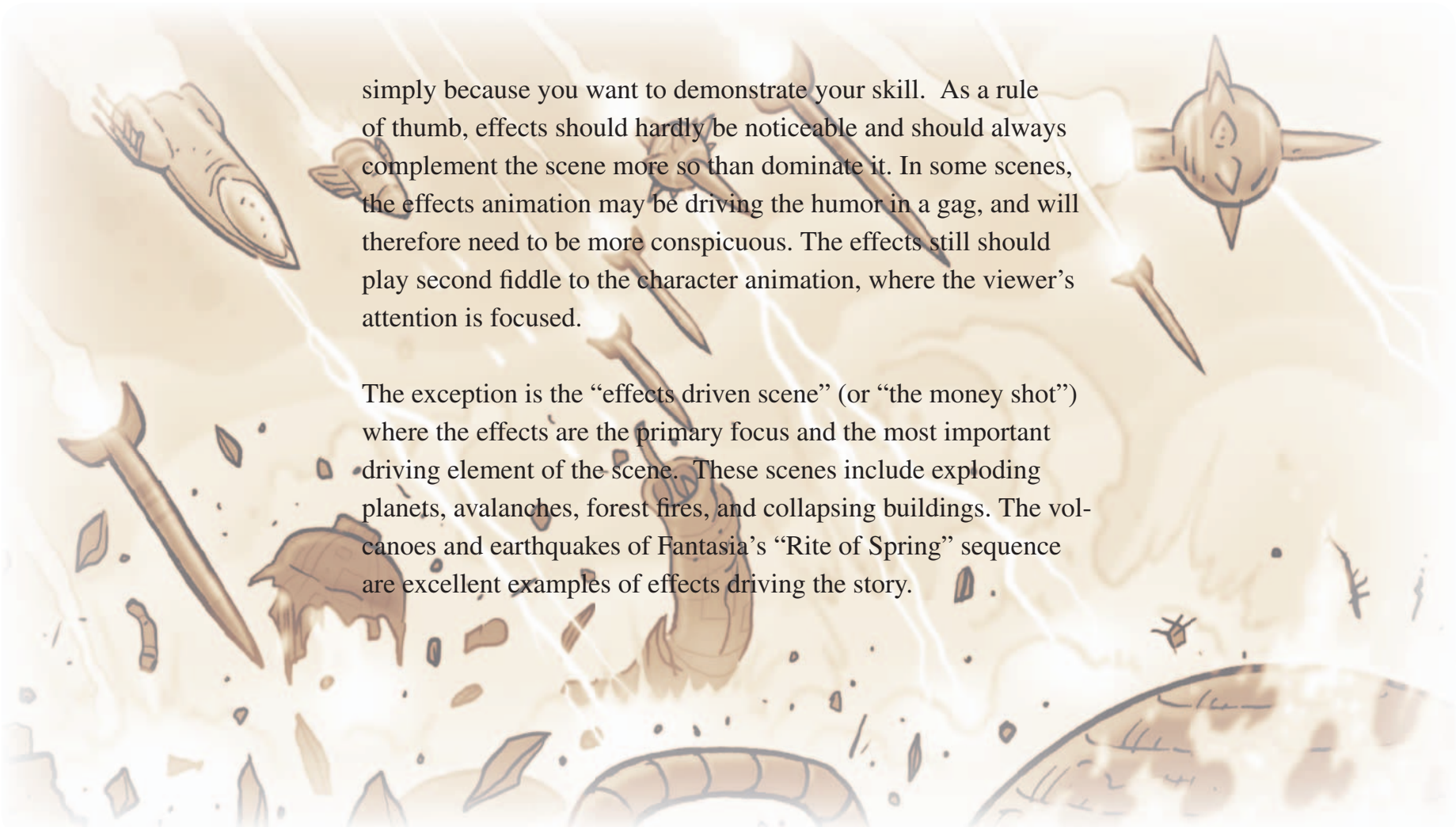


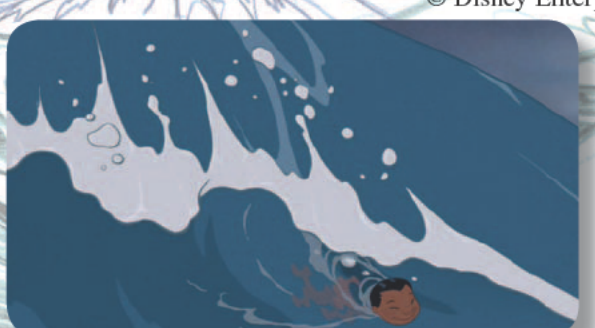
simply because you want to demonstrate your skill. As a rule of thumb, effects should hardly be noticeable and should always complement the scene more so than dominate it. In some scenes, the effects animation may be driving the humor in a gag, and will therefore need to be more conspicuous. The effects still should play second fiddle to the character animation, where the viewer's attention is focused.

The exception is the "effects driven scene" (or "the money shot") where the effects are the primary focus and the most important driving element of the scene. These scenes include exploding planets, avalanches, forest fires, and collapsing buildings. The volcanoes and earthquakes of Fantasia's "Rite of Spring" sequence are excellent examples of effects driving the story.

In all of these different animation scenarios, every special effect drawing must embody the art direction of the film, so that ultimately they appear to have been created by the same artist who drew the characters, constructed the layouts, or painted the backgrounds. This attention to the artistic direction of the film can make the difference between merely adequate effects animation and truly great effects animation.

It was a fascinating process to work on some of Disney's feature films, in which the effects were painstakingly integrated into the overall artistic style of the film. "Mulan", "Hercules," and "Lilo & Stitch" are examples of films containing effects animation that fit flawlessly into the overall picture, a result of working closely with the directors and art director to make sure every element directly reflects the artistic style of the film.

Which brings us to the great importance of stylizing our special effects drawings, not only to better integrate our effects artwork with the style of the film we are working on, but simply bring our drawing to a level of detail which is possible to draw. Upon analyzing a single large wave or splash, you would quickly realize that the amount of detail it contains is staggering. If we attempt to draw every detail, we would be bogged down in no time. But this can work to our advantage! There is a wonderful design challenge to represent a highly complex element with a relatively simple drawing—and there is a beauty in simplicity, as every great artist has discovered through the ages. Hokusai's "Great Wave" painting is a gracefully simple representation of an enormous cresting ocean wave, yet it is powerfully effective.



Lilo bodysurfing on a wave which was carefully designed to have the same gentle, friendly style as Chris Sander's original character designs, from Disney's beautiful feature film "Lilo & Stitch". In this scene the effects certainly are carrying the character. Literally!
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The gorgeous special effects designs in "Lilo & Stitch" were the result of many, many, months of meticulous designing and planning on the part of the effects crew. The result was a film with a consistent design theme throughout all its varied elements.
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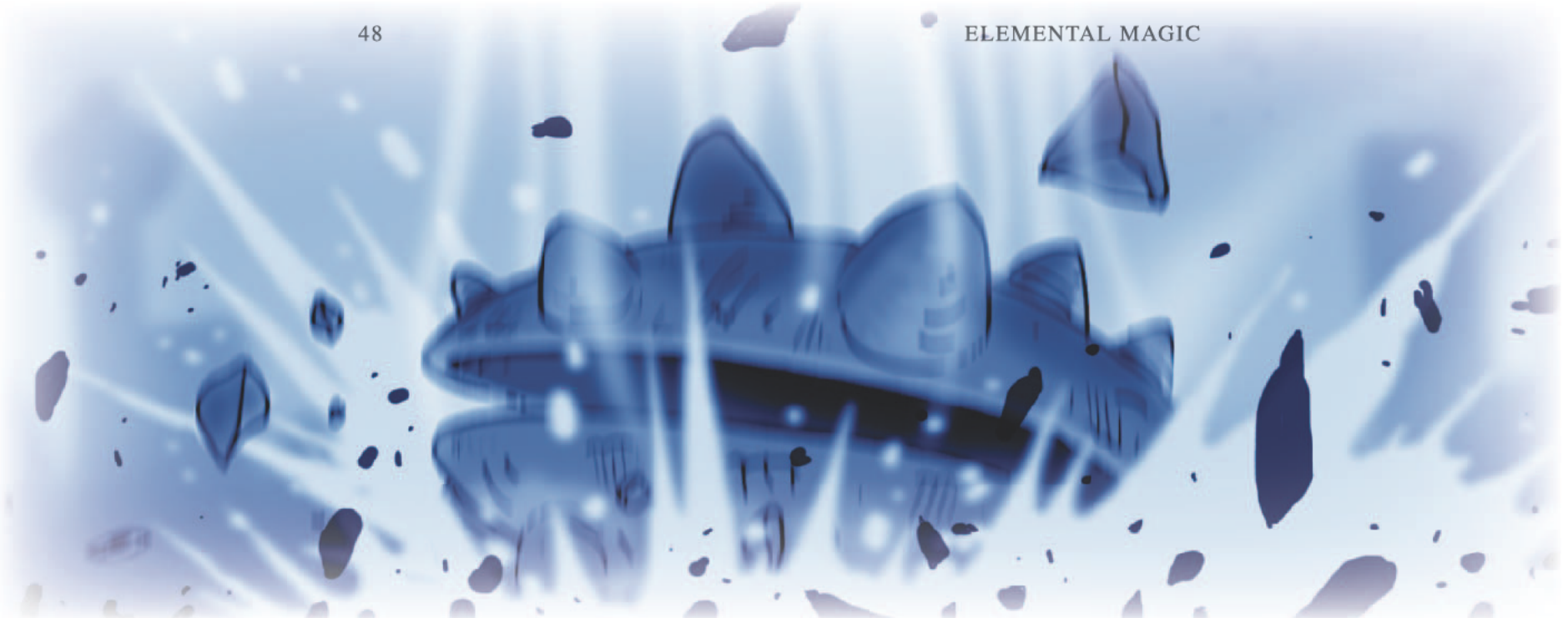
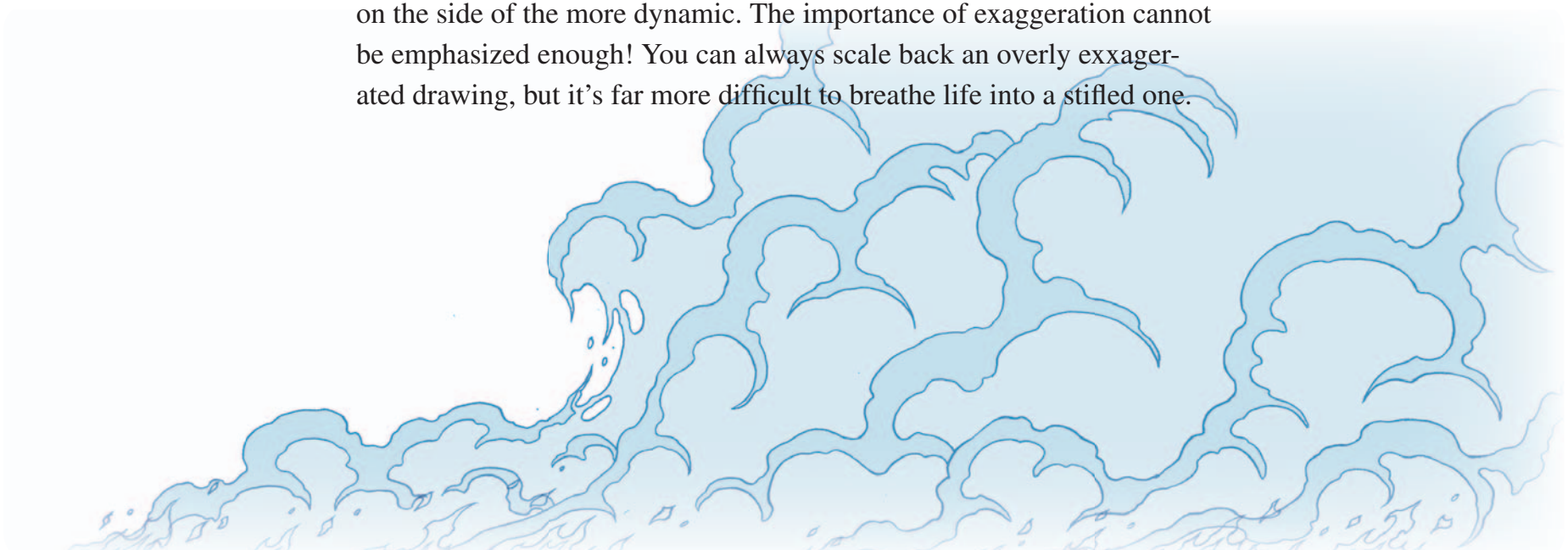


In representing our effects through simplified versions of reality, we are free to focus our attention on the forces underlying the effects, the patterns in motion, the timing and the physics. Details can always be added once our effects animation is working well in all these respects, but if we get lost in the details, we will have lost the natural physics of our effects animation. Stylize, simplify, and find the energy focus!

As far as creating scale and perspective in our effects, we need to exaggerate as much as possible. This can be achieved in a number of ways. If we want to insinuate a massive size, it is necessary to create highly detailed drawings, which will be animated with a ponderous, heavy motion.

This increases the amount of work involved exponentially: more details + slower speed = tons of complex drawings. In many cases, when I've heard a director asking for some wild and crazy effect that is ridiculously grandiose in scale, I will do everything in my power to talk him/her out of it.

Until you have hand-animated something like an enormous avalanche, with so much detail that you can only complete one drawing a day, it is hard to imagine how work-intensive these effects can be. Having said that, always attempt to push your drawing designs a little further, to err on the side of the more dynamic. The importance of exaggeration cannot be emphasized enough! You can always scale back an overly exxagerated drawing, but it's far more difficult to breathe life into a stifled one.

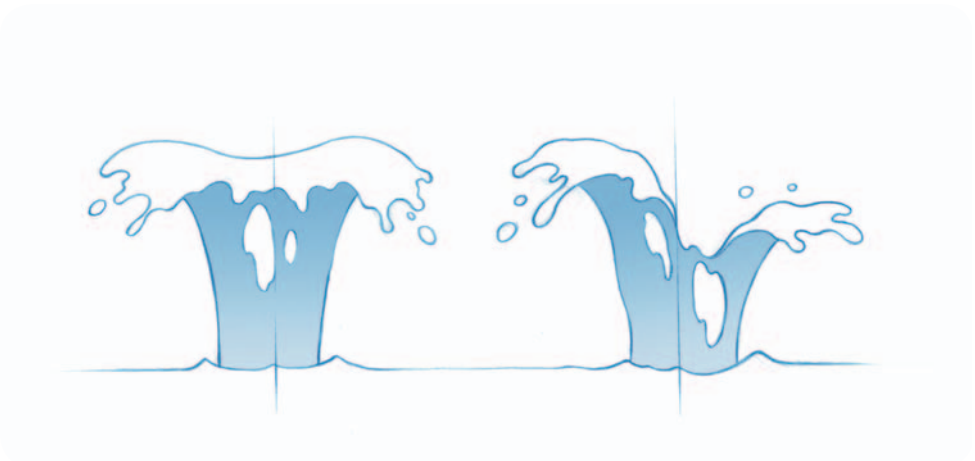


The most common mistake I see effects artists making when they try animating special effects for the first time is they don't push their animation far enough. There seems to be a fear of over-doing it, or an attempt to keep it all under control, but the best advice I often give beginners is to push their effects animation as far as humanly possible, exaggerating the physics, motion, design and perspective. It is easier to pull everything back if you have to, than it is to slowly and gingerly push everything farther and farther. The real payoff to drawing with this approach is the energy that you can get into your drawings by pushing and exaggerating.

Remember, when you animate special effects you are animating energy! Therefore, animate with energy and exuberance!

If you commit to being a serious student of effects animation, be forewarned: it is an extremely difficult and time-consuming craft and it requires patience and a driving passion for understanding nature! On your journey, you will undoubtedly have moments of unmitigated frustration, but don't let it keep you from soldiering on. It's worth it, I tell you, to wake up one day and realize you've become one of those crazy people who are happy to animate crashing waves, incendiary firestorms, celestial cataclysms, and other acts of God!

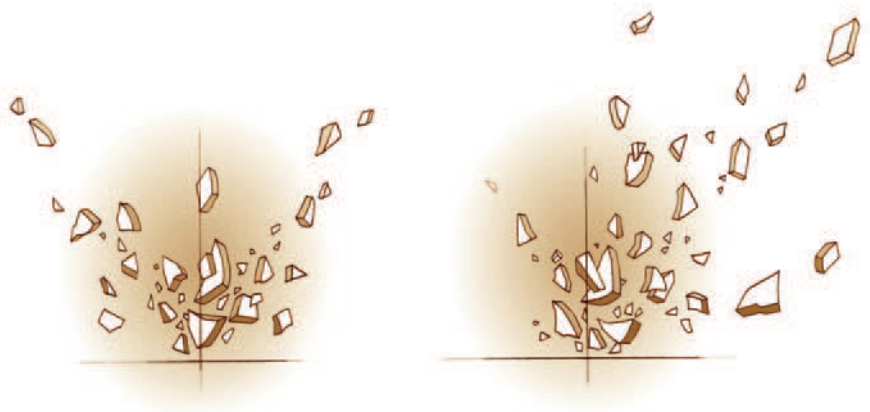
So pick up your pencils and your surfboards, your paper and your pyromania, and welcome to the amazing world of traditional visual effects animation!



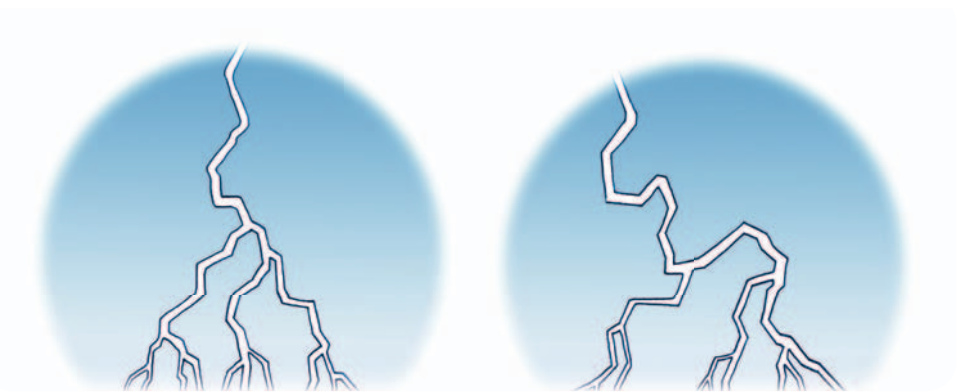
It is important to avoid symmetry when designing or animating special effects. Although in nature we may sometimes observe fairly symmetrical shapes in a splash or a puff of smoke, in order to create appealing, dynamic, and visually interesting special effects drawings we must exaggerate and stylize. The drawing on the far left contains some details which are not symmetrical, but the overall silhouette of the design is quite symmetrical. The overall silhouette as well as the details should be asymmetrical, as in the example on the right.

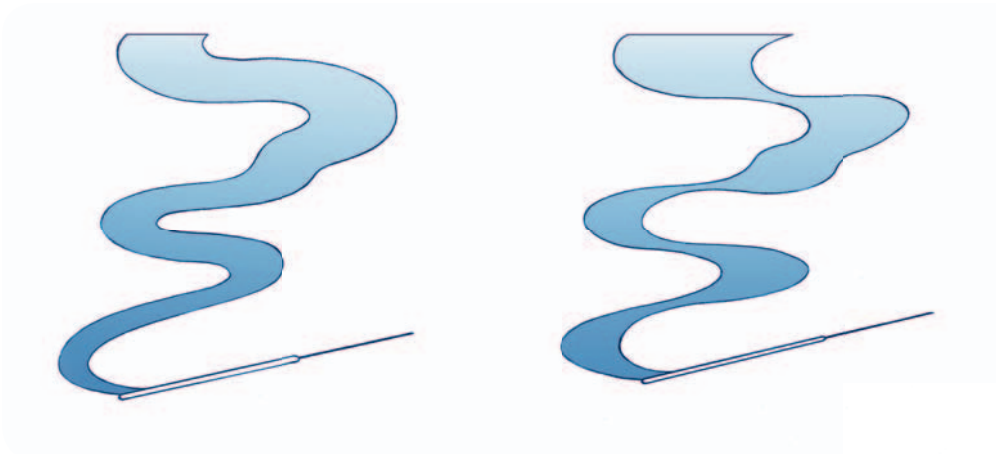
The explosion drawings here illustrate the same symmetry problems and solutions as the splash drawings above. The explosion on the left does have some very minor differences in its details, but overall its silhouette is extremely symmetrical, somewhat resembling a snowman. Although the energy of an explosion actually radiates outward in every direction at almost the exact same velocity, breaking up its silhouette asymmetrically, and overlapping its animation timing, creates a far more interesting and appealing drawing.

Here are two simple drawings of what could be a piece of ice or a ceramic bowl, smashing upon impact with the ground. Again, the example on the left contains elements which aren't entirely symmetrical, but overall the design's silhouette is quite symmetrical. Anyone who has dropped a bowl or a glass on the floor, has witnessed the way a few pieces always end up several feet away from the point of impact. The pattern of pieces left over is not symmetrical, nor is the splashing pattern of pieces emanating outward upon impact.



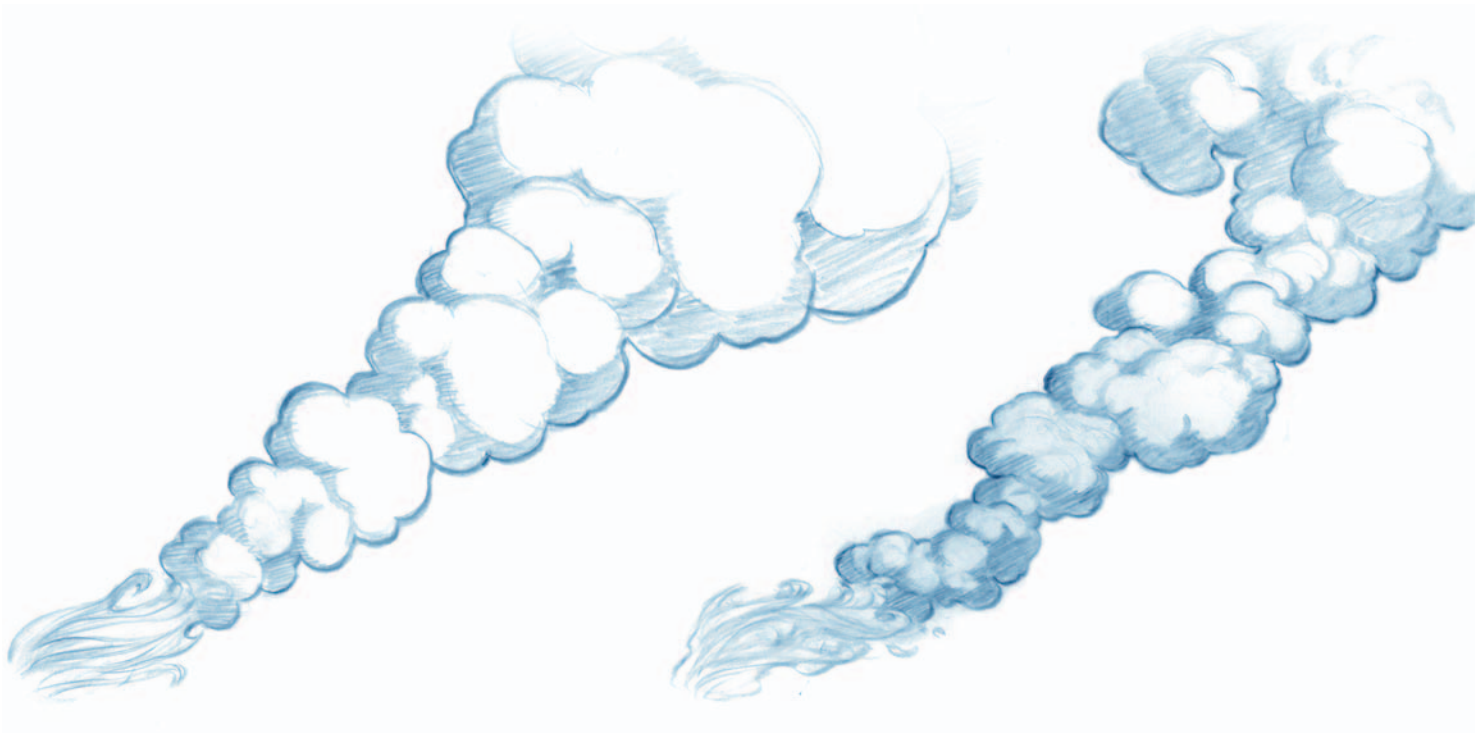
In these two examples we see a bolt of lightning, or electricity. The example on the left is offset a little, but is quite symmetrical. Observing hundreds of photographs of lightning bolts, we may actually see quite a few bolts which are pretty symmetrical looking, like the example on the left. However, when we animate electricity, we want to "push the drawing" and render our drawings far more dramatically than reality. Breaking up symmetry, we avoid "twinning" our shapes, and we push our designs as far as possible to keep things interesting.



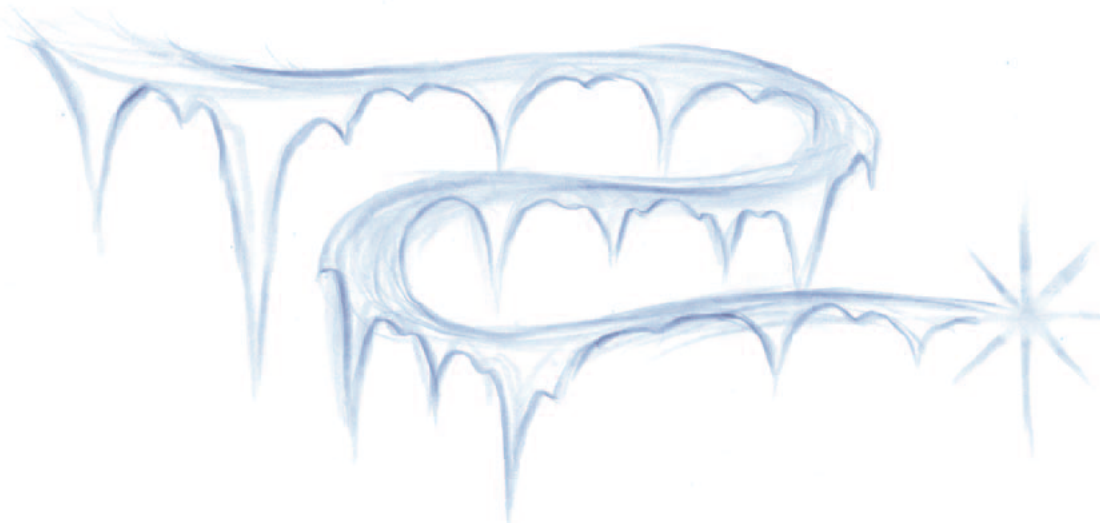


Another important principle important in the designing of good special effects animation is to avoid parallel lines as much as possible. Patterns in nature are rarely if ever truly parallel, and once again, we use exaggeration to add drama to our drawing designs. The incense smoke on the left tapers gently as it rises, but it is far too parallel. Using the same line on the left of the design, the smoke on the right is pinched tighter and spread wider in different areas, making for a much more interesting, realistic, and dynamic shape to animate.

The tonal areas of these two very similar smoke designs illustrate once again where we often see a tendency to draw parallel lines in special effects, and miss out on a chance to add weight and volume to our drawings. The tonal or shadow areas of the drawing on the left, closely follow the outline, and do little to describe the mass of the smoke, whereas the tonal designs on the right vary greatly in width, and describe far more dynamically the actual volumes of the smoke effects. These are simple, yet highly effective drawing principles.



Here is an example of symmetry and parallelism being detrimental to a special effects design. In this case, a large plume of smoke like we might see rising from a burning building. In the drawing on the left we see the smoke expanding upwards and outwards very evenly. If we were to turn this smoke on end so that it was perfectly vertical, we would see that it is almost perfectly symmetrical and quite parallel as well. This is a common problem with smoke effects design, and it is often seen today in digitally created particle effects smoke, especially "canned" or pre-programmed digital smoke effects. If care is taken to introduce more turbulence into our smoke effects, the result will look more like the smoke drawing on the right. Of course we must take into consideration the amount of wind in our scene, as well as how much heat the fire is generating. The asymmetrical qualities of a plume of smoke such as this can vary greatly, but even on a dead calm windless day, a plume of smoke should demonstrate far more turbulence and asymmetry than the drawing on the left.



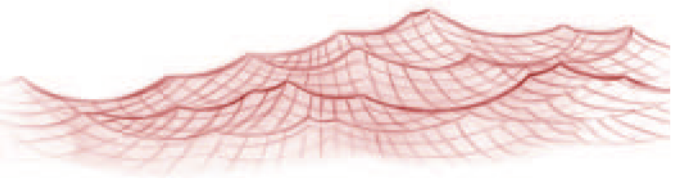
The principles we have discussed in this chapter thus far can be applied to any and all special effects elements, including magical effects like pixie dust. In the drawings here we see a loose, "ruff" drawing, demonstrating good fractal randomness, which avoids symmetry, repetition, and parallel lines. There is a feeling of directional energy in the drawing as well, which gives our pixie dust effects a sense of purpose, as if it is going somewhere for a reason, not just randomly traveling across our field of vision.

Whether we are drawing or animating liquids, waves, splashes, fire, smoke, electricity, lighting, breaking objects, waving grass or a flapping flag, slowly crawling mist or a dynamic explosion, you name it—*these basic principles should be applied to our special effects animation*. These principles apply equally to the creation of digital special effects, if we want to imbue our effects with "The Illusion of Life" to borrow from Frank & Ollie's timeless book on animation.

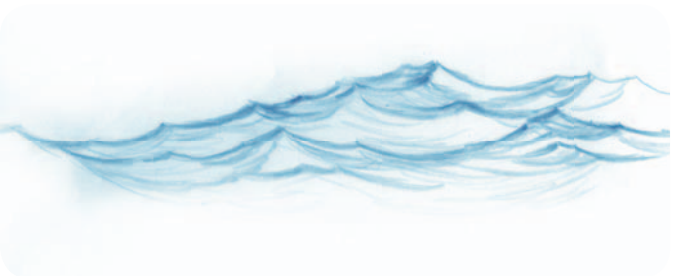
This could be a drawing of ice, rocks, wood, hard-packed snow, smashing porcelain or glass. Reflecting the design principles outlined in this chapter, every individual chunk avoids parallel lines as well as symmetry, and the overall placement of the pieces carefully avoids a symmetrical silhouette as well as demonstrating a great variance of sizes and shapes. Even though we have no idea why these shapes are placed as they are, or what outside energy force is at work, there is a sense of natural energy in this drawing. Underneath the details there is a lot of information—energy informing the drawing and making it speak to us of its origins.

This is actually a drawing of smashing ice chunks. Much observation and research went into examining the subtleties that differentiate the shapes that one might find in the elements mentioned above. Ice breaks up somewhat differently than glass, or wood, or rocks. It also breaks up differently depending on how dense it is, what impurities are in it, and what the water was doing when it began freezing. There is no end to the research we can do to make our effects animation look fantastic!

Every good effects drawing must begin with a solid foundation. This requires an understanding of the environment and its proper perspective. An accurate perspective grid can be immensely helpful at this point.

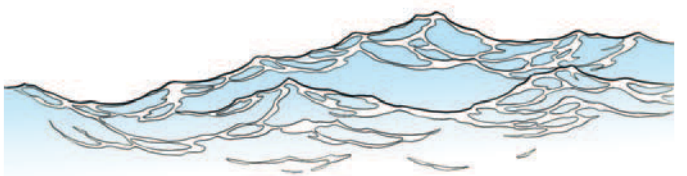


Working with this strong underlying foundation, the effects artist can now relax and feel out the style and details of the drawing, with a loose, easygoing pencil stroke. No worries about details at this stage. The energy is what's important.



Drawing over the loose, naturally flowing energy of the rough sketch, the tighter details can now be added, keeping with the directional flow and fluidity of the rough drawing. All the details can be finessed and noodled with!

The final clean line can now be applied. The style, line quality and look of the drawing at this stage depends entirely on the look of the specific show the animation is needed for. But the steps leading up to this point should be the same.

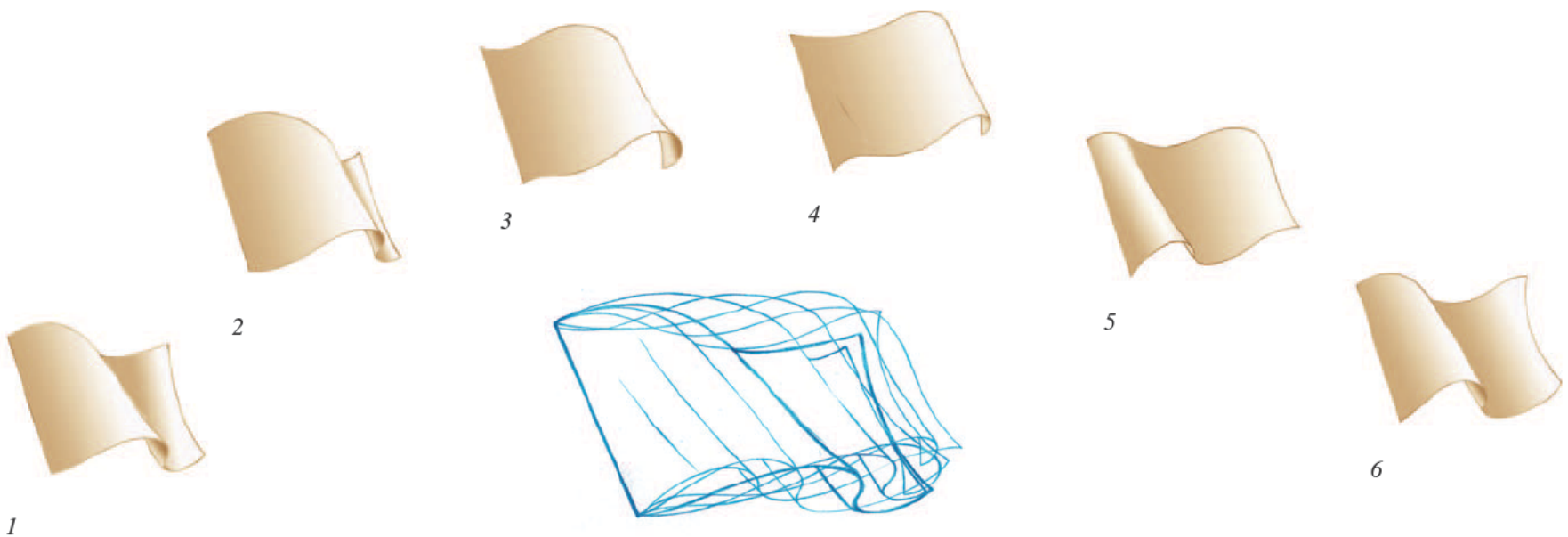


The "Whip" and "Wave" Effects

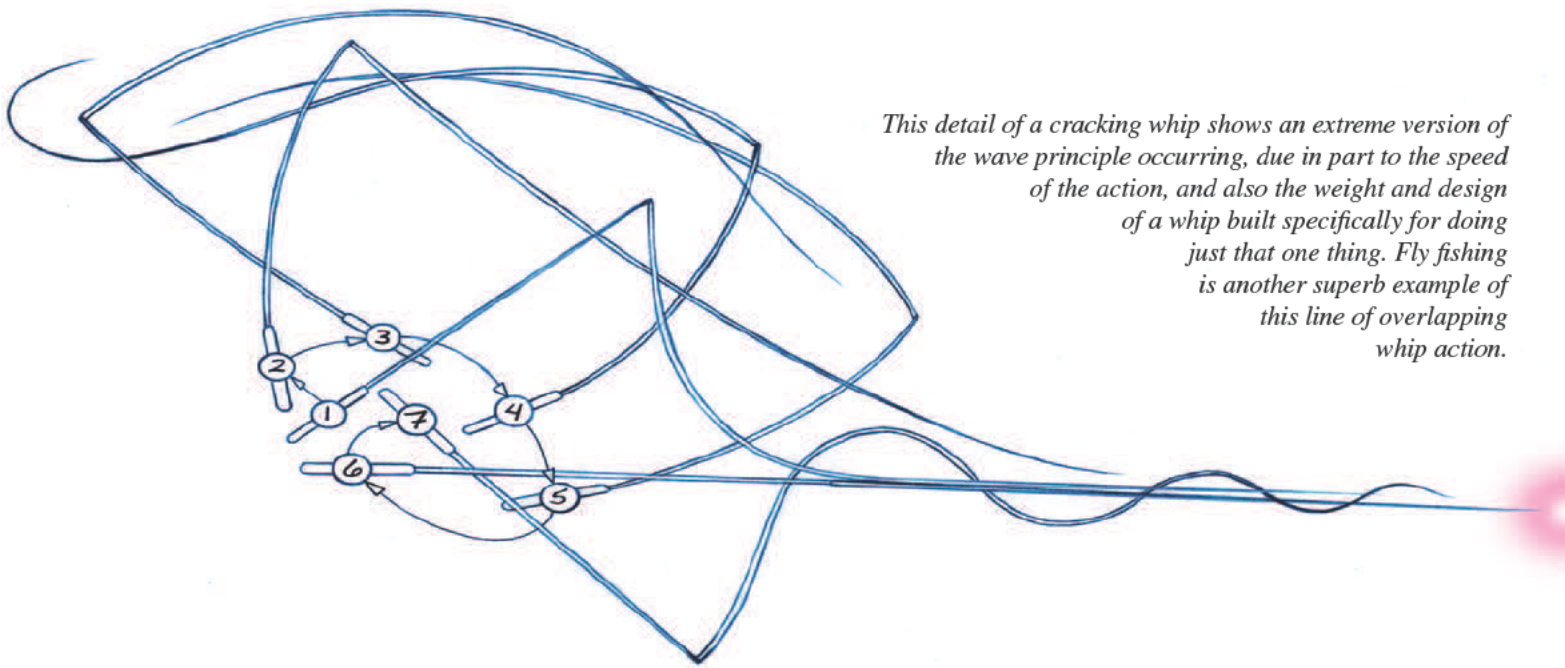
Whether you are animating smoke, fire, water, dust, snow, branches, leaves, a dangling rope, a fluttering cape, a curtain swinging closed, a dog wagging his tail, or a billowing dress, all of these effects have within them the basic whip/wave principle. It is a simple flowing, overlapping action that occurs wherever energy interacts with matter which is not entirely rigid. Just take a piece of rope maybe a few feet long, or a garden hose, lift it up quickly and then snap it back down even quicker, and you will see a wave travel through the rope or hose, just like a wave travels through the water. We can move our arms much in the same way,

We've all seen those imitation flames made out of flimsy fabric, which have a fan blowing straight up from underneath them. This is a perfect example of energy, in this case wind, creating a wave effects in matter. Hot and cool air creates wind that creates the shapes we see in a real fire, which is why this illusion works relatively well. I have actually been fooled by one of these novelties (for a second or two!)

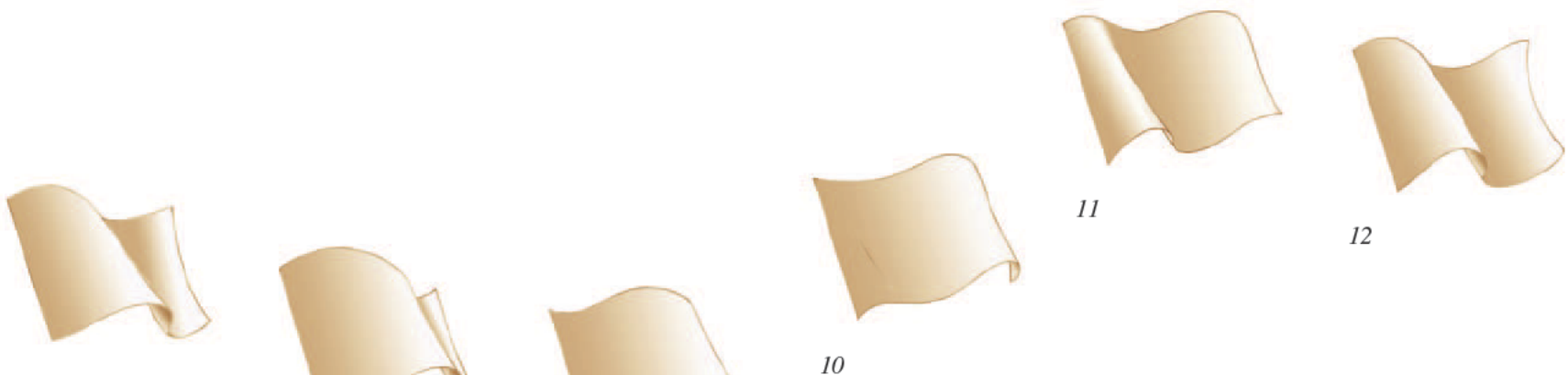
like the dancers who seem to be so rubbery and can make a wave appear to travel across their body. Interestingly, even a perfectly rigid object can appear to move with a wave action, as with the old trick of making a rigid pencil appear to be rubbery and bendy. Simply by shaking a pencil up and down, but also applying a rotational movement with the hand and wrist, the rotation causes the illusion of a wave. Our eye sees the circular motion of our hand transferred to the rigid pencil.



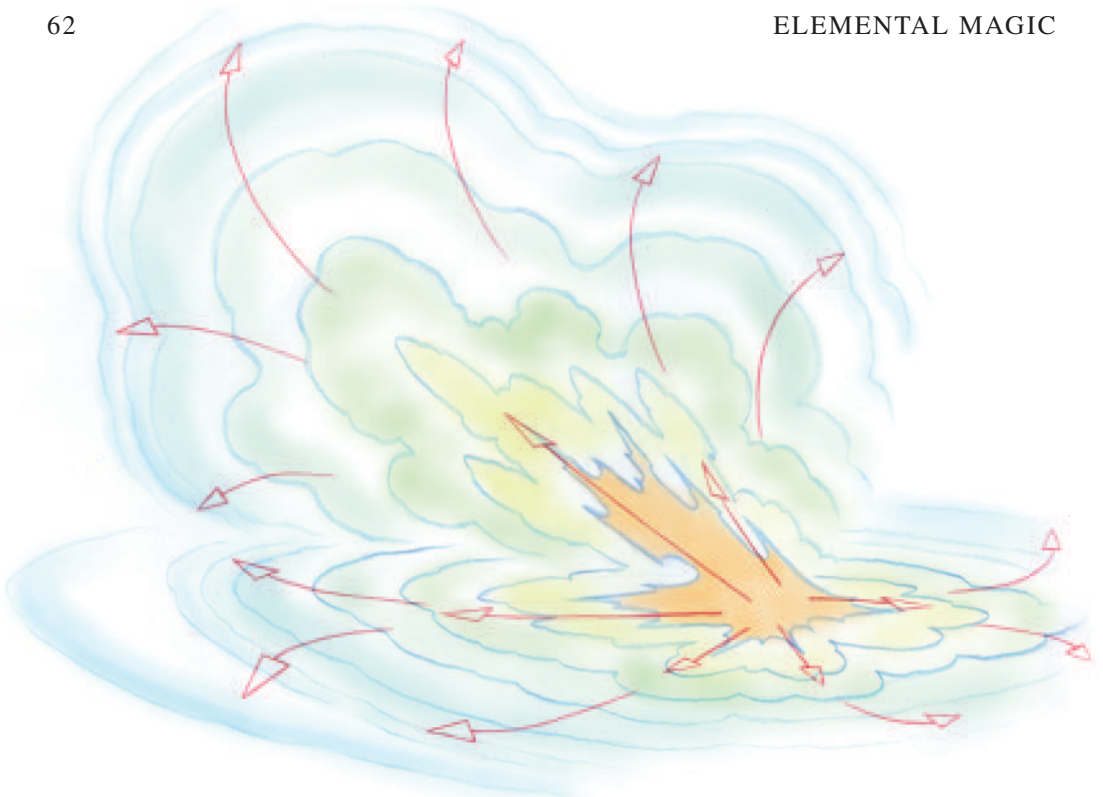
This flag animation illustrates clearly how the energy of wind creates the wave-like motion that causes a flag to flutter so elegantly. This animation is actually a cycle; drawing #7 is actually a repeat of drawing #1, and the subsequent drawings repeat throughout the remainder of the animation.



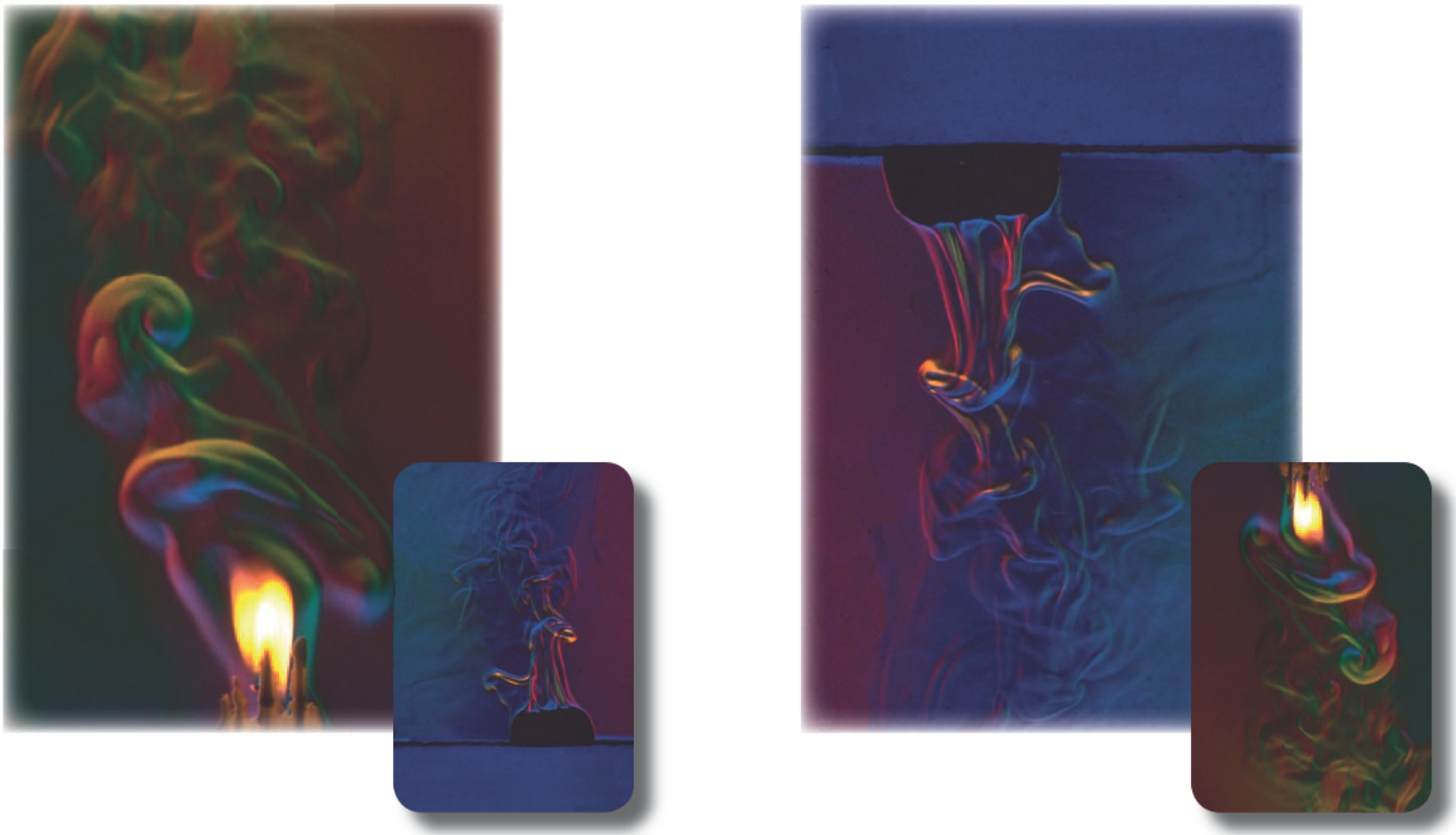
This detail of a cracking whip shows an extreme version of the wave principle occurring, due in part to the speed of the action, and also the weight and design of a whip built specifically for doing just that one thing. Fly fishing is another superb example of this line of overlapping whip action.



7 *8* *9* *10* *11* *12*



The energy that creates any elemental effect—the impact of a splash, or the sudden combustion of an explosion, has a finite lifespan. What starts out explosively, with very strong directional energy, loses energy as it expands. And as the energy dies out, the shapes it creates evolve. They change, interestingly enough, from sharp, angular, energetic designs to soft, flowing, languid designs. A sudden splash ends in subtle, soft ripples. A violent explosion ends with beautiful shapes of gently dissipating smoke. The overall design and dynamic movement of this phenomenon is the unique fingerprint that pure energy leaves on the elements that it moves.



These two Schlieren photos, one of a candle flame, and one of an ice cube melting in water, show a remarkable similarity in the beautiful energy patterns that occur when hot and cold temperatures intermingle, regardless of the context. Rotating both photos 180 degrees and looking at them next to their opposites, shows us just how uncannily similar these patterns really are. We will find these patterns everywhere the various elements react to the cosmic forces at work in the world as we know it.

Into the Digital Realm of Special Effects

Having looked at special effects animation from a classical point of view, what then can we bring to the table when we are animating special effects digitally in virtual 3D space using the incredibly powerful and versatile digital animation tools that we now have at our fingertips?

To begin with, taking advantage of the classical animation design principles contained throughout this book will bring a fresh new look to any and all effects that we endeavor to create digitally. I cannot emphasize this enough, the need for us as effects artists to understand and exploit what it is that makes classical effects animation so fantastic, no matter what element it is that we are animating, and no matter what technique we are using.

Special effects animation all too often (especially in this digital age) is treated as an afterthought, something that we simply have to add the fantastic environments and characters that populate our animated films to, in order to create a little bit of "reality" to a given animated scenario. But this book is all about the *exception* to that halfhearted

approach to animating special effects by default, and whether we are animating effects by hand, or digitally, we can take a cue from the masters of the art and bring that extra magic to every little detail of special effects animation that we create. Whether we have a tiny or massive budget, the classical design and animation principles that drive the highest quality effects animation can be brought into play and elevate our art form across the board!

Indeed, the vast majority of special effects animation throughout our history, has been for the most part an afterthought—"Oh yeah, we are going to have to have some water in these scenes. Do we have anyone around here who knows how to animate water?"

And so frequently some poor character animator (who might have a bit of a knack for drawing splashes and such) would take on the task, usually without a budget, or a plan, or any regard to design, or any consideration of the actual intent or character attributes of the effects animation in the scene. Interestingly though, before we were introduced to computers and digital 3D effects animation tools, there were actually some great advantages to drawing our special effects that are missing for the digital artist today.

I believe that back in the days when our only tools were paper and pencils, some special effects took on a stylized, more cartoony look that worked really well as far as "fitting in" to their environment. In part it

was because the animators were originally character animators, already accustomed to drawing in a cartoony, highly stylized way. It was simply the natural thing for them to do, and as a result, the hand-drawn effects that they created were stylized and fit in quite well with their environments.

Another fact that worked to the art of animation's advantage was the animators' need to stylize and simplify their hand-drawn special effects in order to make it *possible* to draw them. Observing real natural special effects, an artist is overwhelmed with the complexity of even the simplest splash or small fire. To draw these things accurately is almost impossible. And so, by necessity, the earlier effects animators' drawings were simplified and stylized versions of reality that actually looked much more at home in a cartoon than something entirely realistic would have. And so there was a unified look and a more unified set of principles being applied to all the elements of a classically drawn, designed and animated cartoon.

I have puzzled a great deal over the apparent lack of attention that is given today to the importance of these classical animation techniques and principles as they pertain to digital special effects animation. I have come to believe that it is to a large extent the intoxicating power we have using digital tools, to closely mimic the overwhelming detail of natural special effects, that makes us tend to forget about the importance of stylizing, exaggerating and economizing our designs.

Today, the vast majority of young artists getting involved in the creation of digital special effects animation are not spending nearly enough time learning about classical design and animation principles. Although classical technique is emphasized quite a bit in any really good digital *character* animation school, young artists delving into the world of digital special effects are more often than not spending very little time learning about classical techniques.

The emphasis, more often than not, is put squarely on learning how to use the various complex software applications that are used to create digital special effects, and these young artists are missing out on an entire universe of classical effects animation knowledge that would elevate their special effects to a much higher level of artistry—if they were only given the opportunity to learn it. For these young artists, the natural thing to do is to look for the solution to their effects animation problems, somewhere in the highly complex 3D software that has been so heavily emphasized in their animation educations.

Keep in mind also, that digital tools open up a world of detail that was never possible for classical animators to draw by hand. Freed from these constraints of detail and complexity, today's digital effects artists are a lot less likely to even consider simplifying or stylizing their special effects. It feels good to be able to make cool effects that look realistic—but do they really look great in a highly stylized cartoon world?

In today's digital animation industry, when a young digital effects artist is asked to create a new special effect, he or she does not have a wealth of classical training on which to draw for inspiration. In most cases if any research is done, it will be through the observation of that effect in nature (which is great), or finding a similar digital special effect previously created in another film.

The emphasis is generally put on making the effect look as realistic as possible, and so the elements of stylized design that go into the environments and characters, and the exaggerated animation techniques that are taught to the digital character animators, are missing in the digital special effects that we all too frequently see. All too often, the results are cold, lifeless, and predictable digital versions of special effects reality superimposed into a highly stylized cartoon world.

So let's take a look at a hypothetical animation sequence in a computer-generated 3D cartoon, and see how we could approach such a task in a way that honors the legacy of special effects animators who pioneered these techniques, thus raising our art form back to the the level of artistic mastery that is its heritage.

If we have a sequence in a film in which a house is ablaze, the sets will be painstakingly designed, the well-designed characters will be animated doing whatever it is they have to do, and then usually, the fire and

smoke will be added—without any of the design considerations given the environment and characters. As long as it looks like fire (sort of), it can be approved by the director and pushed on through the production pipeline.

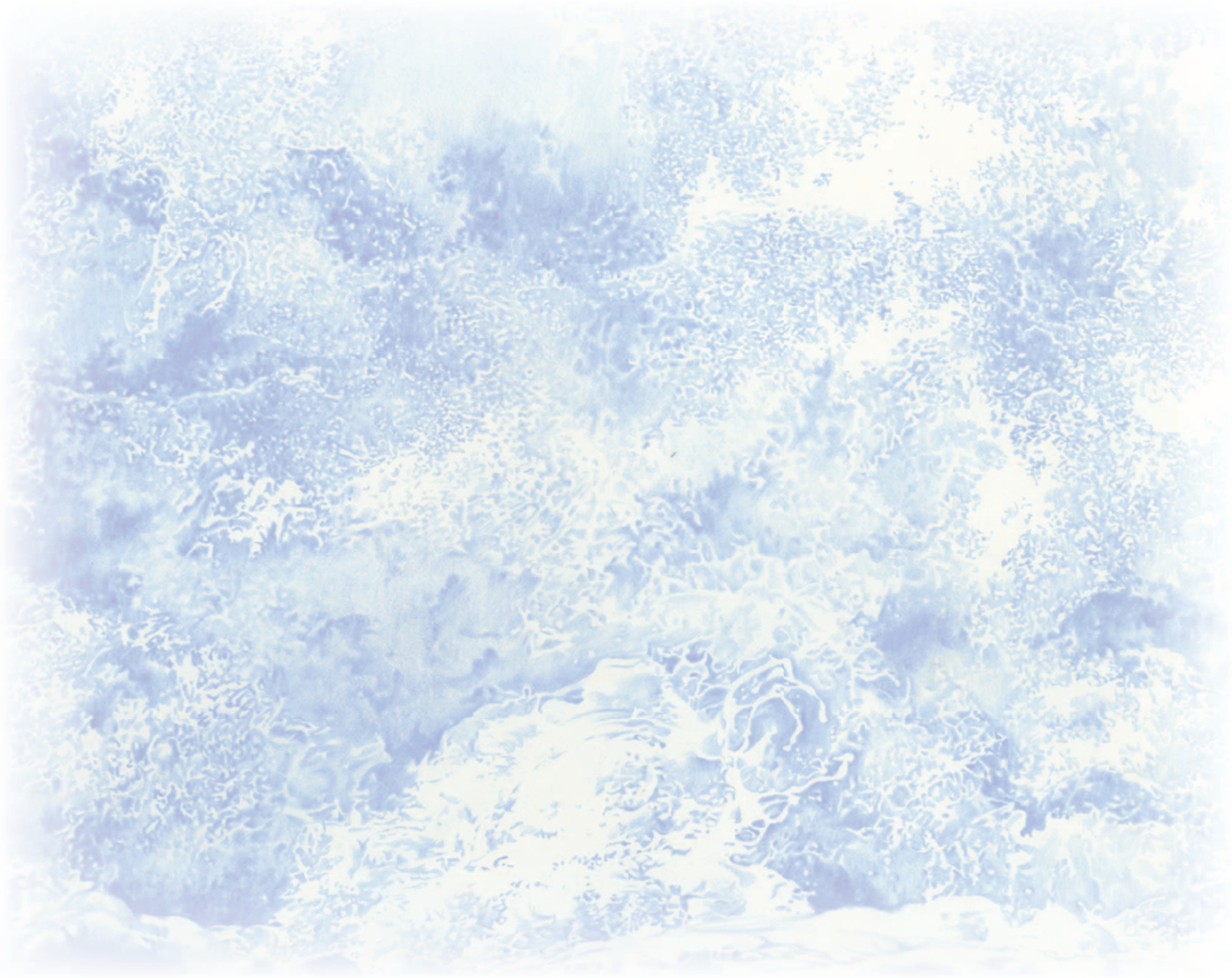
The same can also be said if the fire brigade arrives and begins hosing the fire down. The water, smoke and steam will probably be given just enough attention to ensure that they "work" adequately, and help to tell the story just enough. As long as the audience knows what they are supposed to be looking at that is probably going to be good enough for the special effects animator and the director .

But with an awareness of the classical special effects animation knowledge contained within the pages of this book, why not look at the design of the environments and characters in the film, and make at least some sort of attempt to design our special effects in a way that they integrate as well as possible into the overall look of the film? Animation is, if anything, the art of exaggeration and stylization, so why not apply those principles to not only the design and look of our environment and characters, but equally to our special effects animation?

Armed with the tricks of exaggerated animation timing and classical design principles, a digital effects artist can far better integrate his or her effects into a digital 3D or 2D world. (The term 2.5D frequently comes up in my conversations with animation professionals, as the vast majority of animation projects today utilize both 2D and 3D techniques, digital and hand drawn, intelligently integrated to best to realize a director's animated vision.)

Add to that a solid understanding of the underlying structures and energy patterns inherent in all special effects elements, and the well-informed digital effects artist of today does not have to lean on the computer for inspiration, but rather to bring the inspired imagination and vision of a classically trained animation artist to the computer.

We must never forget that the real *super computer* is our imaginations. The cold digital boxes in front of us filled only with plastic and wire, zeros and ones, and a bit of electricity, are really only information storage devices, like a blank piece of paper awaiting the stroke of a pencil. And our audiences around the world, the countless millions of people who are inexplicably drawn to and have been enormously entertained by the magic of animation for the past 80 years, need us to remember that.



Chapter 3

The Specific Effects Categories

Over the course of my career, I have animated an unbelievably broad range of elements, all of which fall under the umbrella of what can be considered “special effects”. To best illustrate just how diverse this art form really is, let me share with you a short list of some of the things I have been called upon to animate over the years.

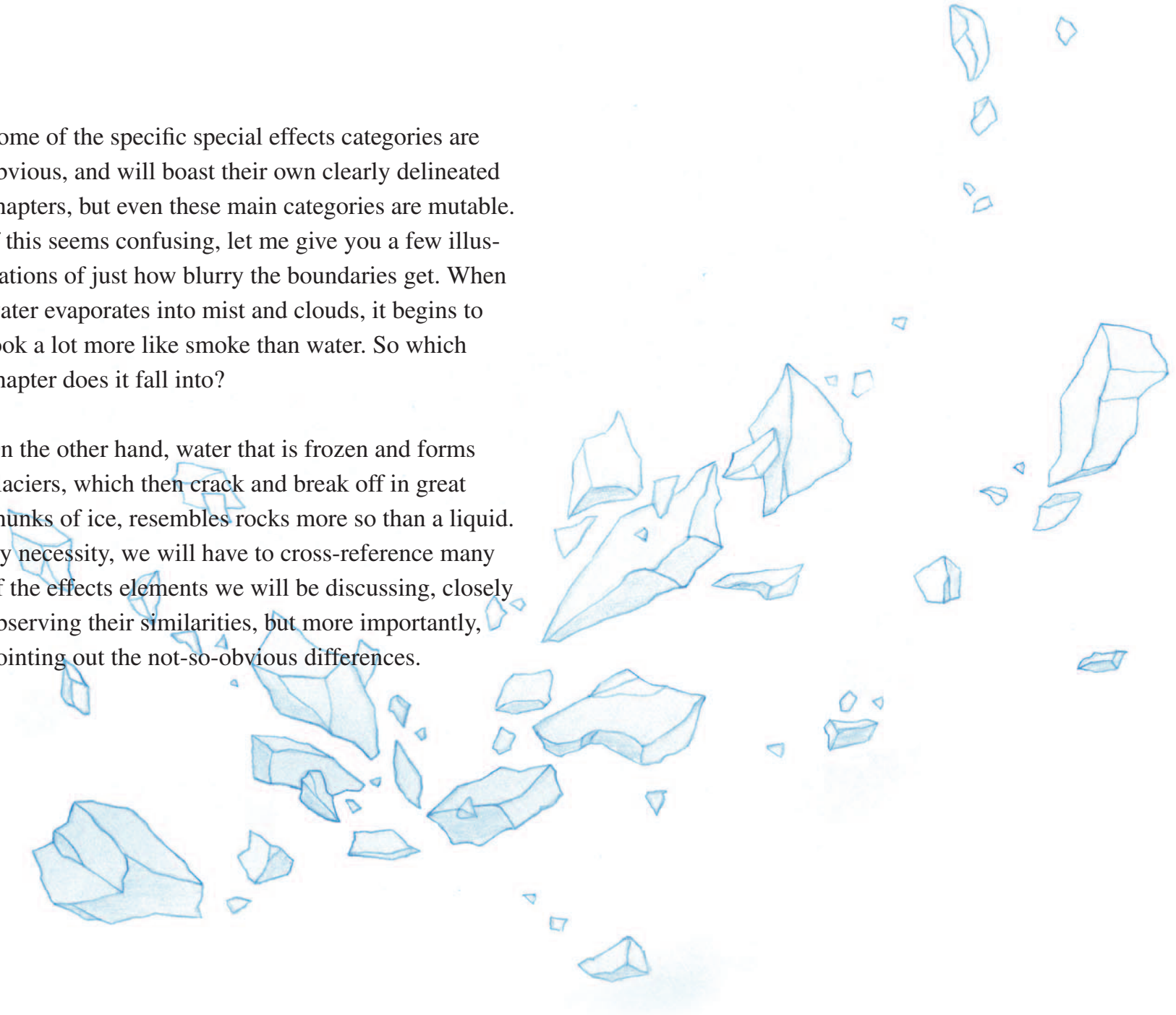
First of all—water. Lots of it! Splashes, ripples, waves, bubbles, fountains, white water, geysers, water hoses, raindrops, rivers, waterfalls, oceans, ice, snowflakes, avalanches, and slush.

And then there are all the other possible elements: Vapor, alcohol, and clouds. Gas, oil, syrup, oatmeal, mud, slime, and saliva. Chairs, cars, bags of flour, chains, whips, guns, spaceships, candle flames, campfires, bonfires, burning buildings, blowtorches, sparks, electricity, and lightning. Pixie dust and evil magic powders, flags and curtains, dust and dirt. Cigarette smoke, smoke signals, smoking guns, ray guns, exploding planets, bursting bubbles, laundry detergent, soup, salad, sandwiches, wine, whiskey, beer, coffee, bottles breaking, windows smashing, wood splitting, tires spinning, skateboards and running shoes.

Boats and bombs, ships and saddles, boots, pillows, handbags, chicken legs and tacton toes, forks and knives, cups and plates, tornadoes and whirlwinds, leaves and branches. Exploding houses, surfboards surfing, bicycles, motorcycles, water pistols, guitars, ukuleles and violins. Strings, ropes, threads, beds, and spider webs, just to name a few.

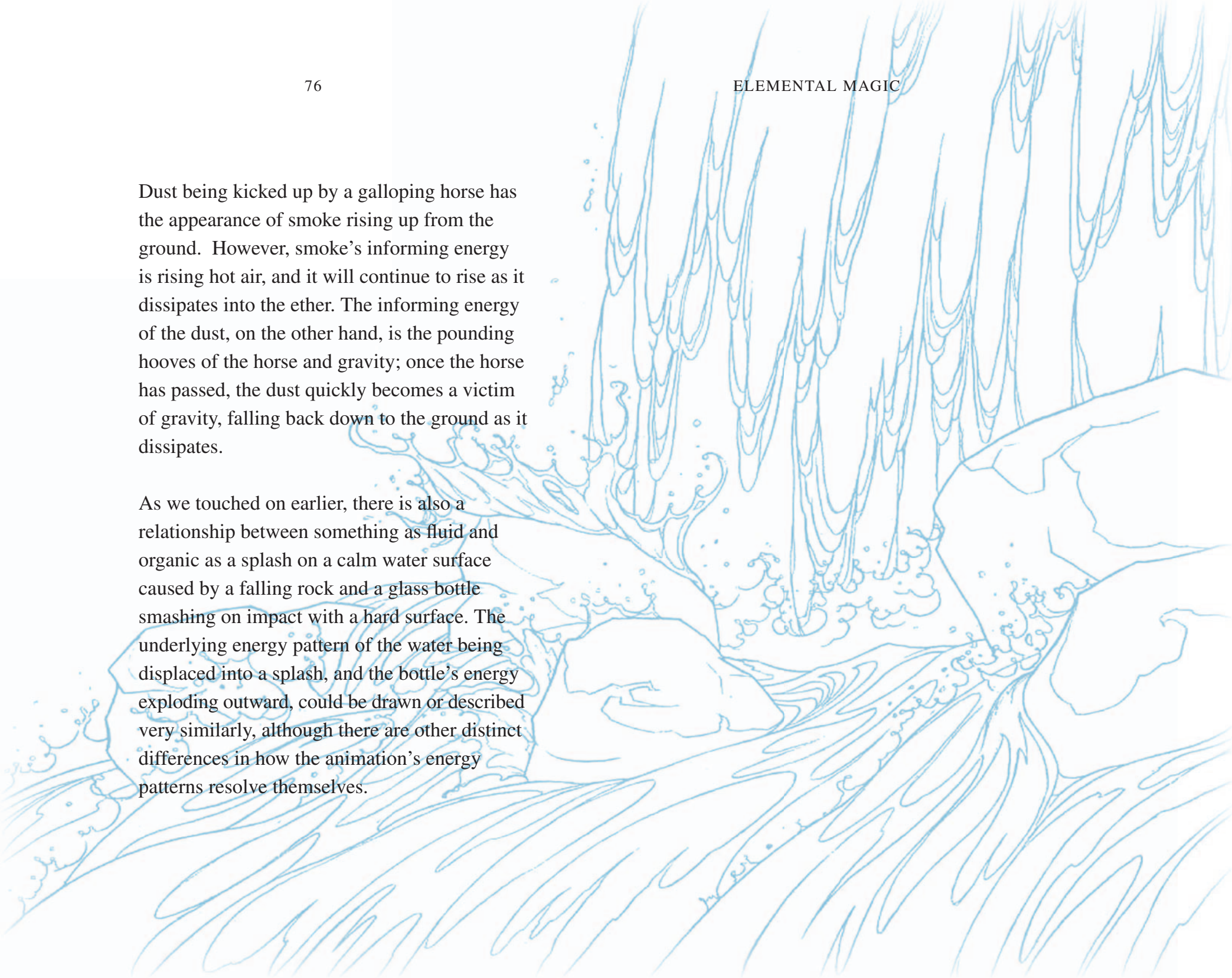
Some of the specific special effects categories are obvious, and will boast their own clearly delineated chapters, but even these main categories are mutable. If this seems confusing, let me give you a few illustrations of just how blurry the boundaries get. When water evaporates into mist and clouds, it begins to look a lot more like smoke than water. So which chapter does it fall into?

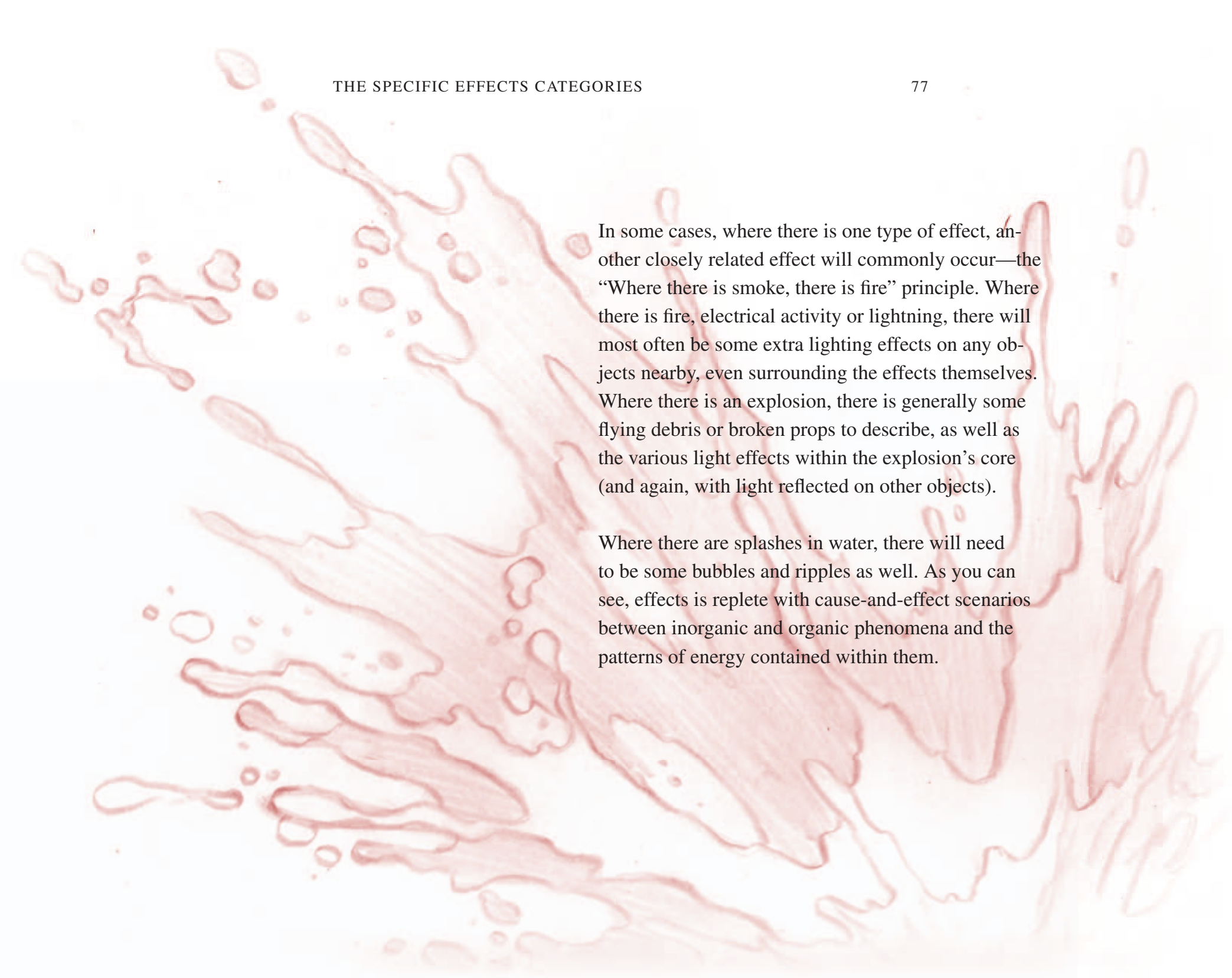
On the other hand, water that is frozen and forms glaciers, which then crack and break off in great chunks of ice, resembles rocks more so than a liquid. By necessity, we will have to cross-reference many of the effects elements we will be discussing, closely observing their similarities, but more importantly, pointing out the not-so-obvious differences.



Dust being kicked up by a galloping horse has the appearance of smoke rising up from the ground. However, smoke's informing energy is rising hot air, and it will continue to rise as it dissipates into the ether. The informing energy of the dust, on the other hand, is the pounding hooves of the horse and gravity; once the horse has passed, the dust quickly becomes a victim of gravity, falling back down to the ground as it dissipates.

As we touched on earlier, there is also a relationship between something as fluid and organic as a splash on a calm water surface caused by a falling rock and a glass bottle smashing on impact with a hard surface. The underlying energy pattern of the water being displaced into a splash, and the bottle's energy exploding outward, could be drawn or described very similarly, although there are other distinct differences in how the animation's energy patterns resolve themselves.





In some cases, where there is one type of effect, another closely related effect will commonly occur—the "Where there is smoke, there is fire" principle. Where there is fire, electrical activity or lightning, there will most often be some extra lighting effects on any objects nearby, even surrounding the effects themselves. Where there is an explosion, there is generally some flying debris or broken props to describe, as well as the various light effects within the explosion's core (and again, with light reflected on other objects).

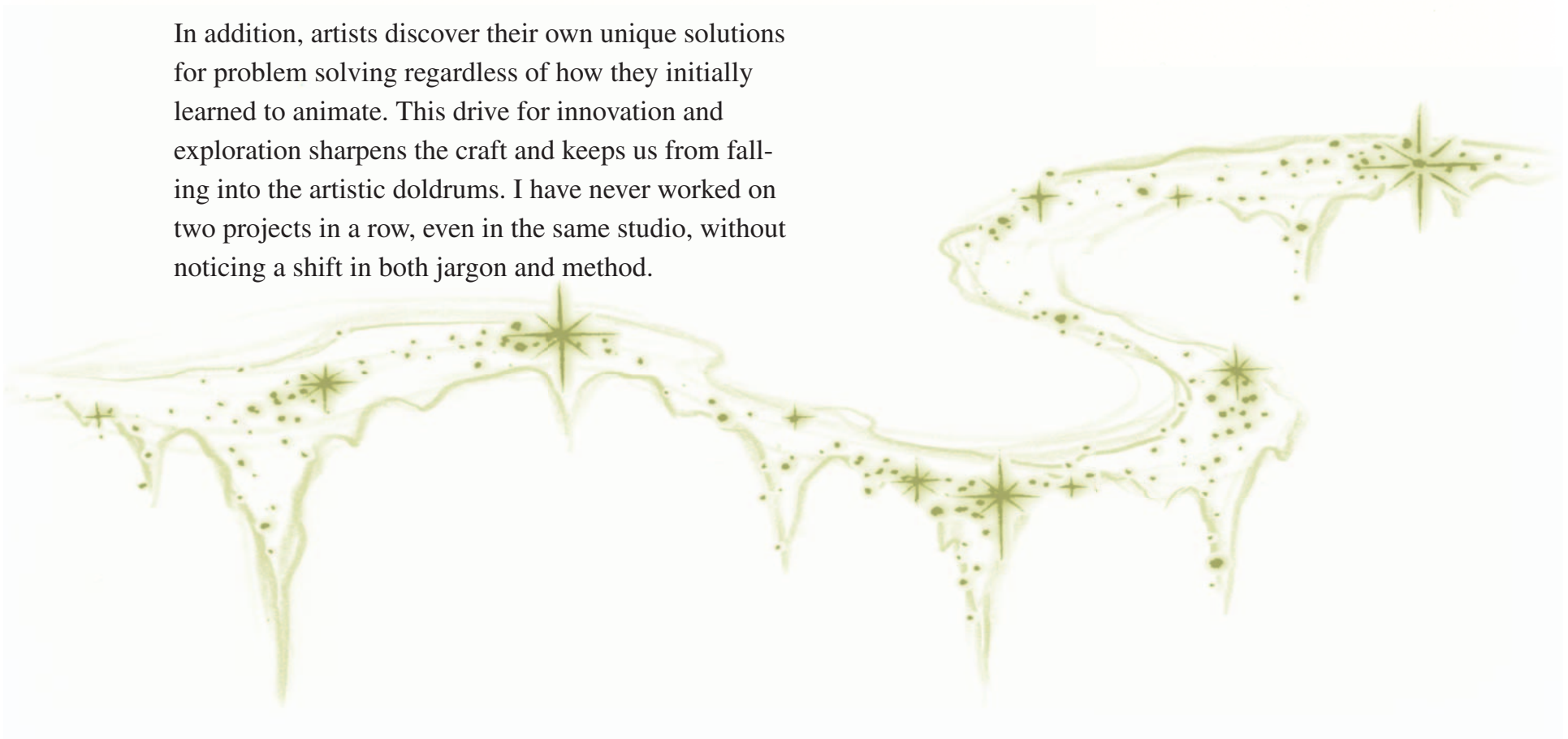
Where there are splashes in water, there will need to be some bubbles and ripples as well. As you can see, effects is replete with cause-and-effect scenarios between inorganic and organic phenomena and the patterns of energy contained within them.

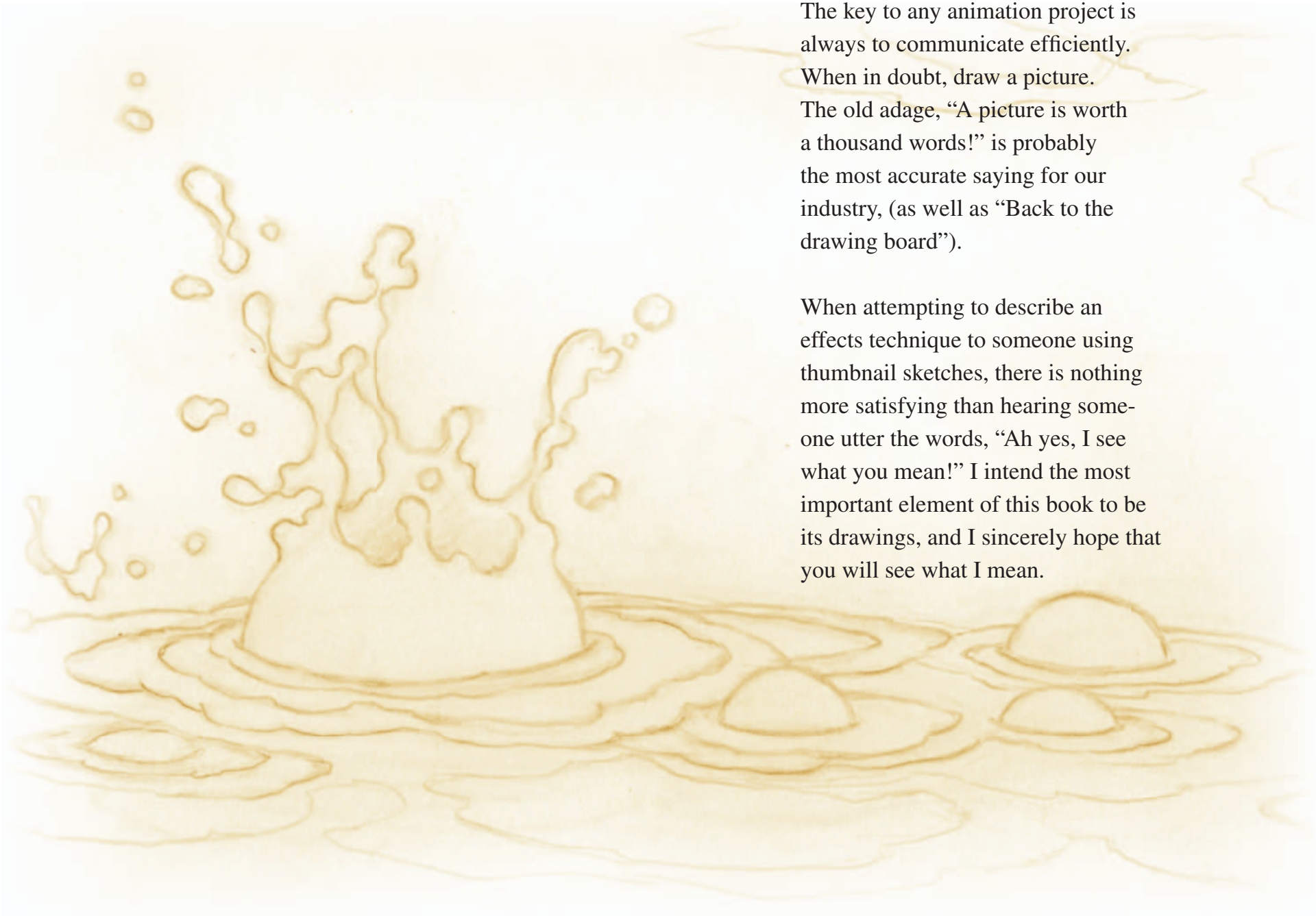
Imagine, for example, a person throwing water from a bucket on a campfire: The water will splash (some of it evaporating to steam, some of it creating overlapping pools of water on the ground), the smoke will become more thick, the flames will quickly dissipate, the embers will stop glowing, the light reflected on other objects will diminish, and the water droplets will run down the surface of the bucket.

This is only a description of changes in state; we haven't even addressed how the forces would interact. While the shapes of the elements themselves may appear random, if not scattered, as they change states, the main force in the scene will determine the path of that chaos. The water coming out of the bucket is the most forceful path of action, which will cause the fire and smoke to reverse arcs once the water hits them. Special effects animation is full of these kinds of forces of nature at work.

It is fascinating, frustrating, and challenging stuff to master! The categories I have committed to are not carved in stone, they are simply the fairest approximation of a logical categorical breakdown that I could imagine. Each animated film, and every special effects animation crew, will come up with its own effects language particular to the needs of the project they are working on.

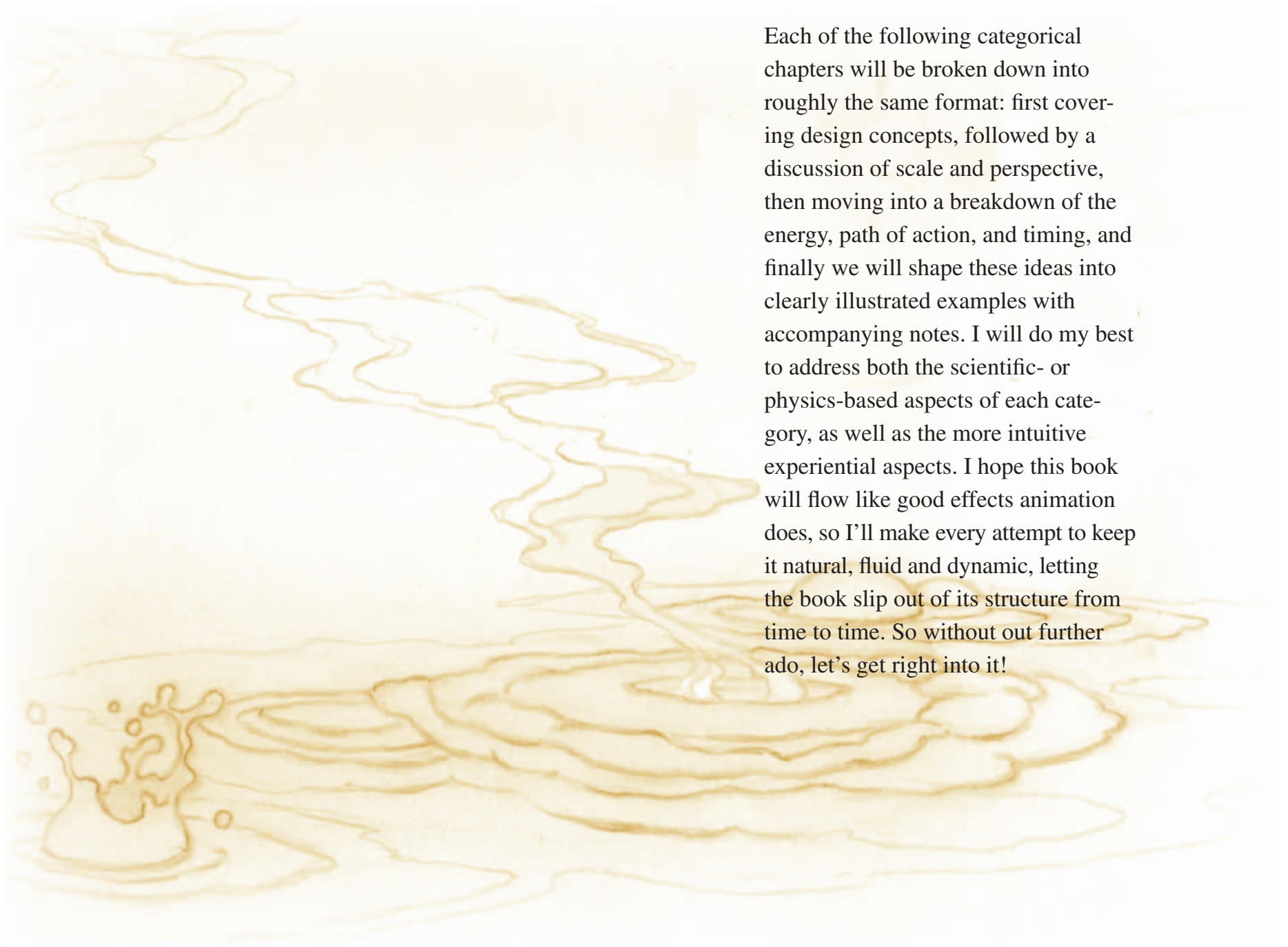
In addition, artists discover their own unique solutions for problem solving regardless of how they initially learned to animate. This drive for innovation and exploration sharpens the craft and keeps us from falling into the artistic doldrums. I have never worked on two projects in a row, even in the same studio, without noticing a shift in both jargon and method.





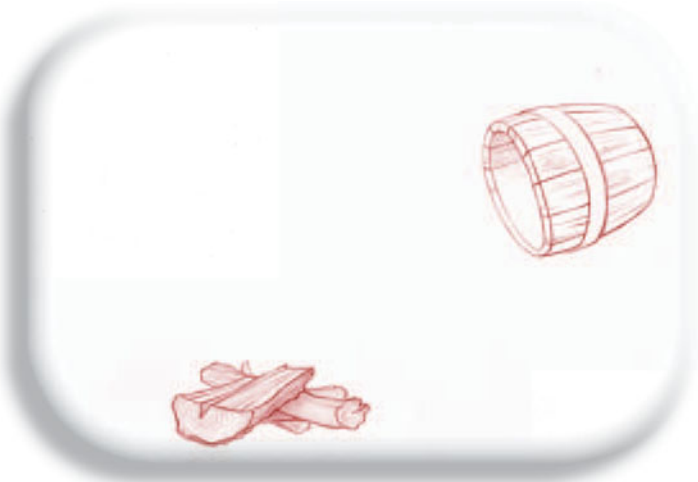
The key to any animation project is always to communicate efficiently. When in doubt, draw a picture. The old adage, “A picture is worth a thousand words!” is probably the most accurate saying for our industry, (as well as “Back to the drawing board”).

When attempting to describe an effects technique to someone using thumbnail sketches, there is nothing more satisfying than hearing someone utter the words, “Ah yes, I see what you mean!” I intend the most important element of this book to be its drawings, and I sincerely hope that you will see what I mean.



Each of the following categorical chapters will be broken down into roughly the same format: first covering design concepts, followed by a discussion of scale and perspective, then moving into a breakdown of the energy, path of action, and timing, and finally we will shape these ideas into clearly illustrated examples with accompanying notes. I will do my best to address both the scientific- or physics-based aspects of each category, as well as the more intuitive experiential aspects. I hope this book will flow like good effects animation does, so I'll make every attempt to keep it natural, fluid and dynamic, letting the book slip out of its structure from time to time. So without out further ado, let's get right into it!

Let's imagine a director decides he or she wants to see a character in his or her film pouring water out of a bucket onto a campfire. Sounds simple enough, right? But what sounds like a relatively simple scenario, can quickly turn into a great deal of work for a special effects crew. First of all, the effects artists might have to come up with a design for the wood on the campfire, as well as the bucket, depending on whether or not the layout department or a special props unit has already taken care of that.

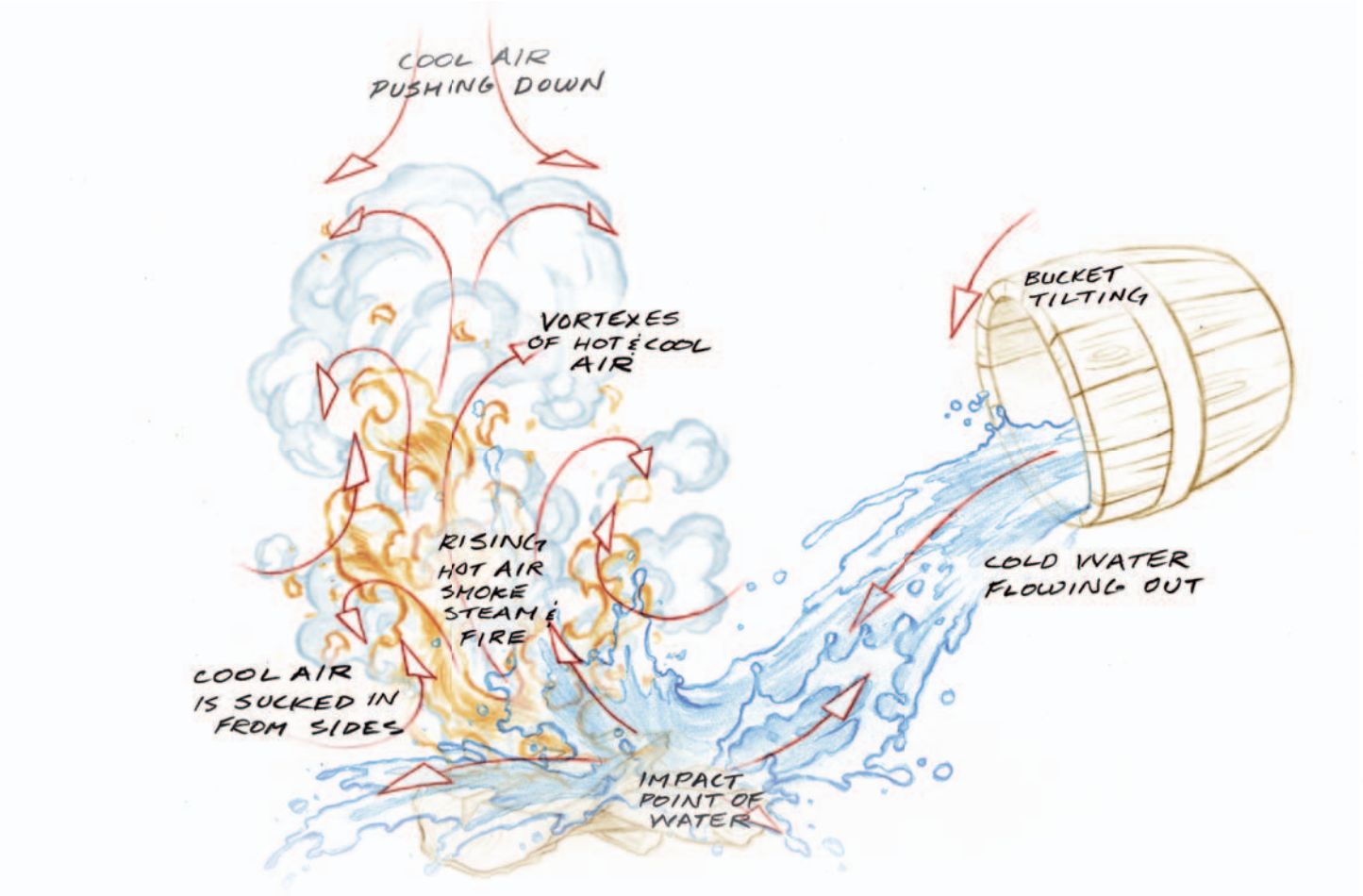


Once a character animator has animated a character pouring the bucket, and the bucket has been carefully added to fit into the character's hands, an effects animator will have to design and animate the water flowing out of the bucket. Depending on the style of the film, this effect may vary greatly in complexity and detail, but even the simplest water design is an interesting challenge to animate! The effects animator also has to animate the fire and any embers, smoke and steam that will be needed to make the scene convincing. Depending on

artistic preference, an effects animator might decide to animate the water first, and then animate the fire and smoke reacting to it, but there is no particular order carved in stone that an effects artist must abide by. In this respect effects animation can be an inexact science, but it is a problem-solving riddle, and choosing which element to animate first can greatly reduce the difficulty of the task.

To best decide which element comes first, an animator needs to think about the cause and effect that these elements will have on each other as they interact. The fire may be burning placidly, when suddenly a bucket full of cold water hits it, causing total physics mayhem! Hot air and cold air collide creating vortexes of energy, water evaporating into steam, and embers flying everywhere! The energy released by this clashing of elements is chaotic and unpredictable, but at the same time it contains a marvelously designed energy pattern that is intuitive and natural.

The final product, with all the elements working together, looks like a single event in time. But the animator has had to take a lot into consideration to get to this point. Each element has been carefully planned out and executed according to the laws of physics, and the needs of the scene from a storytelling viewpoint. To better understand what's going on in this scene, we can look at all of the forces at work on the next page.



First of all, the bucket is tilted by the character. Then gravity sucks the water out of the bucket and it pours out onto the fire. On contact there is an explosion of water colliding with the firewood, and another explosion of energy released from the rapid heating up of the water, and the rapid cooling down of the fire. As the hot air, smoke, fire and steam, escapes and rises, it all collides with a layer of cool air above it, which causes the mushroom-cloud-like effect.

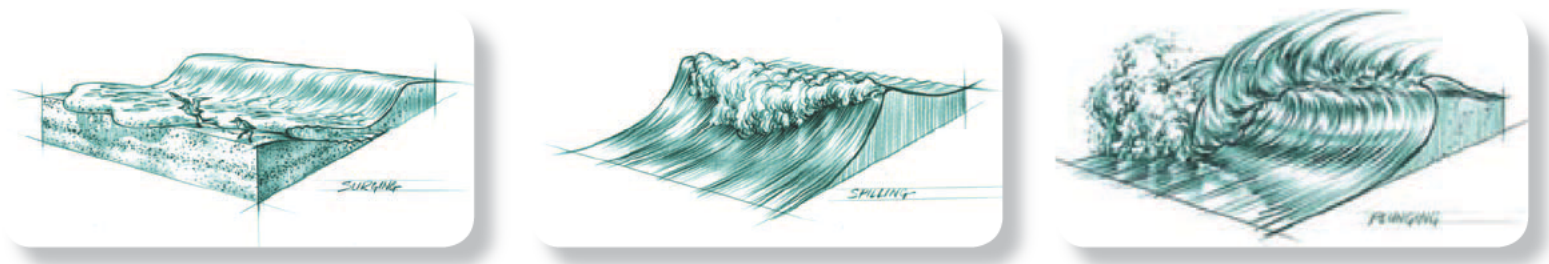
Chapter 4

Liquids

This is perhaps the most important chapter in this book. In the same way that Richard Williams spent much of his energy breaking down walking and running characters in order to best describe the basic principles of character animation in his cornerstone animation book, “The Animator’s Survival Kit,” I will focus a lot of energy on the design and physics principles that come into play when animating water and other liquids—principles which apply to some degree no matter what kind of effects you are attempting to animate, and no matter what medium.

A flapping flag, curling smoke, or a leaf flying in the wind—all of these effects contain principles that we will need to learn if we first master (or at least attempt to master) the natural principles of animating liquids.

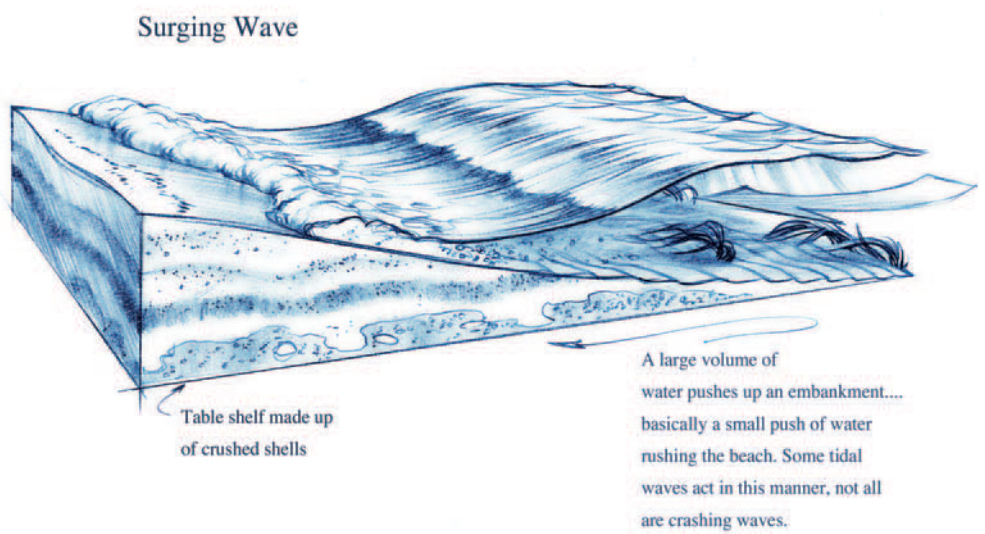
Animating liquids, usually water, is generally considered to be one of the most complicated, difficult and specialized effects animation jobs. Water comes in a staggering array of shapes, sizes, and varieties, each one with its own unique set of physics laws, energy patterns and animation techniques. Splashes, ripples, waves, rivers or streams, waterfalls, containers filled with water or any other liquid, pouring liquids, fountains, rain, etc. And each one of these forms of water in motion has in

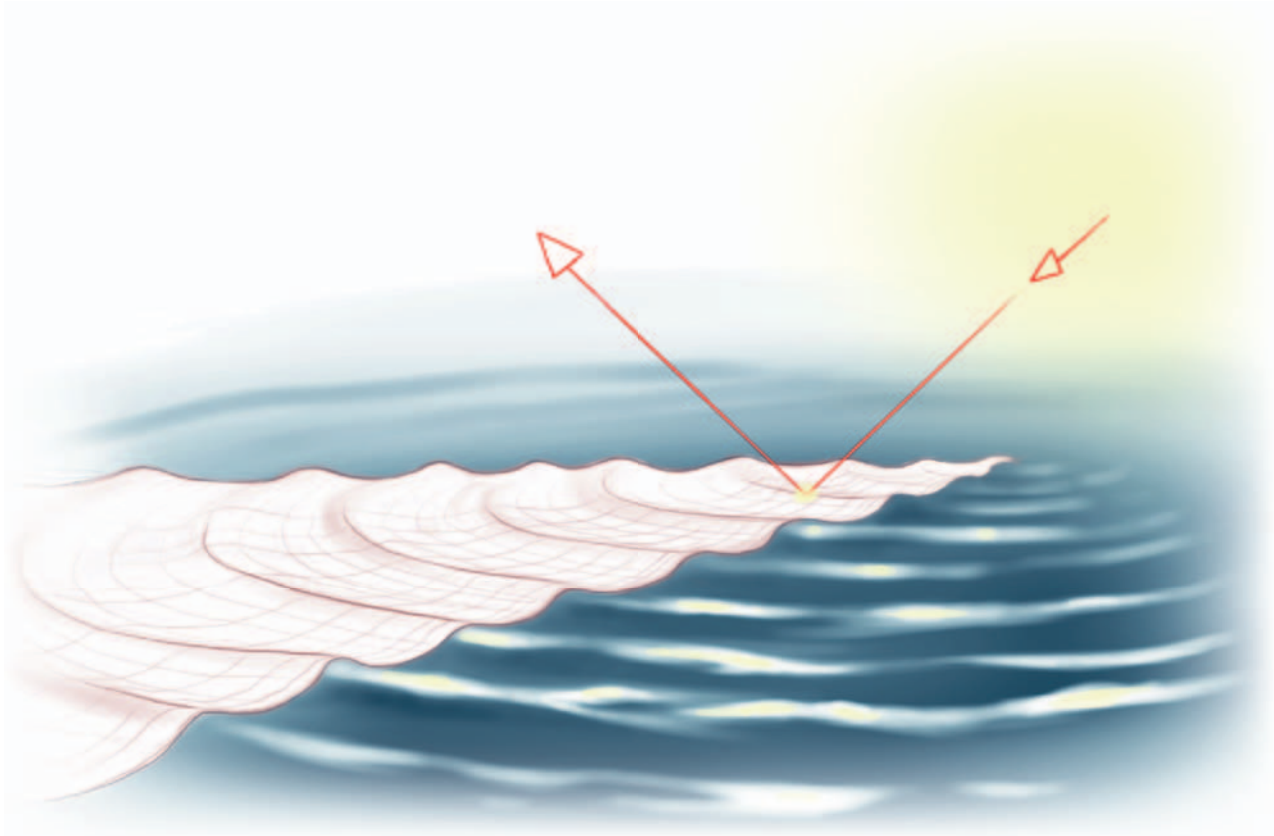


addition countless variations. A seemingly simple classification like "waves" for instance, can take on an almost infinite number of shapes and sizes. Surging, spilling, rolling and plunging waves, ocean swells, tidal waves, (tsunamis), or choppy water whipped up by a brisk wind. A wave kicked up by a boat or ship. Smaller waves may be traveling across the surface of a larger wave, and waves tumbling over on themselves create foam and bubbles. The wave variations go on and on.

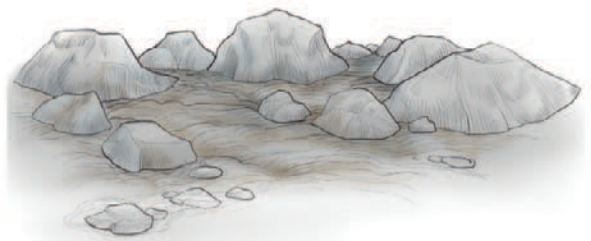
The same can be said about splashes, which vary widely from a single small pebble or raindrop landing on a calm flat water surface to something as immense as a several thousand ton glacier calving and falling into the ocean.

After over two decades of animating water I am still learning every time I animate another splash, no matter how simple. Imitating and animating such a sublimely formed natural phenomenon is a process that never gets old for me. It is important as always, to remember the basic golden rules of effects design: avoid repetition, twinning, and symmetry, and keep your silhouettes dynamic and interesting.

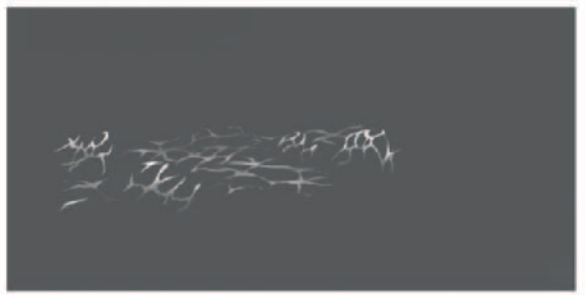




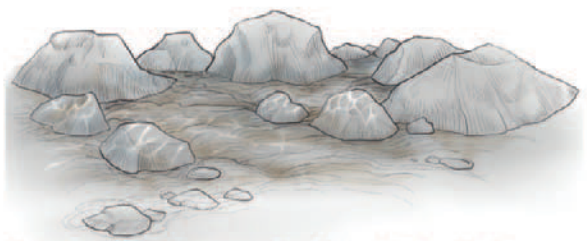
It is important to understand that what we are really looking at when we see shapes in the water is reflected light. Ripples, splashes, currents, bubbles formations, or droplets, do not really create any kind of lines in the water at all. If we learn to “see” water designs in this way, it is easier to understand how they undulate and move. The tiniest change in water surface shapes can radically effect how light bounces off of it. This is a big part of what makes water look so magical, ethereal, and difficult to describe.



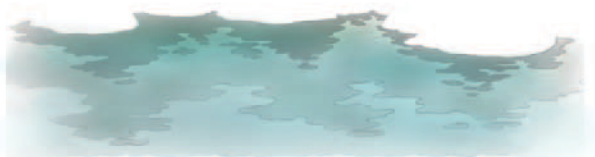
1.) A simple drawing of rocks and the bottom surface



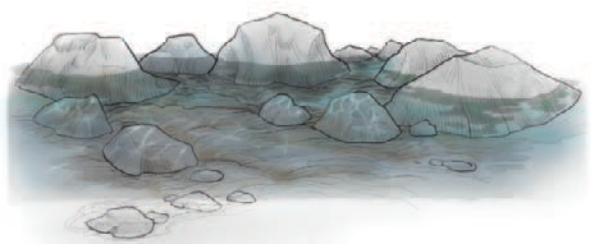
2.) Add refracted light



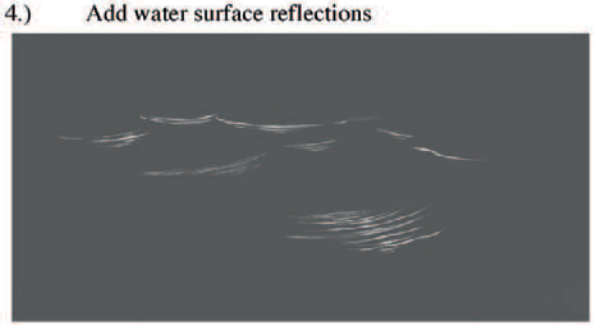
3.) Rocks and bottom with refracted light added



4.) Add water surface reflections

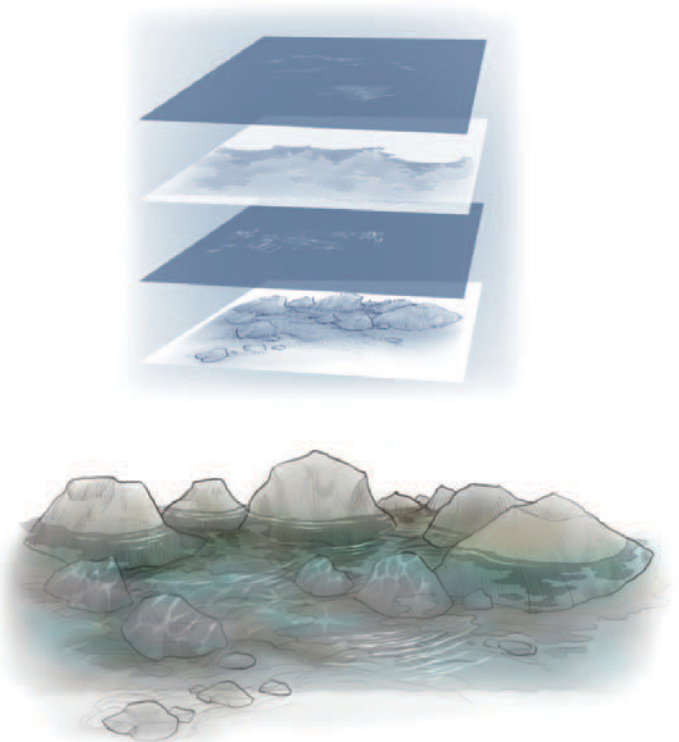


5.) Surface reflections added to overall composition



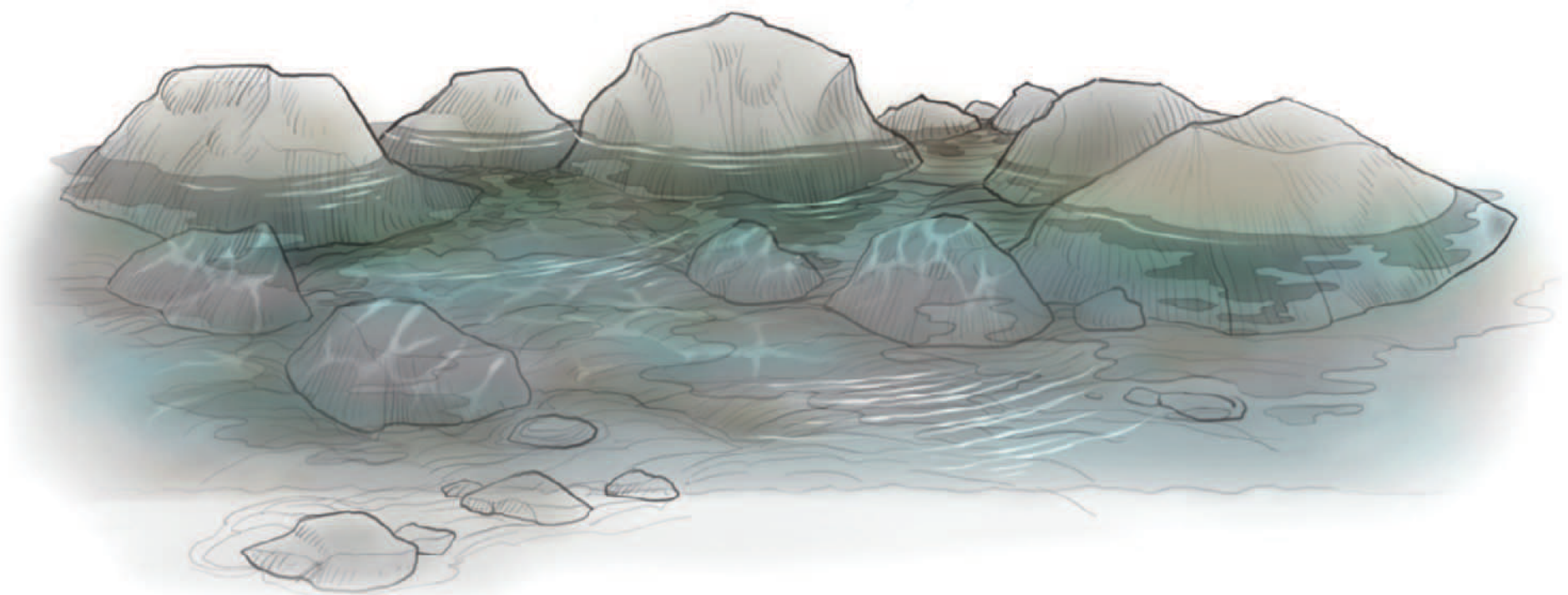
6.) Add refracted light on water surface ripples

Since water is essentially clear and colorless, what we are looking at when we see a water surface is a combination of reflections and reflected and refracted light. In this series of drawings, I have taken a piece of calm, rocky shore line and broken it down into four simple and distinct elements. The rocks and ground, the refracted light on the bottom, the reflective water surface, and the ripples reflecting light.

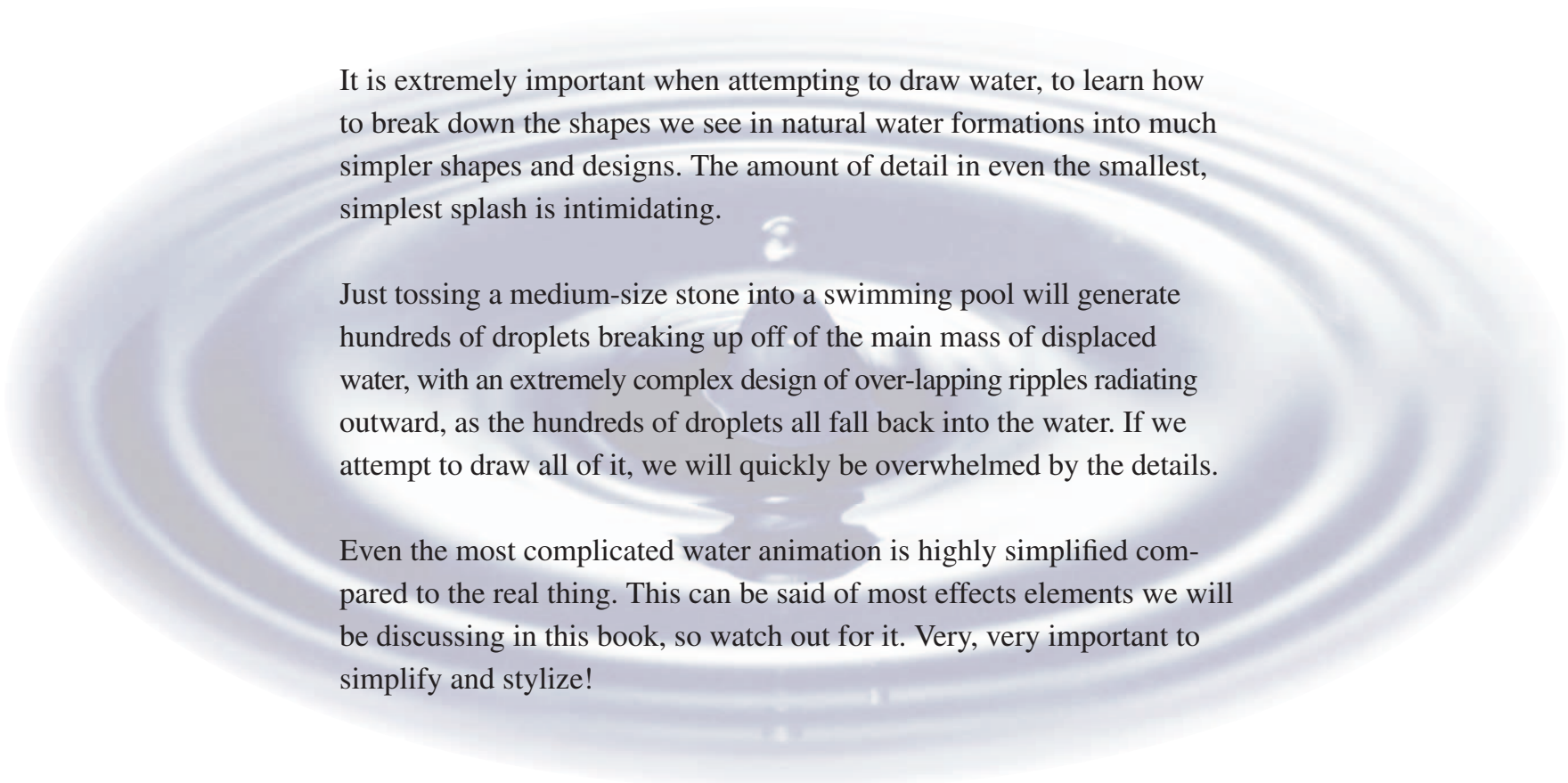


Here I have illustrated how the layers are sandwiched together to create the final image. This step of the process is what is referred to as “compositing.” In the earlier days of animation, these levels were actual pieces of artwork you could hold in your hand; the process was much as it is illustrated here, the layers were more or less simply laid over the top of one another, although there were tricky camera techniques to achieve certain transparencies and glowing effects. The same is true today, although the artwork is either scanned or is computer-generated and then the layers are sandwiched together using software specially designed for this task.

THE FINAL COMPOSITION OF COMBINED ELEMENTS CREATES A SENSE OF DEPTH AND REALISM USING ONLY FOUR RELATIVELY SIMPLE LEVELS OF ARTWORK.



Here is an absolutely wonderful example of the way water and light play together to produce incredible visual effects. This gorgeous painting by Phil Roberts illustrates surface reflections and underwater lighting effects superbly. Of course achieving a look like this in an animated film would be quite a daunting undertaking, but by breaking it down into the relatively simple elements described in the preceding pages, we can build an animated scene like this step-by-step.



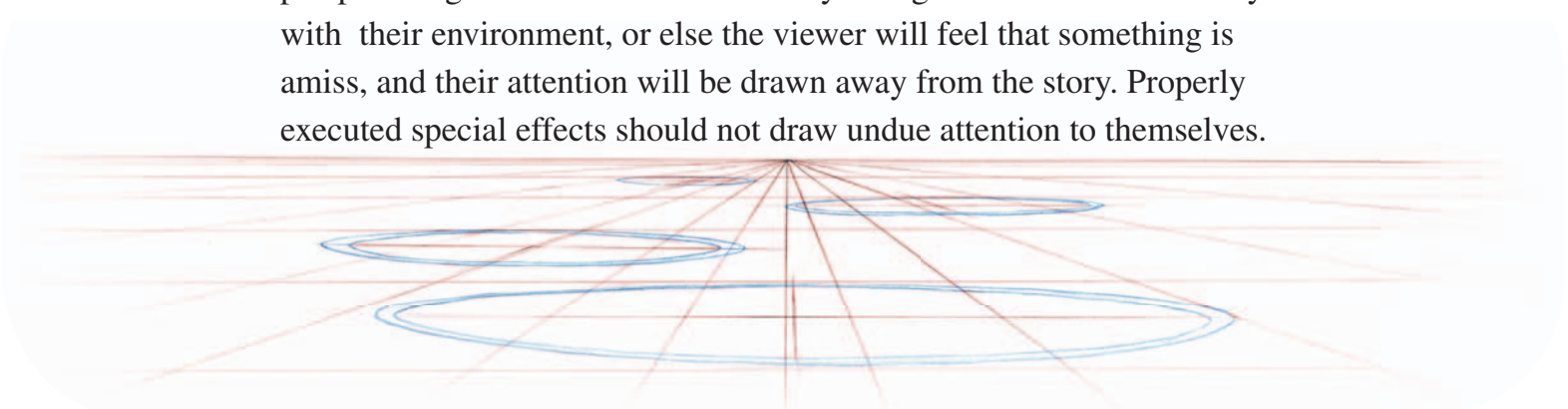
It is extremely important when attempting to draw water, to learn how to break down the shapes we see in natural water formations into much simpler shapes and designs. The amount of detail in even the smallest, simplest splash is intimidating.

Just tossing a medium-size stone into a swimming pool will generate hundreds of droplets breaking up off of the main mass of displaced water, with an extremely complex design of over-lapping ripples radiating outward, as the hundreds of droplets all fall back into the water. If we attempt to draw all of it, we will quickly be overwhelmed by the details.

Even the most complicated water animation is highly simplified compared to the real thing. This can be said of most effects elements we will be discussing in this book, so watch out for it. Very, very important to simplify and stylize!

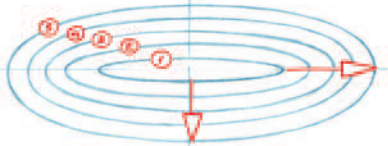
Perspective is extremely tricky to master when animating water. Such apparently abstract shapes don't at first glance appear to fall under the rules of perspective as they are generally taught, working with vanishing points and a predetermined perspective grid against which a given image is constructed. However quite the opposite is true. I have seen a lot of fantastic water animation that is plagued with fundamental perspective drawing mistakes, in high-quality animation films.

How a splash integrates with the perspective of its environment is extremely important if we expect our animation to feel natural. We must determine our perspective grid before attempting to animate any water surface or splash, and then determine how the circular or elliptical shapes that underlie the structure of our water designs sit upon that perspective grid. Our effects must always integrate themselves credibly with their environment, or else the viewer will feel that something is amiss, and their attention will be drawn away from the story. Properly executed special effects should not draw undue attention to themselves.

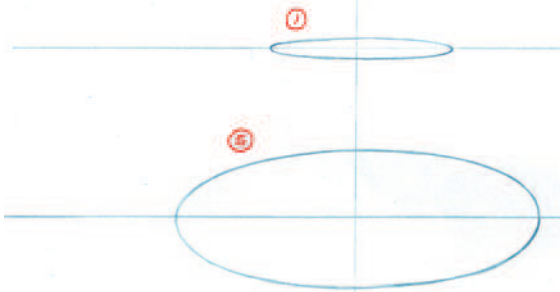


A perspective grid must be established when animating effects which occur on a surface. The ellipses of ripples caused by a splash must always conform to this perspective grid.

DON'T DO IT THIS WAY!

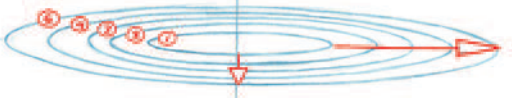


If we animate ripples expanding uniformly in every direction, we lose the correct angle of the perspective.

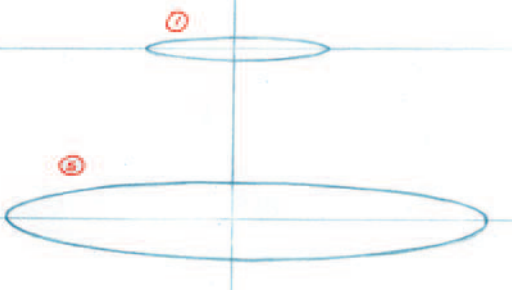


Ellipse 1 is quite narrow vertically, and describes a sharp perspective on which the ellipse is sitting. Ellipse 5 has expanded incorrectly and now sits on a much flatter perspective.

DO IT THIS WAY!

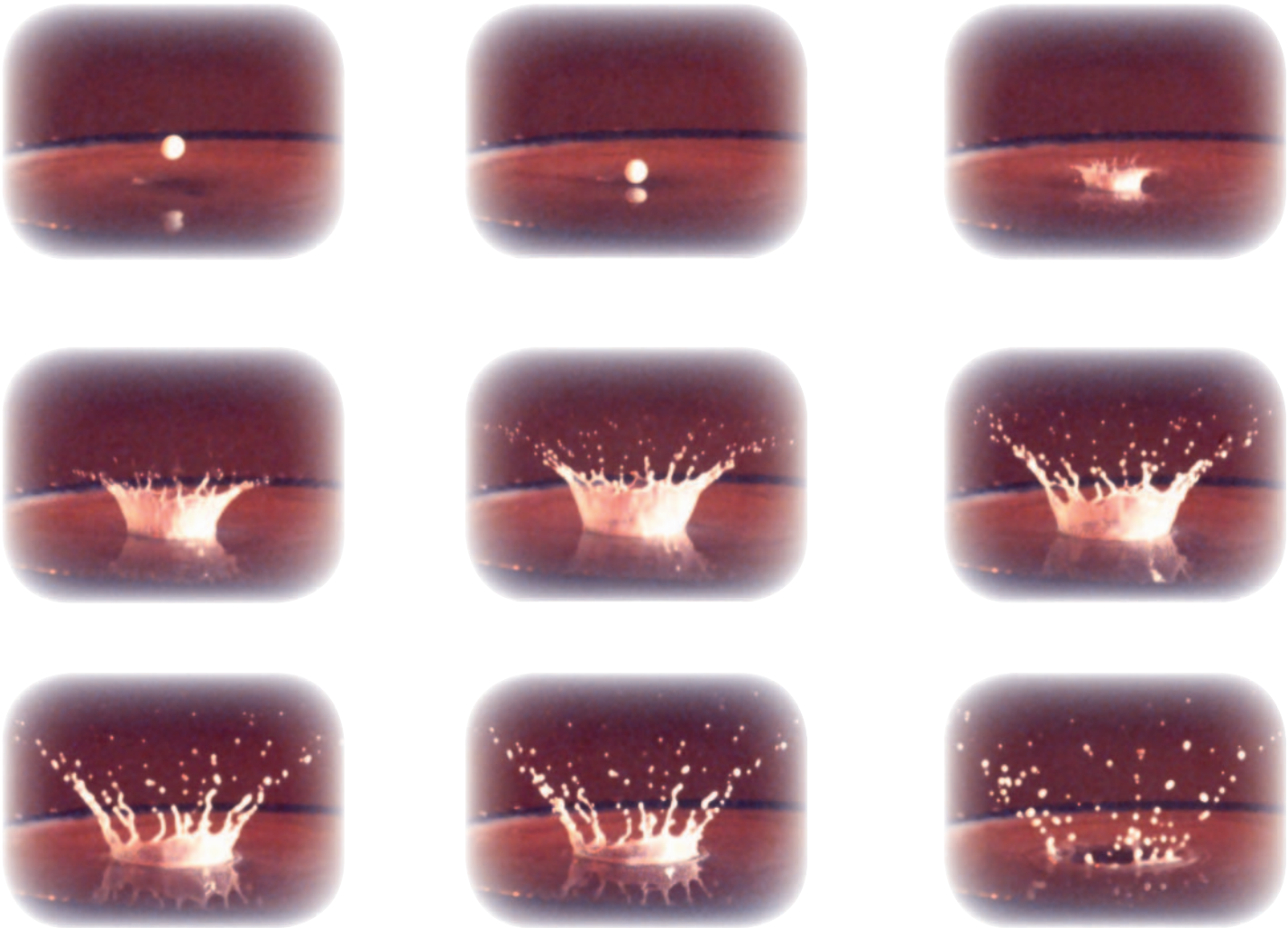


Ripple ellipses must expand farther horizontally than vertically, in order to stay true to the perspective.

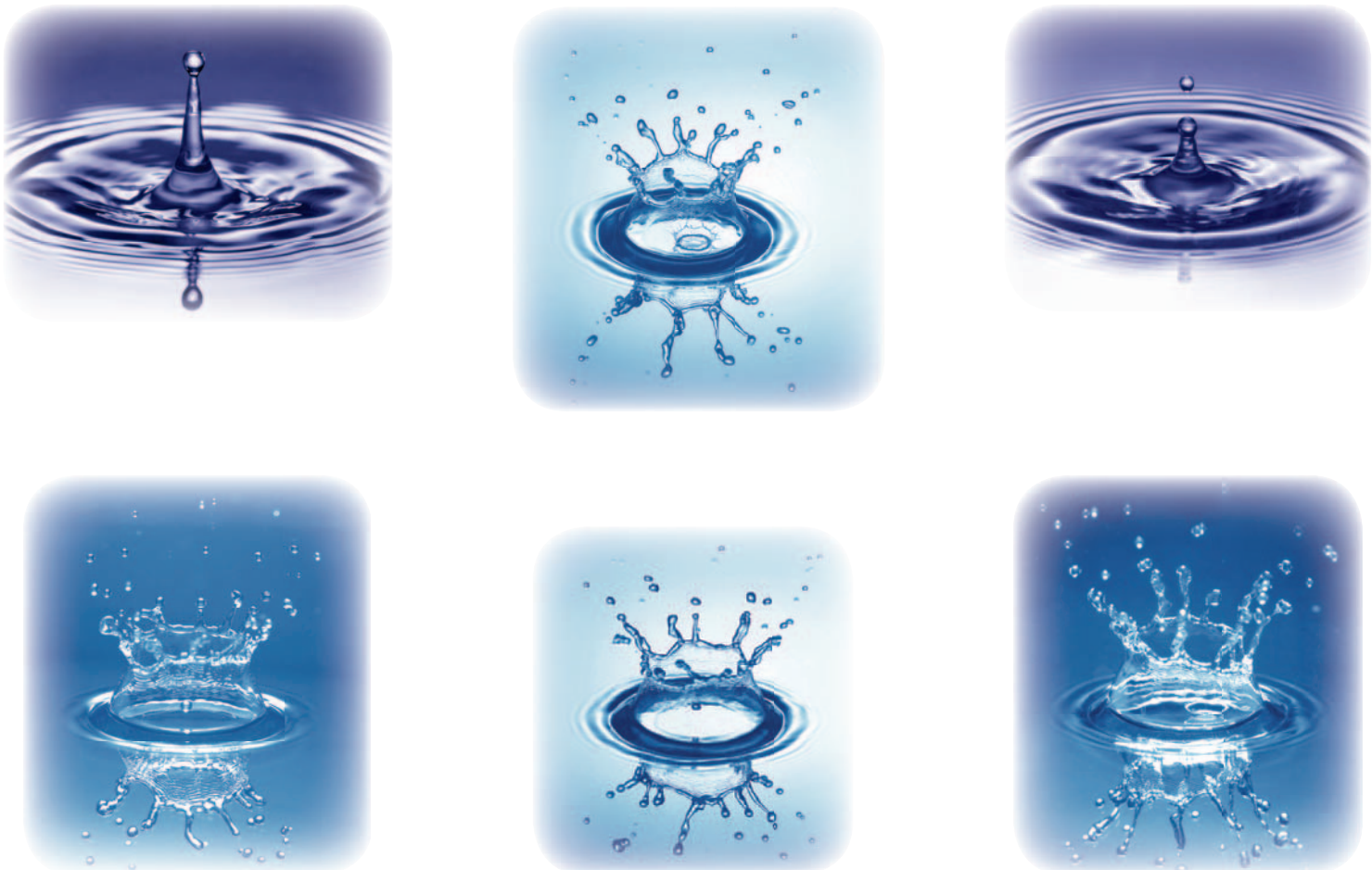


Here we have the same ellipse 1, and ellipse 5 has stayed true to the perspective, and is simply a larger version of the exact same ellipse as the smaller one.

Also extremely important to remember is that as our elliptical shapes animate—the perspective must be retained from drawing to drawing. This is where the mistake is often made, even by some of the best effects animators in the business. It is incredibly easy to allow our ellipses to flatten out in perspective, as they expand away from the center of a splash, if we don't pay attention to the underlying perspective grid and follow some basic principles of how ellipses expand in perspective. Now let's look at the forces at work in the anatomy of a splash, and break it down into simple steps.



This high-speed photography sequence of a drop of milk landing in a bowl of milk, illustrates very well just how complex even the smallest splash can be. In order to draw this phenomena, we need to stylize and simplify. At the same time the effects animators have an advantage, because they can also exaggerate the design and physics of a splash, to make it more snappy, flashy, punchy, and dynamic!



In these beautiful high-speed water droplet splash photos by Andrew Davidhazy, we see a brilliant array of perfect designs. These photos are random and do not represent a sequence of action. Each splash, either the initial splash or the secondary splash, is as unique as a fingerprint! It is important to keep in mind always, that while we do learn a great deal from observing reality, it is the special effects animator's magic to be able to exaggerate and thus almost improve upon nature's staggering beauty.

Raindrops are the simplest, shortest splashes. In these sequences, only four drawings are used. In some cases only three or even two drawings can work reasonably well. Personally I always like to go a little bit farther even with something as simple as a raindrop. This is a straight on side view. The drops on this page would be hitting a solid surface.

It is plain to see, that these are much simpler designs than something like the photos of the milk drop on the previous pages. This is especially important when you are animating lots and lots of raindrops, but it also looks cleaner and neater than getting too detailed with your splashes. This is a slightly downward view of a raindrop.

This is a straight down shot of a raindrop splashing on a solid surface. Again, this formula is not carved in stone. If I am animating a lot of raindrops I might do the majority in three drawings, some in four drawings, and the occasional one in five or even six drawings. This would introduce a nice amount of variety to the look and implied scale of the raindrops.

A splash's proportion, or size, is a very important thing to consider when animating a splash. The #1 splash here is roughly the size of a big raindrop splash, or a small pebble being dropped in the water. It should only last about half a second, or 12 frames of film and the ripples resolve almost immediately. The secondary splash of this tiny splash, is no more than a tiny droplet on top of a little jet, shooting up for 4 to 6 frames. The second, splash #2, is about the size of a splash created by a baseball or a fist-sized rock. It should last about 24 to 36 frames, or a little over a second, and its ripples could carry on as long as two or even three seconds. Its secondary splash should be substantial.

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The #3 example is a medium to large splash, as might be created by a basketball-size rock. It should last about 2 or 3 seconds; its ripples could carry on outward away from the point of impact for several seconds before resolving, and its secondary splash might be as big as its initial splash. The key to the size and duration of a splash, is how much water is being displaced.

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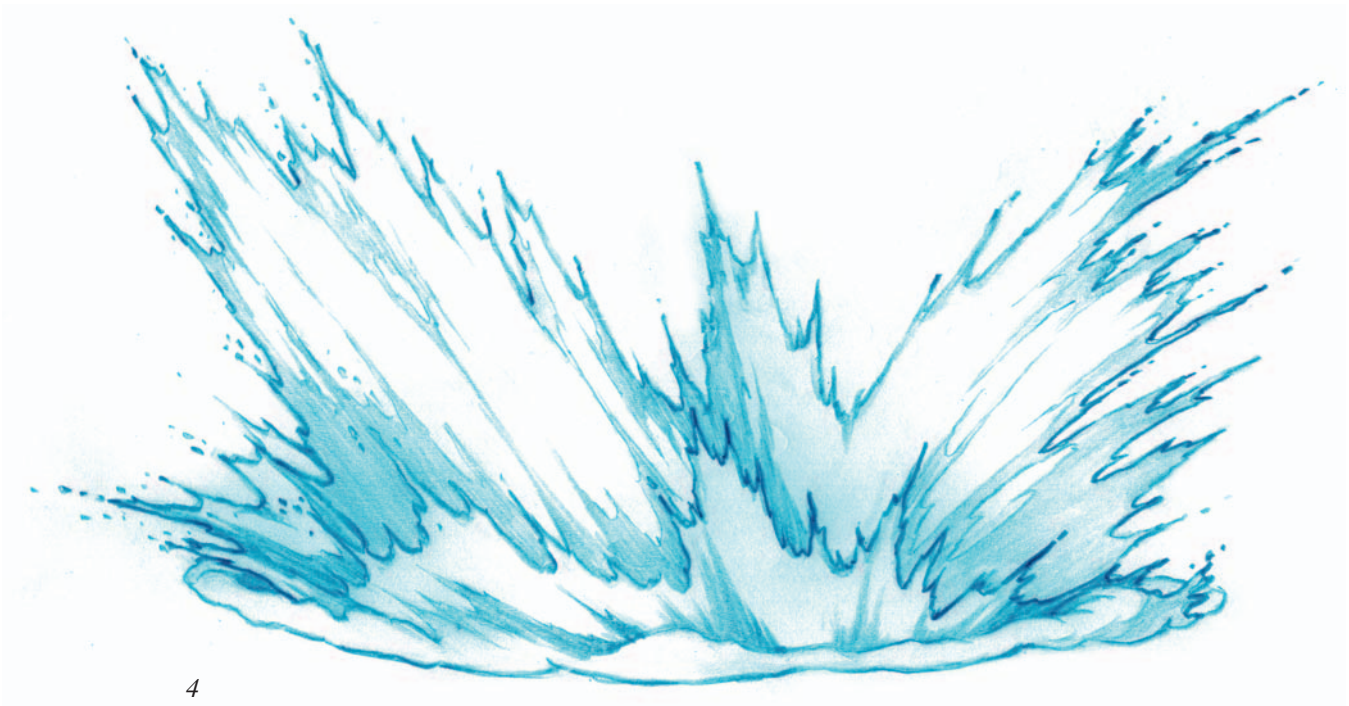

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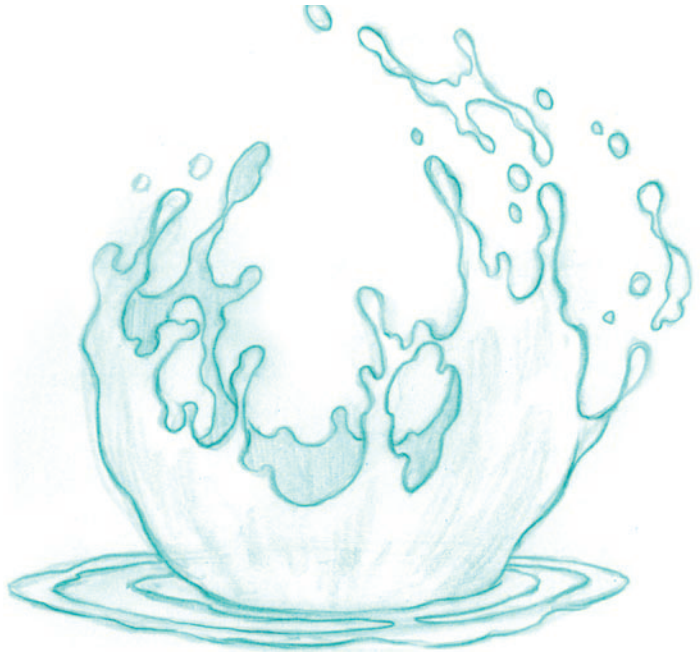
The #4 splash is absolutely enormous—the kind of splash that might be created by a breaching whale or a chunk of glacier ice falling into the ocean. The sheer volume of water being displaced is tremendous, and to really get that feeling of immense volume across to the viewer, the effects animator needs to make this splash move relatively slowly after the initial explosive impact. The secondary splash will also be huge and a splash like this doesn't just generate ripples, it generates waves, which can follow through for several seconds. This kind of effects animation takes a great deal of finesse, labor, and patience to do properly. There are no fast easy ways to make something this big look convincing.



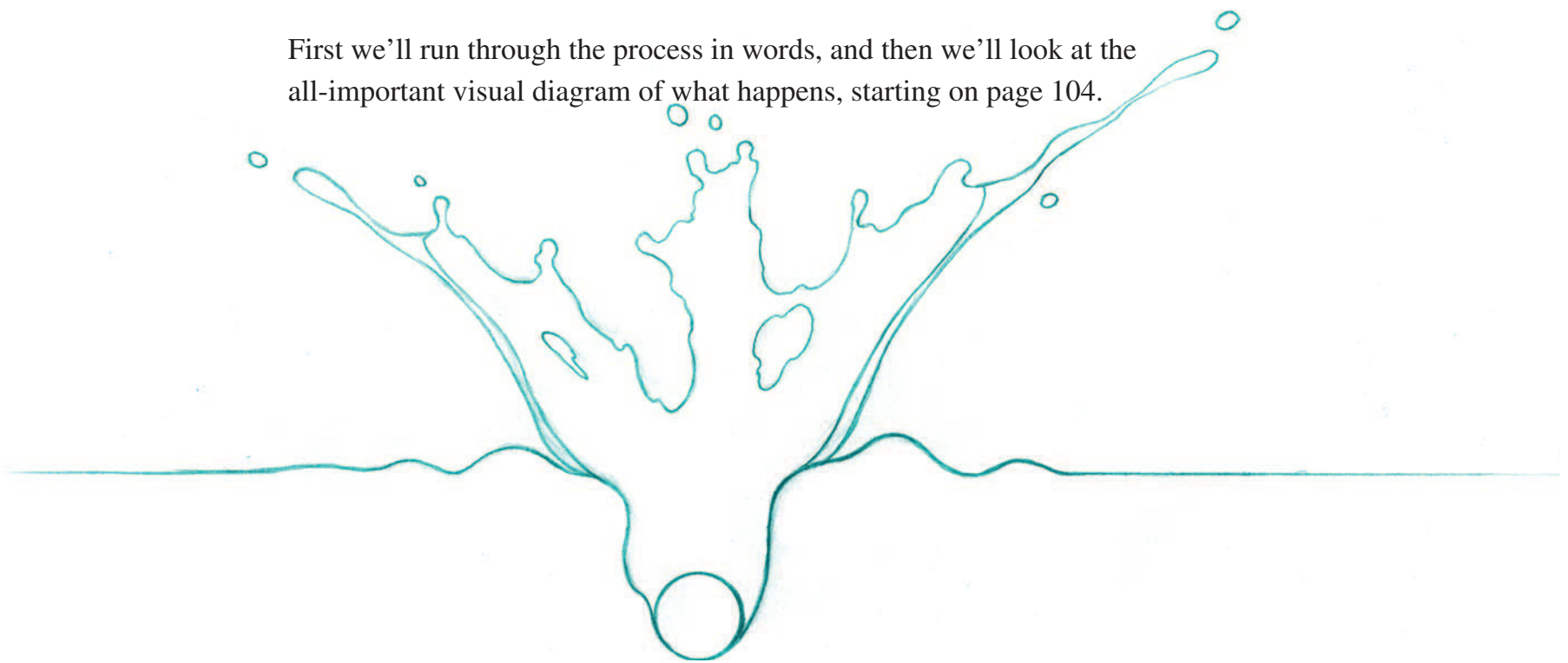
A splash's design is a wonderful thing! One of my favorite aspects of animating special effects like water is that there is always room for quite a bit of personal design interpretation, even if you are bound by a film's design to stay within some fairly strict design parameters. It is interesting to note that after working with certain highly talented effects animators over the years, I am able to recognize their effects drawings with only a cursory glance. A bit of an artist's personality can go into every effect. All of the splashes on these two pages are roughly the same size. The first one on the left here, is somewhat bowl-shaped, and fairly realistic. The splash to the right, is almost the opposite shape of the first one, and its details are far more globular and slightly less realistic. This is similar to the style of effects drawings that we designed for the feature film, "Lilo & Stitch", at Walt Disney Feature animation in Florida. Not only is it pleasant to look at, it is also a far easier style to animate, and therefore cheaper to produce. Less full, "cartoony" animation might go even farther in this direction, simplifying water design to its most basic forms, in the best interests of economy—both visual economy—and fiscal economy!

highlights and transparencies to heighten the sense of realism (and the film's budget!). The edgy design on the right, is very similar to the effects I designed on a television show called "Silverwing." The overall design of the show was very angular and hard-edged, and I designed all the effects, water, smoke, fire, etc. to reflect that design hook throughout the show. Designing effects is a lot of fun! Every scene is a new design challenge, and it never gets old.

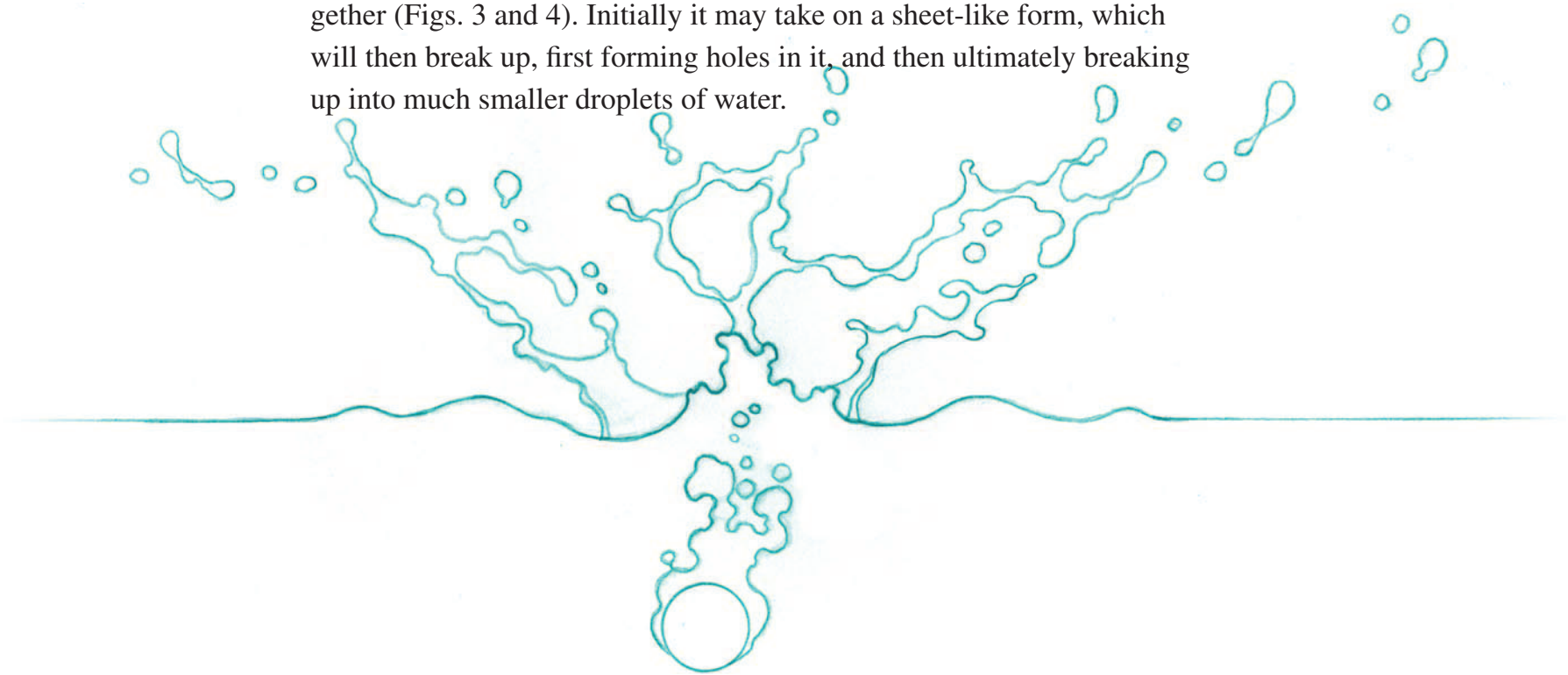
This splash is a far more elegant and less cartoony approach to designing a splash. This style might be more suitable to a more dramatic, stylistically sophisticated film, like Walt Disney's "Mulan" or Dreamworks "Prince of Egypt." Being more realistic and far more detailed, naturally is also more time-consuming and expensive to produce. With effects this realistic looking, it is almost a given that the director is going to want additional

The primary force at work in any splash is generally an object entering and displacing the water, forcing it to move out of the way, or pushing it aside. The energy of the object entering the water, its size, shape, density and velocity, all contribute to the splash's attributes, as well as the tendency of all liquid to displace itself on a molecular level quickly and efficiently as governed by its viscosity and the forces of gravity. In the case of a medium-size splash, let's say a baseball landing in the middle of a swimming pool, there is actually quite a violent reaction by the water as the baseball collides with it.

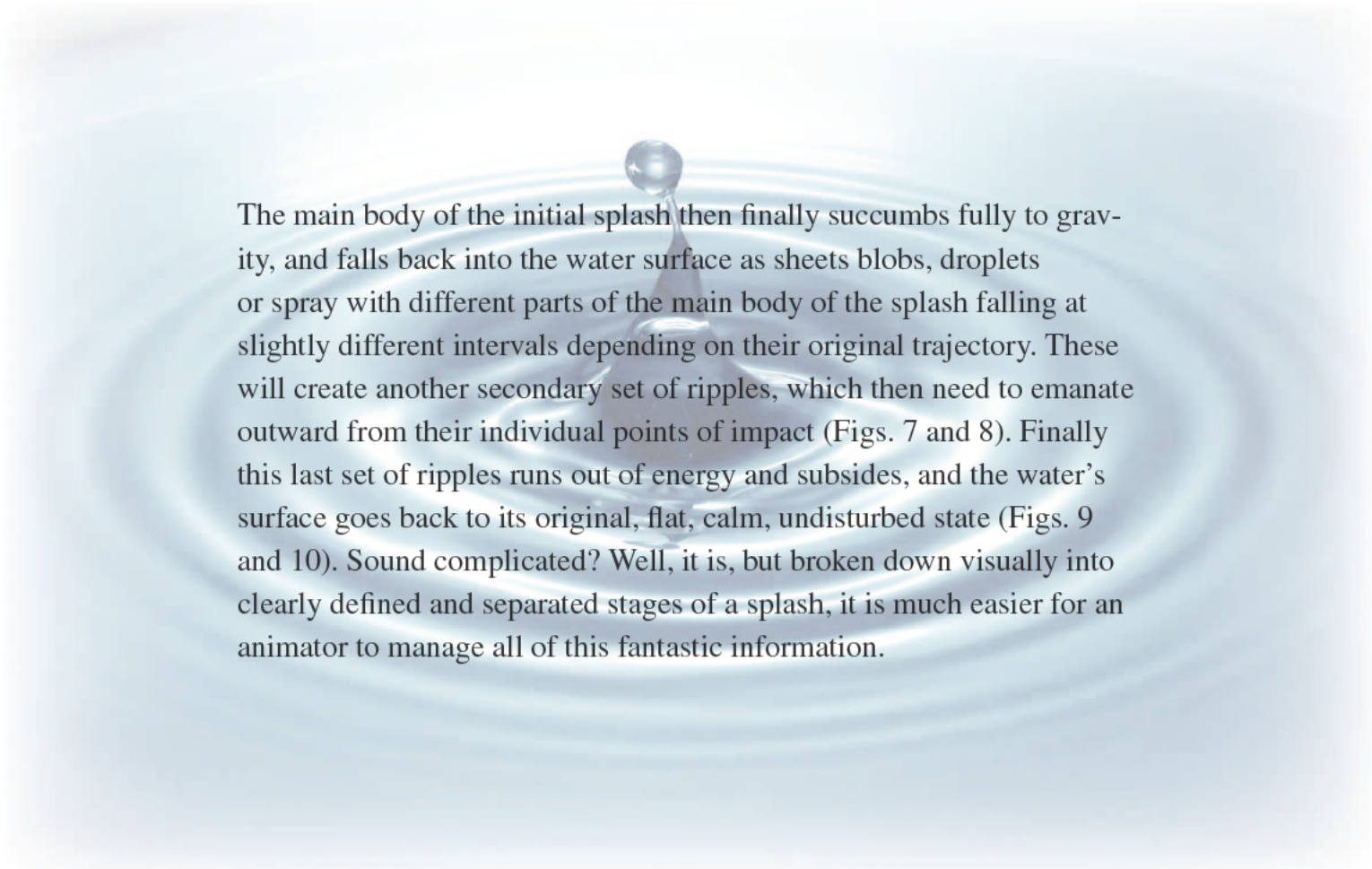
First we'll run through the process in words, and then we'll look at the all-important visual diagram of what happens, starting on page 104.



Initially, the energy and mass of the ball entering it forces the water forming the actual splash violently outwards and upwards (Fig. 1). The ball drags a bubble of air underwater with it, which breaks apart almost immediately into smaller air bubbles. Waves or ripples emanate outward in a circular pattern from the point of impact. As the main body of the splash, the water that was first displaced continues its trajectory up and away from the point of impact (Fig. 2), it begins to succumb to gravity, and at the same time it begins to be pulled apart by the effect that movement and energy have on its ability to hold its viscosity together (Figs. 3 and 4). Initially it may take on a sheet-like form, which will then break up, first forming holes in it, and then ultimately breaking up into much smaller droplets of water.



During this process, the mass, sheet, or droplets of water will reach an apex in their trajectory path, where they will slow down before beginning their continued path of motion downward, back toward the water's surface (Figs. 4 and 5). We refer to this slowing in and out at the apex as "hang time." It is a clear illustration of the force of gravity at work. This hang time is an important aspect of any splash animation, and will largely determine the scale and dramatic effect of our splash. As this apex hang time is occurring, ripples continue to emanate outward from the initial point of impact, slowing down and diminishing in size and intensity as they go. During this mere fraction of a second, water also rushes back into the air hole that was punched into it by the ball in the first place; this often creates what we call a secondary splash or jet, as the water fills the hole and is propelled upward, however, this secondary splash does not always occur and is generally much smaller than the initial splash. Its timing will also differ from and overlap the timing of the initial splash (Figs. 5 and 6).



The main body of the initial splash then finally succumbs fully to gravity, and falls back into the water surface as sheets blobs, droplets or spray with different parts of the main body of the splash falling at slightly different intervals depending on their original trajectory. These will create another secondary set of ripples, which then need to emanate outward from their individual points of impact (Figs. 7 and 8). Finally this last set of ripples runs out of energy and subsides, and the water's surface goes back to its original, flat, calm, undisturbed state (Figs. 9 and 10). Sound complicated? Well, it is, but broken down visually into clearly defined and separated stages of a splash, it is much easier for an animator to manage all of this fantastic information.

(This is a straight-on horizontal side view of a splash as if it were cut in half)

Fig. 1

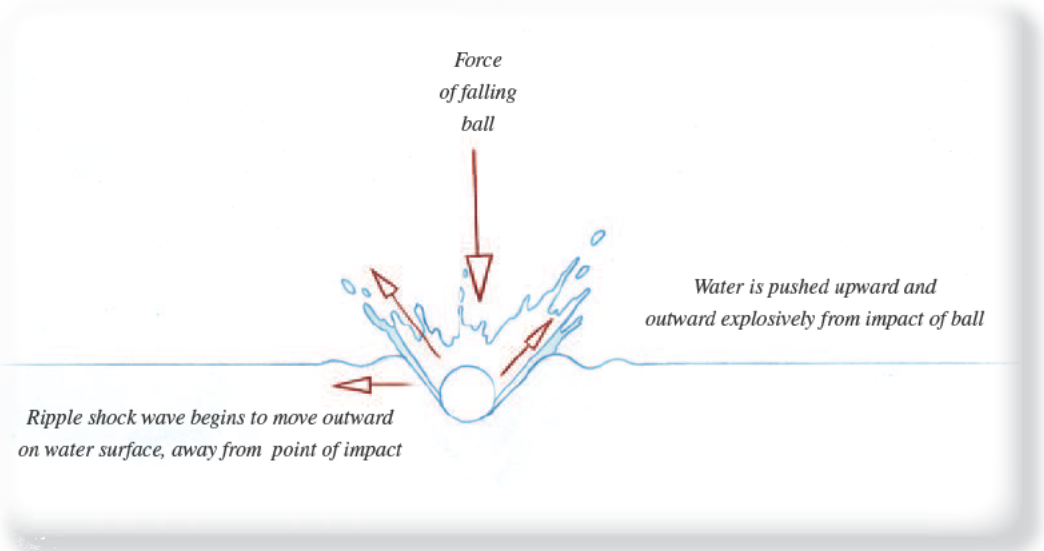


Fig. 2

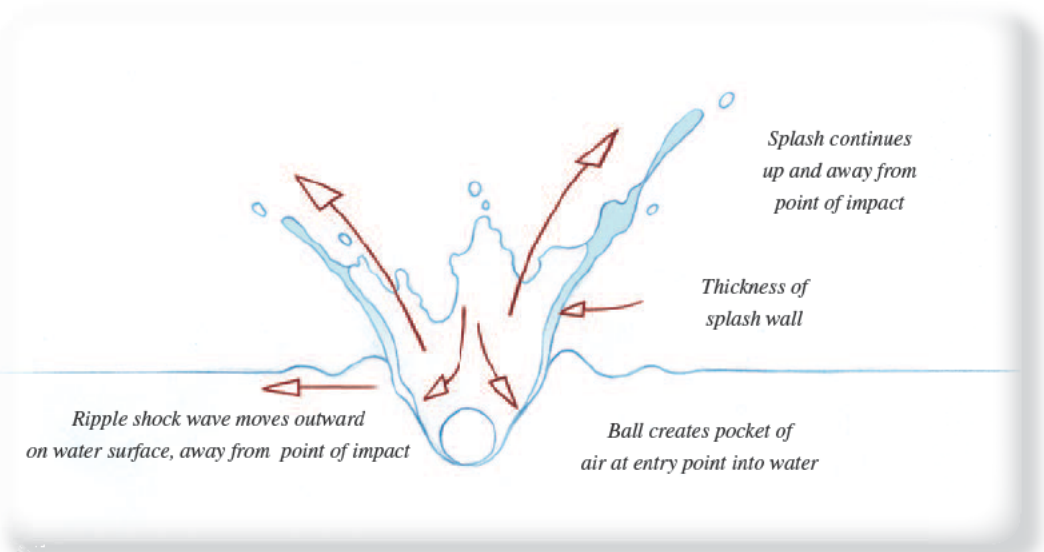


Fig. 3

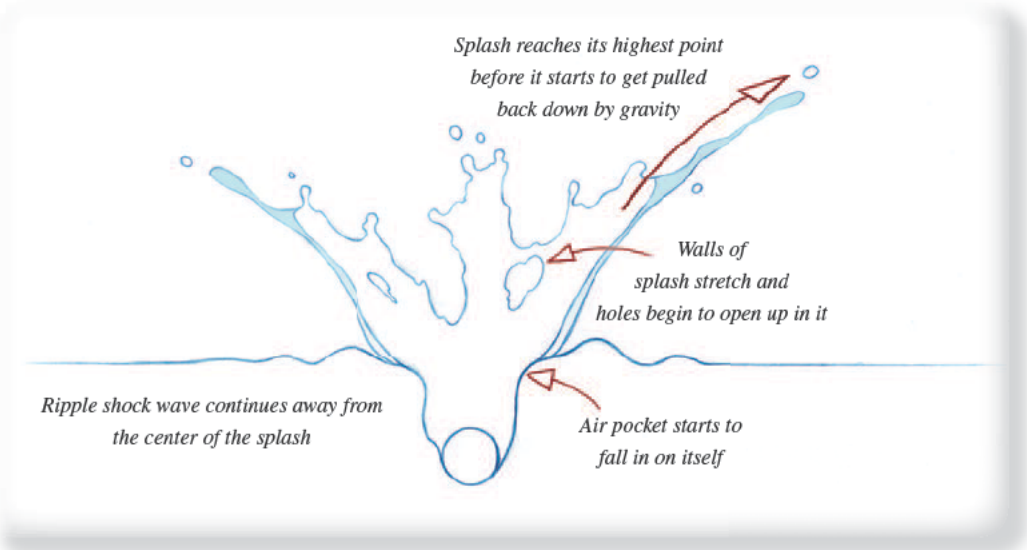

Splash reaches its highest point
before it starts to get pulled
back down by gravity
Walls of
splash stretch and
holes begin to open up in it
Ripple shock wave continues away from
the center of the splash
Air pocket starts to
fall in on itself

Fig. 4

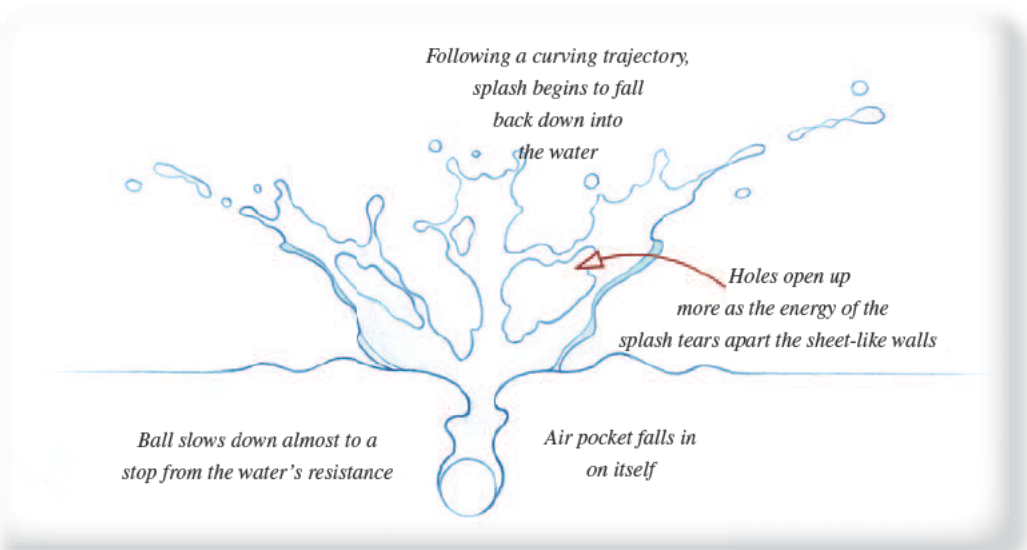

Following a curving trajectory,
splash begins to fall
back down into
the water
Holes open up
more as the energy of the
splash tears apart the sheet-like walls
Ball slows down almost to a
stop from the water's resistance
Air pocket falls in
on itself

Fig. 5

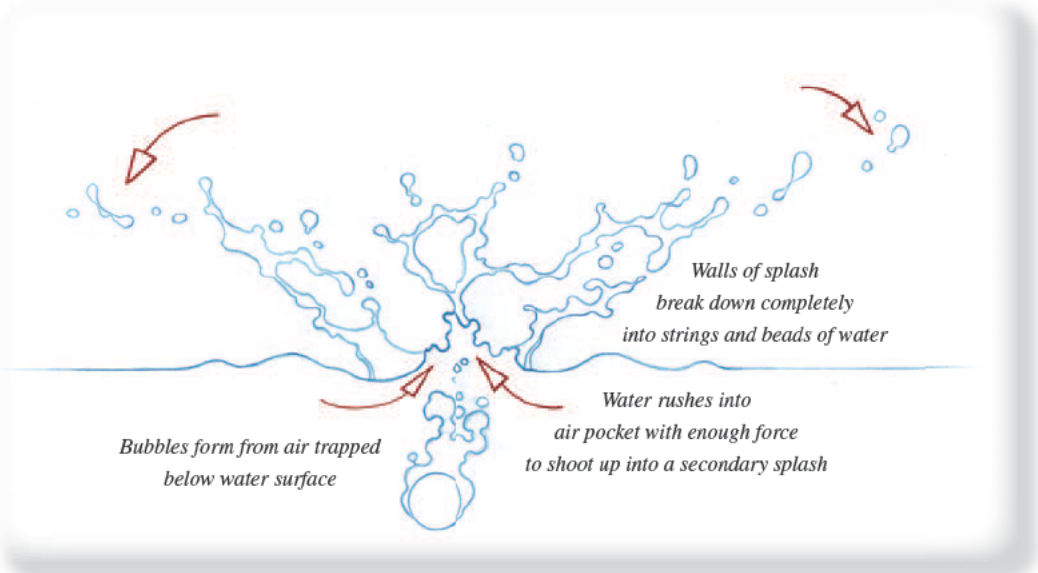

Walls of splash
break down completely
into strings and beads of water
Water rushes into
air pocket with enough force
to shoot up into a secondary splash
Bubbles form from air trapped
below water surface

Fig. 6

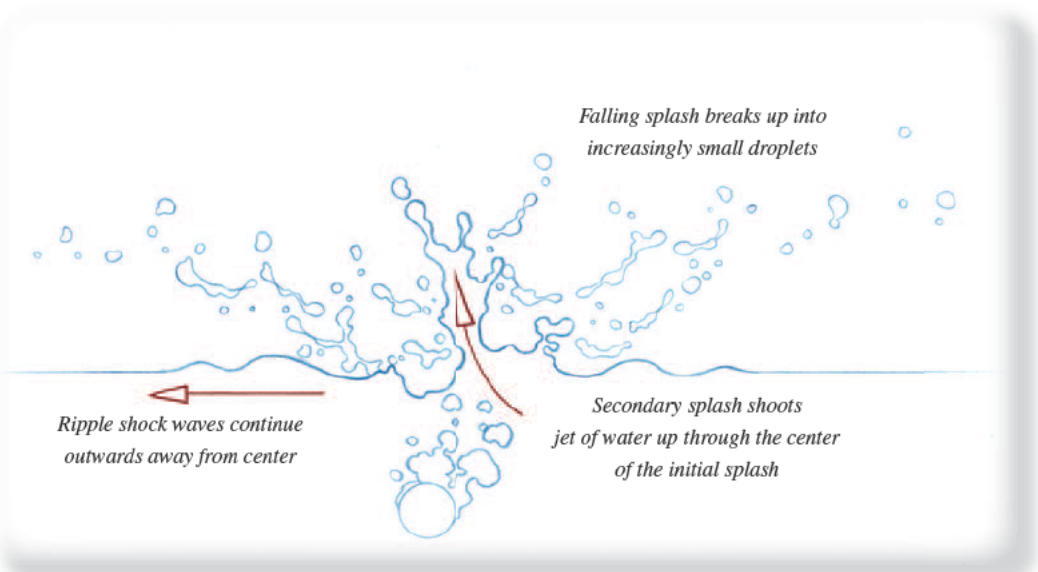

Falling splash breaks up into
increasingly small droplets
Secondary splash shoots
jet of water up through the center
of the initial splash
Ripple shock waves continue
outwards away from center

Fig. 7

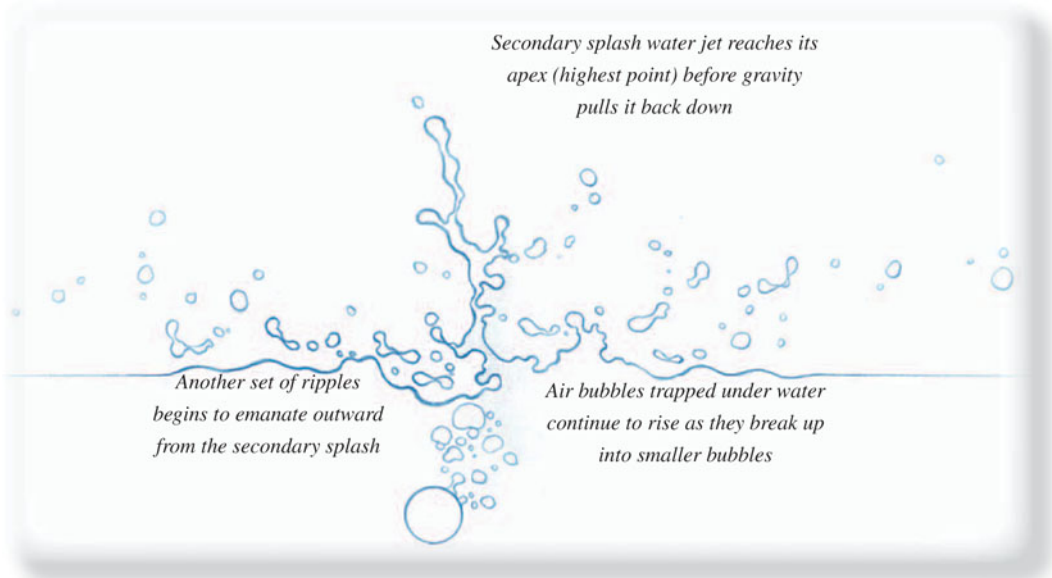

Secondary splash water jet reaches its apex (highest point) before gravity pulls it back down
Another set of ripples begins to emanate outward from the secondary splash
Air bubbles trapped under water continue to rise as they break up into smaller bubbles

Fig. 8

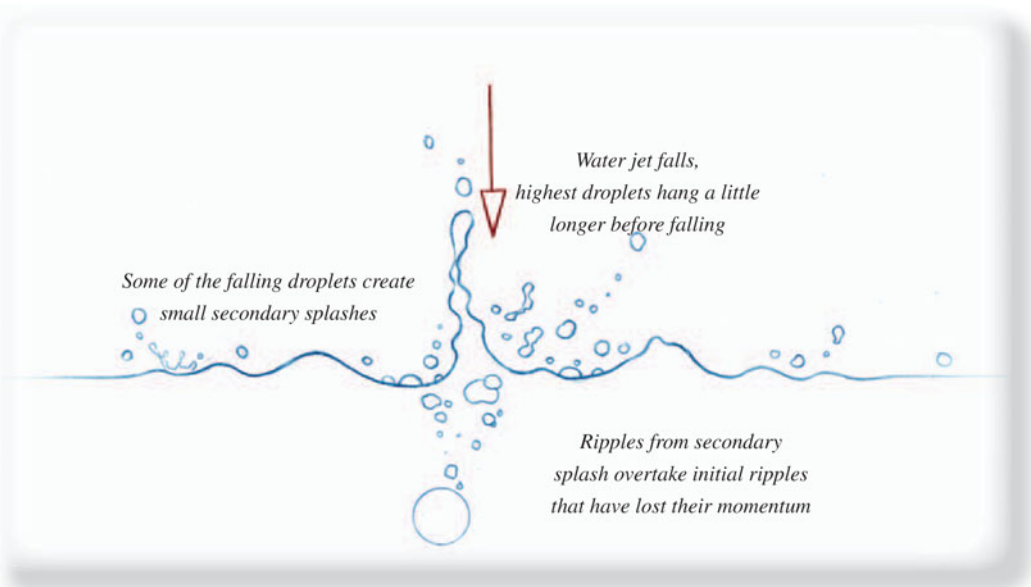

Water jet falls, highest droplets hang a little longer before falling
Some of the falling droplets create small secondary splashes
Ripples from secondary splash overtake initial ripples that have lost their momentum

Fig. 9

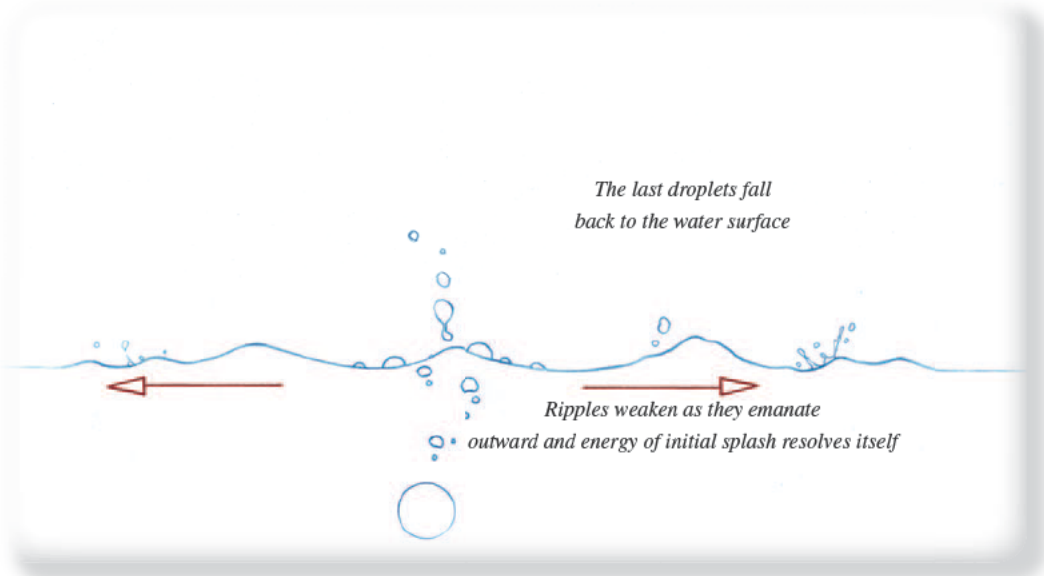
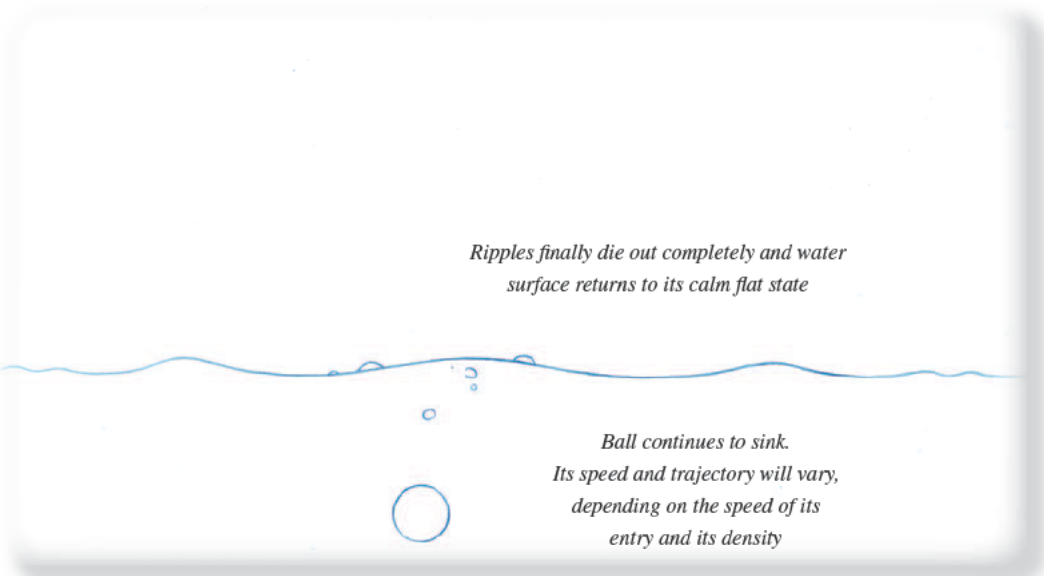

The last droplets fall
back to the water surface
Ripples weaken as they emanate
outward and energy of initial splash resolves itself

Fig. 10


Ripples finally die out completely and water
surface returns to its calm flat state
Ball continues to sink.
Its speed and trajectory will vary,
depending on the speed of its
entry and its density

Here's a real splash that was created by tossing a rock about the size of a baseball into a river. The walls of the splash are breaking up, and the secondary splash, or jet, is just starting to shoot up the center.

Here is a photo of a well-developed secondary splash, or jet, at its highest apex. You can see the leftover ripples from the initial splash, still emanating outward. Depending on the size, shape, weight, trajectory and entry angle of the object that caused the splash, the secondary splash can vary greatly in size relative to the initial splash. In some cases it is much larger than the initial splash, but it is generally a much smaller splash.

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A small water or soap bubble bursting creates its own tiny splash. A bubble can form extremely quickly, suddenly appearing on a water surface, or it can expand slowly as it reaches its breaking point. When it does reach critical mass, a tear first appears on the top of the bubble, bursting outwards very suddenly (4, 5, 6). After the initial tear in 4, the bubble breaks down very fast as in 5.

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As the bubble breaks down completely, the force of its bursting creates a unique little splash, which then resolves itself much as a larger splash created by an object which has been thrown into the water, just on a much smaller and lighter scale. In this bubble animation, I added a secondary splash caused by one droplet which lands after the main splash has resolved (15 through 18).

This splash is very similar to the more technically drawn diagram of a side view of a splash on the preceding pages. A splash of this size and intensity could have been created by an object about the size of a golf ball being thrown into a calm pool. In drawing #1 we see the initial impact, which should have an explosive energy to it, appearing as an already well developed little splash shape. If this stage is animated too slowly, with the splash growing gradually in the first 6 to 8 frames, it will not have the necessary impact, it will look slow and unreal. As the splash continues its trajectory upward, its wall begins to become too thin for the

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water molecules to hold it together, it begins to stretch and tear, and we see holes begin to open up in the thin sheets of water. Always keep in mind the directional energy that originally created the splash! As the walls of the splash tear apart into droplets and fall back to the water's surface, we see a secondary splash, or water jet, shoot straight up the middle of the splash's point of impact, as water rushes in to fill the hole created on impact. Drawings #10 through #15 show the secondary splash shooting up and falling back down, creating yet another tiny splash. The overlapping timing of these different splash elements is what gives good splash animation its dynamic and entertaining quality.

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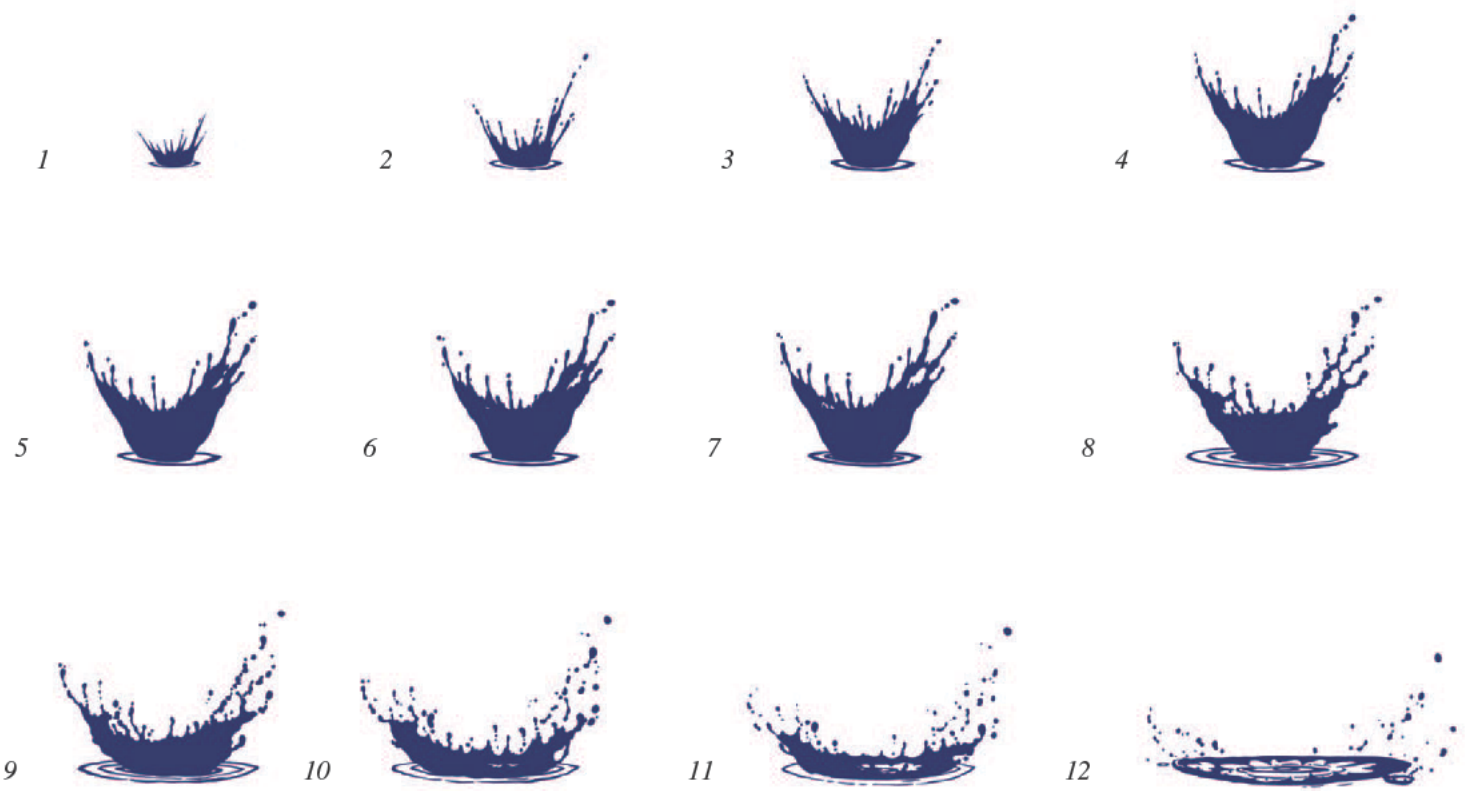

The splash on these pages is a medium-size splash, as if we dropped a baseball in the water. I have silhouetted this animation to show more clearly the importance of asymmetry when designing a splash. The sequence of separate events that make up this splash are basically identical to the smaller splashes in the preceding pages. The initial impact, drawings #1 through 4, in which the shapes are most pointed and directional. This must not be too slow or effect will lack impact.

The next stage is the apex, or highest point that the water will reach as it shoots upward. The drawings get closer and closer together as the splash slows to its highest point in drawings #5 through 7. The walls of the splash stretch and begin to tear. From drawings #8 through 12, the splash falls back down, continuing to tear apart, accelerating as it goes the drawings getting farther and farther apart as they get closer to the surface. From drawing #12 through #18, the splash does its final resolve, with the main ripple that was caused by the initial impact slowing down as it spreads out.

Care must be taken to assure the elliptical perspective is kept consistent. Several droplets create tiny secondary splashes as they land, giving a nice overlapping feeling to the overall timing of the splash. Although there may have been around fifty droplets visible around drawings #9 through 12, only four or five actually splash down, but that is enough to suggest that they all landed. Adding too many landing drops will just make your animation overly busy!

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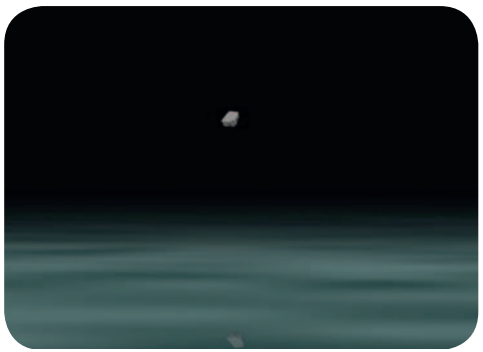

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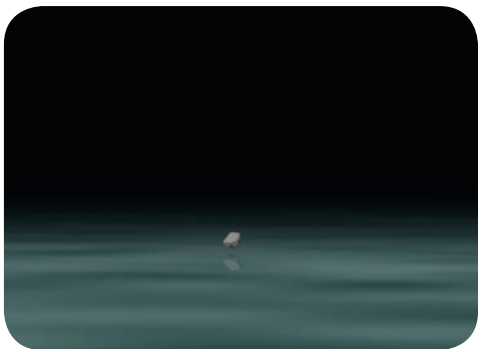

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Here is the same splash from the preceding pages, in full color with reflections, bubbles, and additional secondary splashes. We can see clearly here, how overlapping actions moving at different speeds and directions add to the dynamics.

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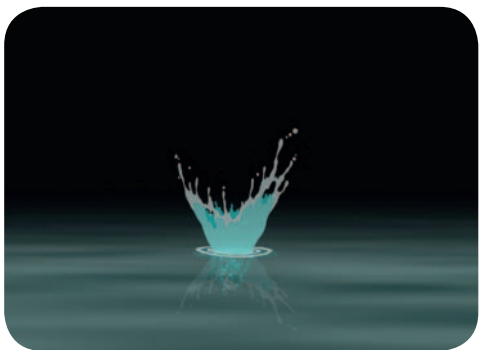

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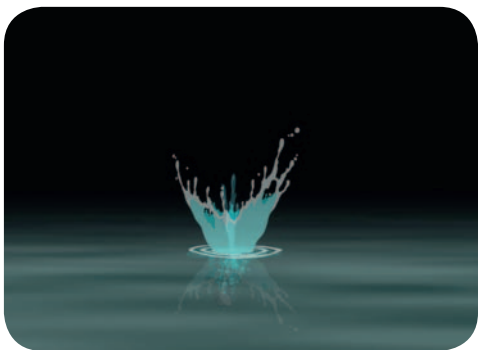

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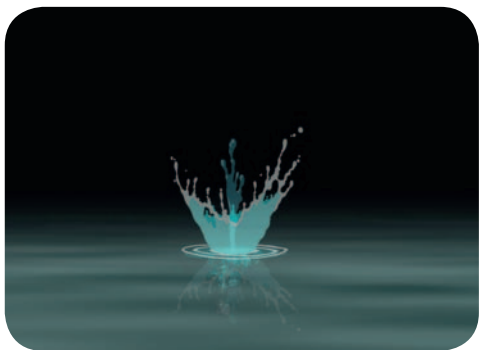

As the splash reaches its highest point and begins to fall, a secondary splash shoots up, creating overlapping action. The pocket of air that was pulled underwater by the rock, breaks up into individual bubbles.

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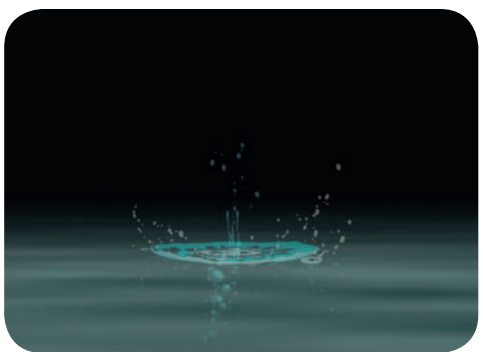

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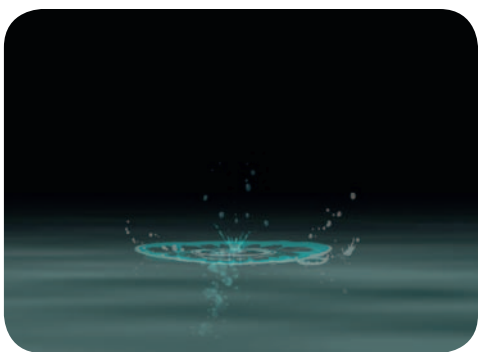

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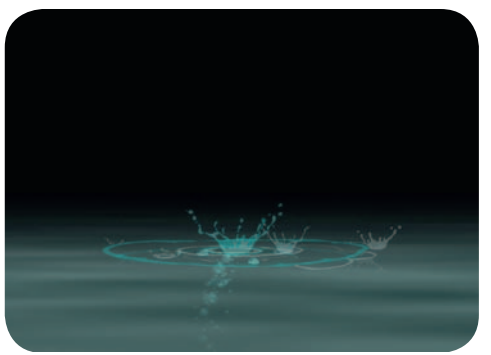

The final droplets from the initial splash reach the surface, and the droplets from the secondary splash accelerate downward. There are small additional splashes from the falling secondary splash, and the bubbles reach the surface, popping as they hit the air.

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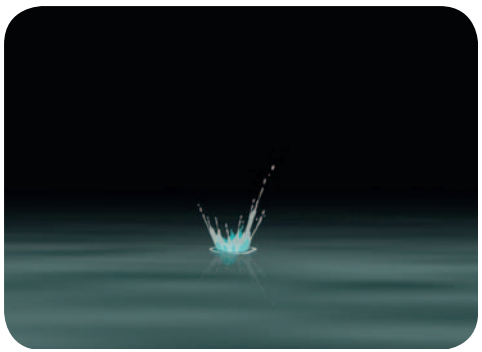

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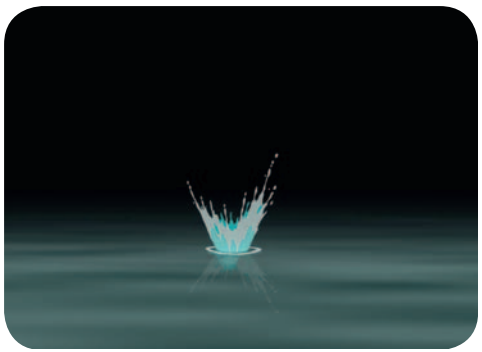

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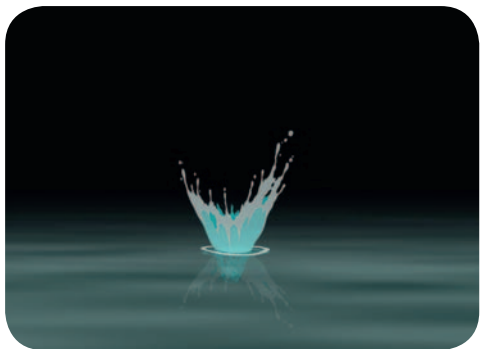

As the rock hits the water, it pulls a pocket of air under the water. The initial splash is very spiky and explosive, but it begins to slow down quickly at it shoots upward, taking on softer, more liquid shapes. Holes begin to tear in the walls of the splash.

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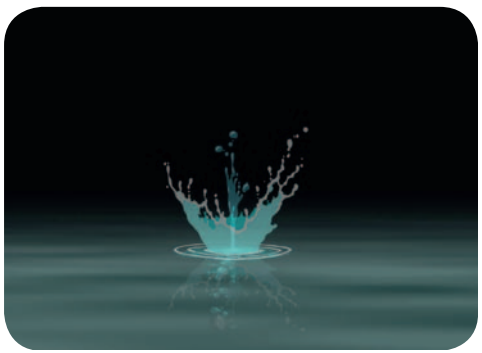

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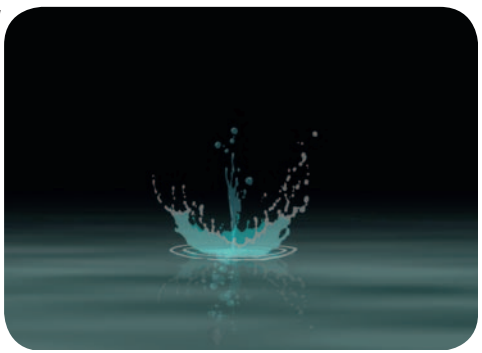

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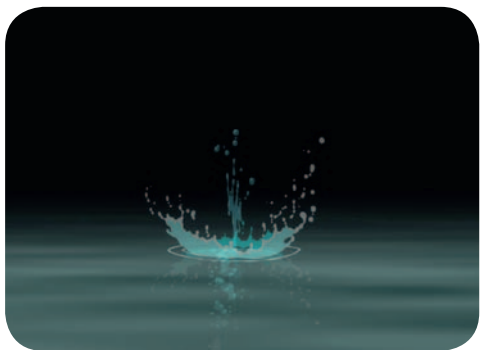

Now the walls of the splash really tear apart, as the the initial splash accelerates downward, and the secondary splash reaches its highest upward point. The bubbles from the rock make their way wiggling to the surface.

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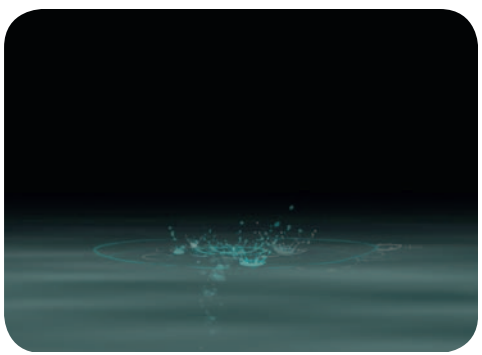

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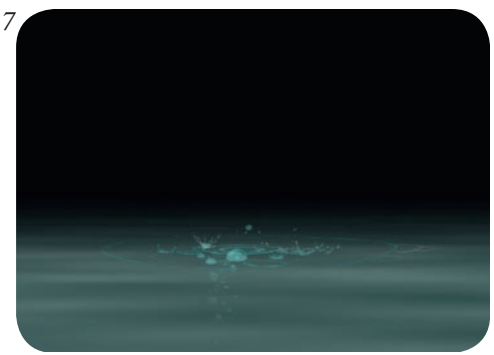
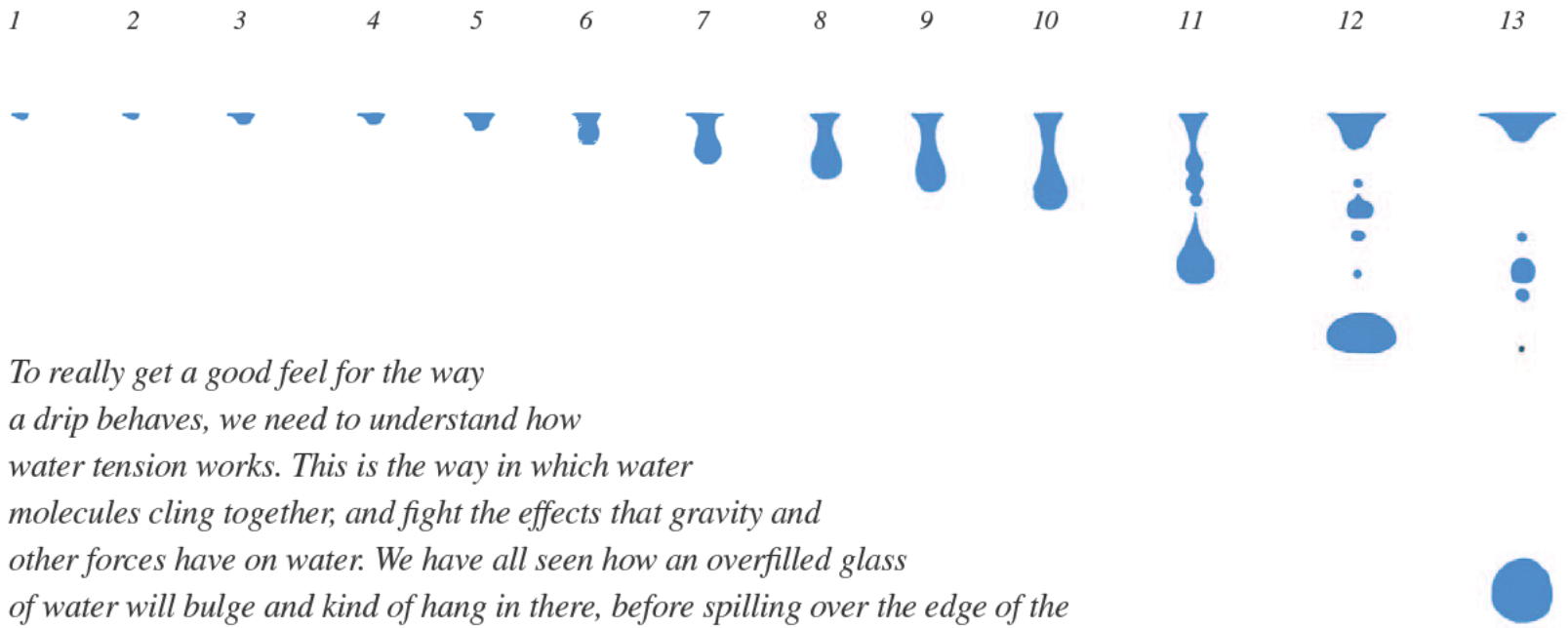


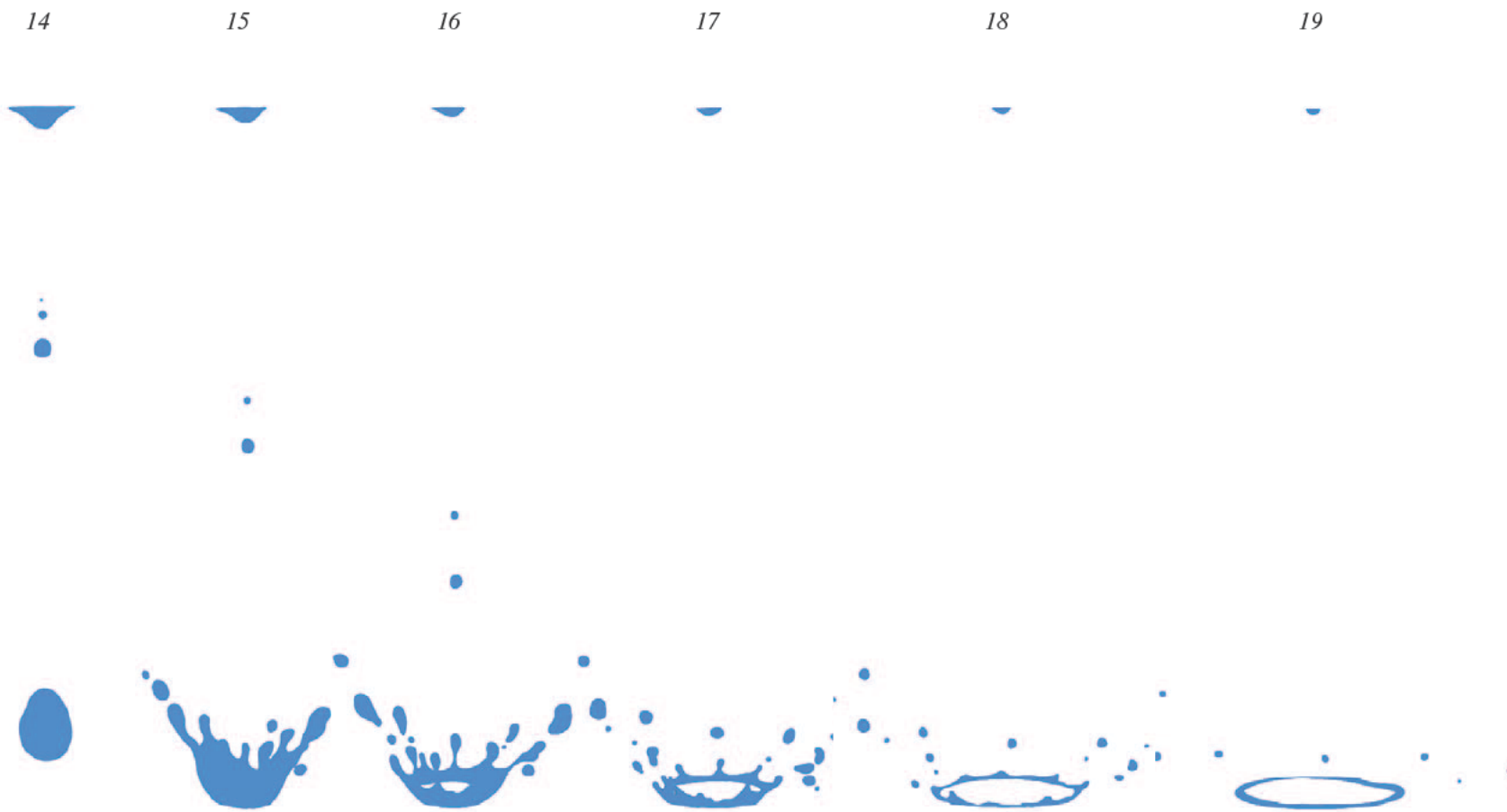
18

The ripples of the initial splash and the secondary splash fade away as the last bubbles to reach the surface pop and leave tiny ripples that resolve quickly.

Here is a sequence of a typical drip like we might see coming out of a tap that isn't quite turned off.



To really get a good feel for the way a drip behaves, we need to understand how water tension works. This is the way in which water molecules cling together, and fight the effects that gravity and other forces have on water. We have all seen how an overfilled glass of water will bulge and kind of hang in there, before spilling over the edge of the glass. That is water tension, and it is responsible for all the fascinating shapes that occur when water is pushed around, splashed, falling or flowing. In drawings #1 through 9, we see the water tension holding a small volume of water together as it builds up and increases in size, until the effects of gravity on its weight are stronger than the water tension holding it together. It stretches and strains to the breaking point in drawing #10. Immediately upon snapping, the effects of water tension take over again, pulling the molecules together to form a beaded string of droplets. It is important to vary the size of these droplets, for a more dynamic-looking effect. As the largest droplets falls first, it stretches and then squashes as it falls in drawings #11 through 17. If we were to watch this sequence in slow motion on live action film, we would see that the droplets jiggle like jello as the water tension attempts to pull them back into a perfectly spherical shape, and they recover from the snap of the stretching water breaking off. The splash when the droplet hits is exactly like the splash of a raindrop, although the droplet probably didn't have the velocity that a raindrop does, thus its splash is somewhat more subtle and globular than that of a raindrop.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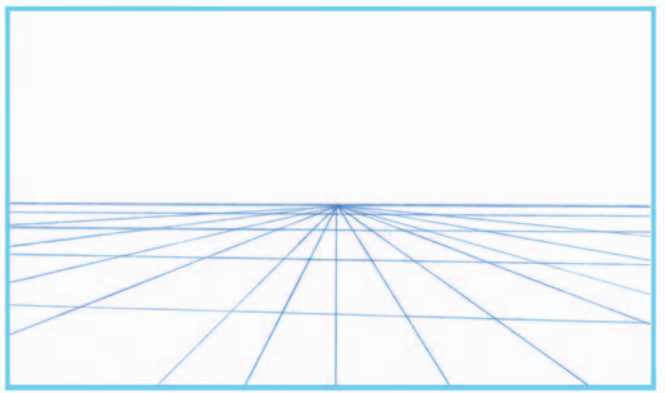
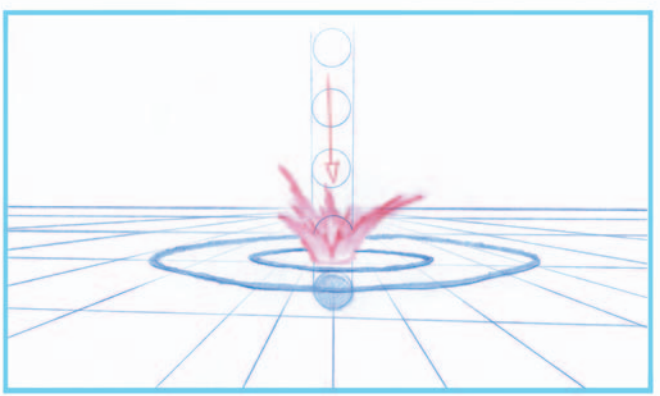
19

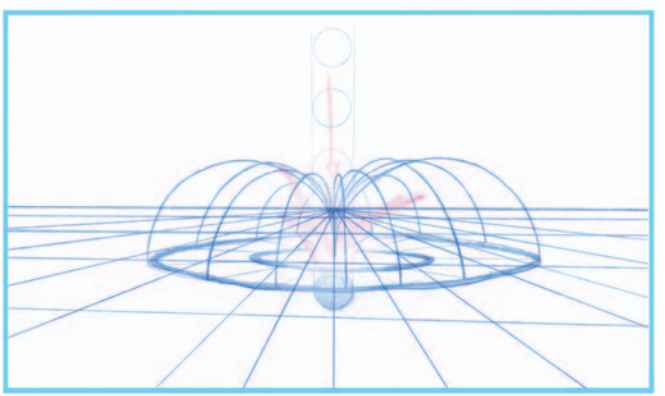
In the following splash sequence, the splash is roughly the size that would be created by a small cannon ball, or a rock about the size of a small coconut, being dropped into the water. This approach to drawing/animating a splash, is clearly not an attempt at realism. This particular design and approach owes its look and physics to the Disney effects animator's animation of the 1930s, 40s and 50s, and is very similar to splashes seen in films like Dumbo and Peter Pan.



Of course, the first thing an animator must do, before forging ahead with animating an effect, is have a plan. The very first step in this plan is a simple perspective grid, inside a frame, or field. This is the stage on which the animation is set, and it needs to work with whatever other elements there are in the scene's layout.



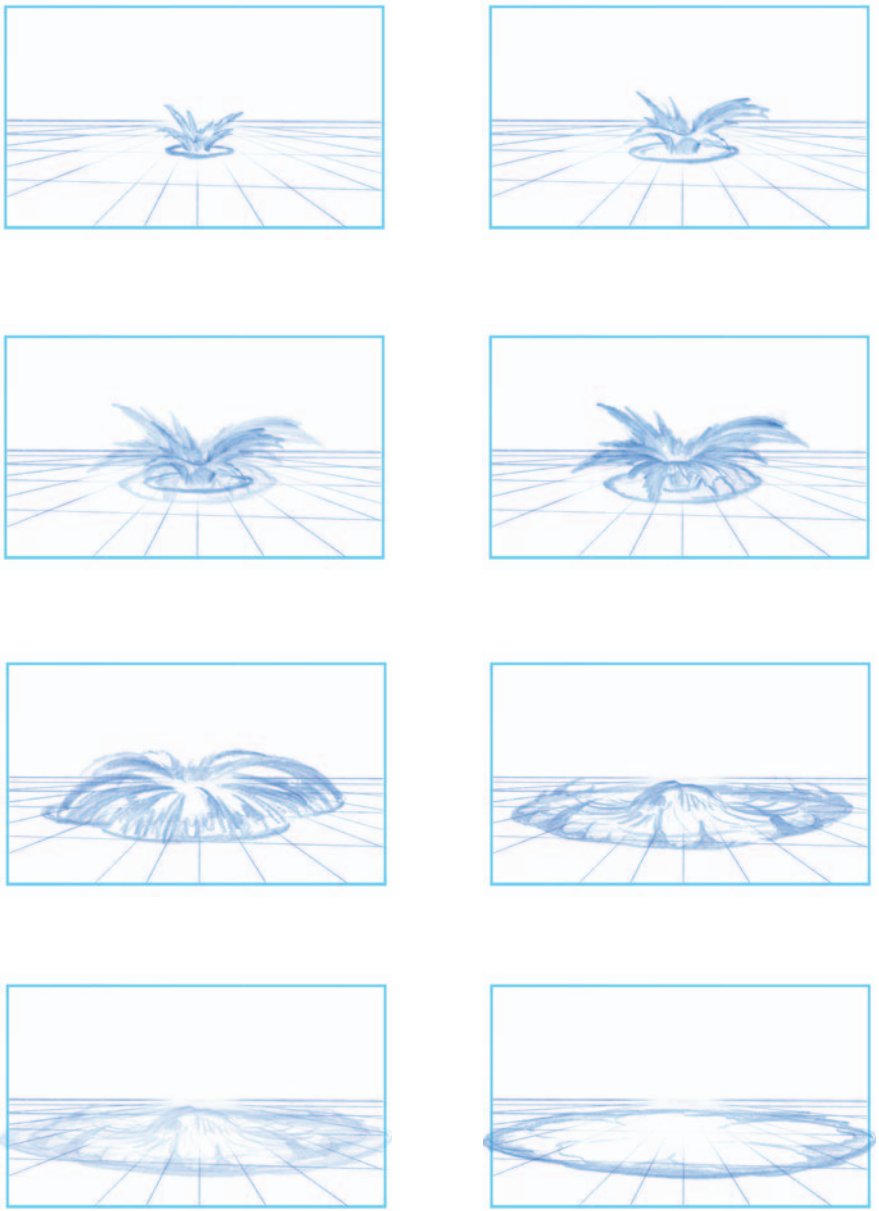
The next step is to begin considering the parameters of the shot. What is causing the splash? How fast is it moving? What is it made out of? Is it light or heavy, dense or porous? How much water would this object displace, and how big will the splash have to be? How much of the frame should it fill up as it dissipates?



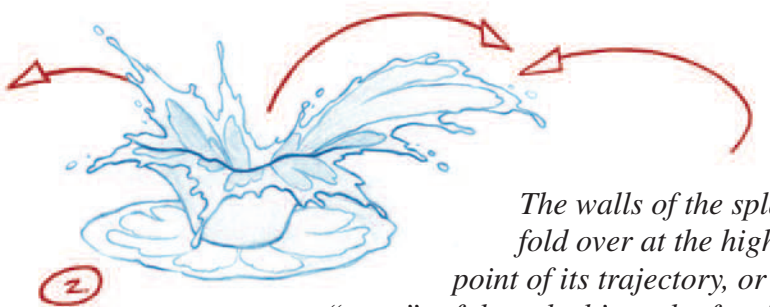
A path of action for the effect should be established before we begin animating. This serves as your guide as you animate, and can be very helpful in keeping the proportions of the effect accurate. Always consider as well, what the intention of the effect in the scene is. Is it the star of the scene, or does it only have a supporting role? Know what the scene requires in every way before starting!

There are as many subtly different approaches to animating a special effects scene, as there are animators. There is not necessarily a right or wrong way, as long as your final animation looks good! But there are fundamentally sound ways to proceed that will give you the greatest chances of success. By far the most tried-and-true technique for hand-drawn effects animation, is to roughly animate to begin with. Avoid thinking about the details of your effect until the overall motion is working well. Draw with a loose hand, think of the energy rather than the design, and push your drawings dynamically as if they were actually moving on your page. Don't be tentative at this point! The finely detailed and clean splash drawings on the following pages, got their energy and direction, from dynamically drawn, rough preliminary drawings.

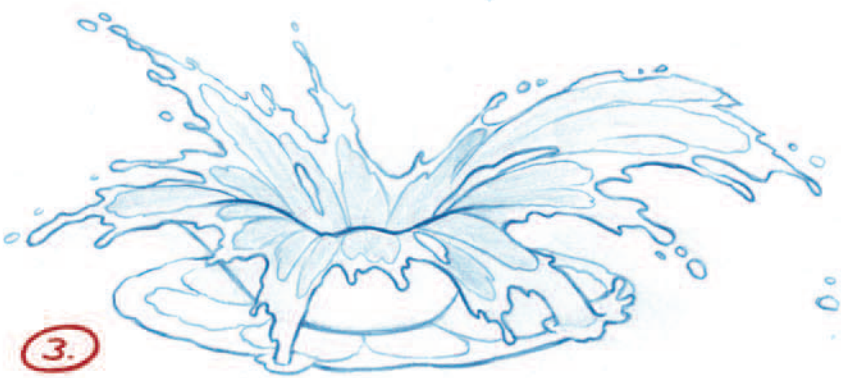
There are several things to always consider when you are animating a splash. What is causing the splash is one of the most important. That factor alone will have a lot to do with how much water will be displaced, how high the splash will shoot up before it succumbs to gravity, and how big its secondary splash will be. Always remember to think about what else is going on in the scene. Is the effect important to the telling of the story? Is there another action going on in the scene that is more important? Knowing what is causing the splash, what the splash's role is, and then integrating the effect with the overall style of the film is a great recipe for successful splash effects animation!



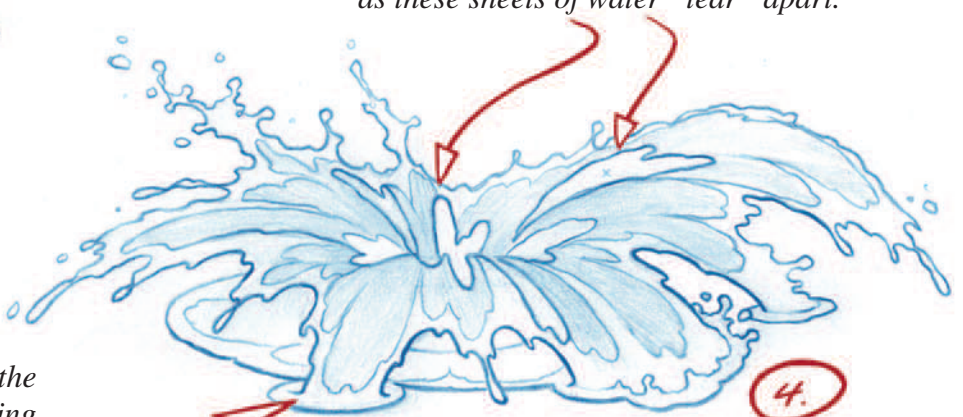
This is a very classical, stylized approach to animating a splash. The initial splash shoots up and outward, like a blossoming flower.



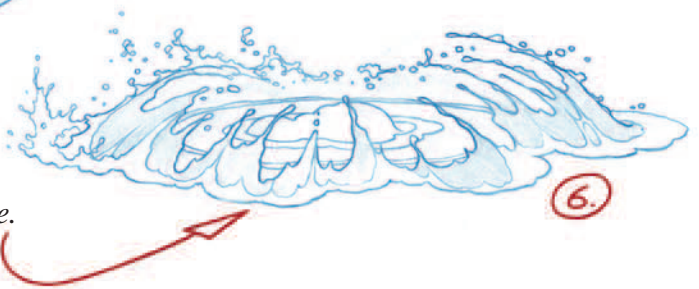
The walls of the splash fold over at the highest point of its trajectory, or the "apex", of the splash's path of action.



As the walls of the splash stretch outward, away from the point of impact, holes begin to open up, as these sheets of water "tear" apart.

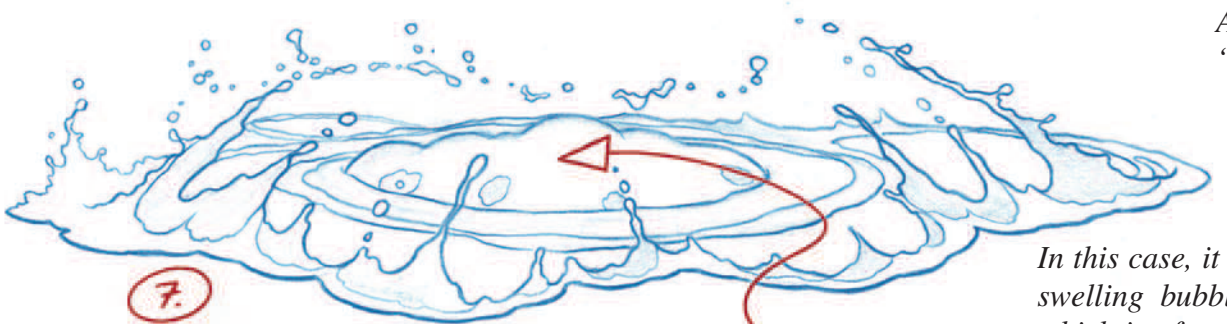


The leading edges of the initial splash fall back to the surface first, and are absorbed into the surrounding water surface, creating a new set of ripples.

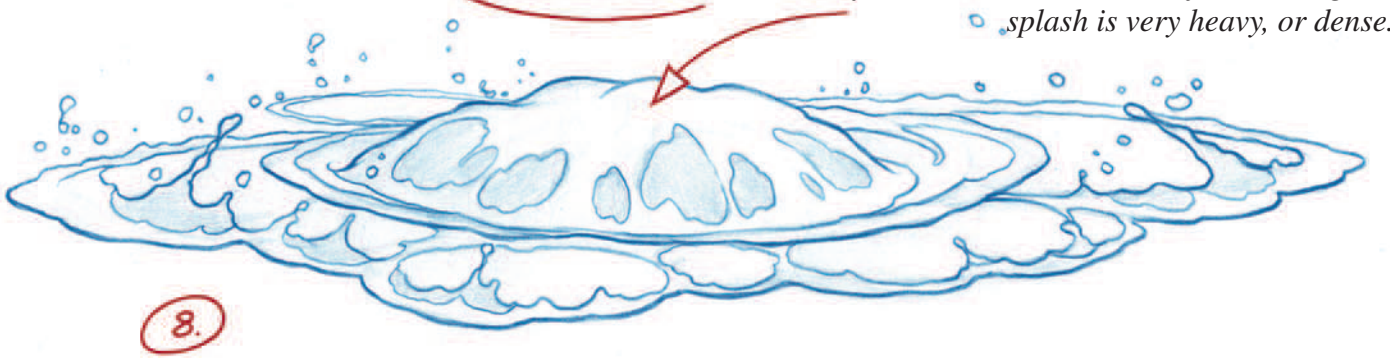


The "sheets" of the splash fold over at the apex as they stretch and tear, and they are finally absorbed back into the surrounding water surface.

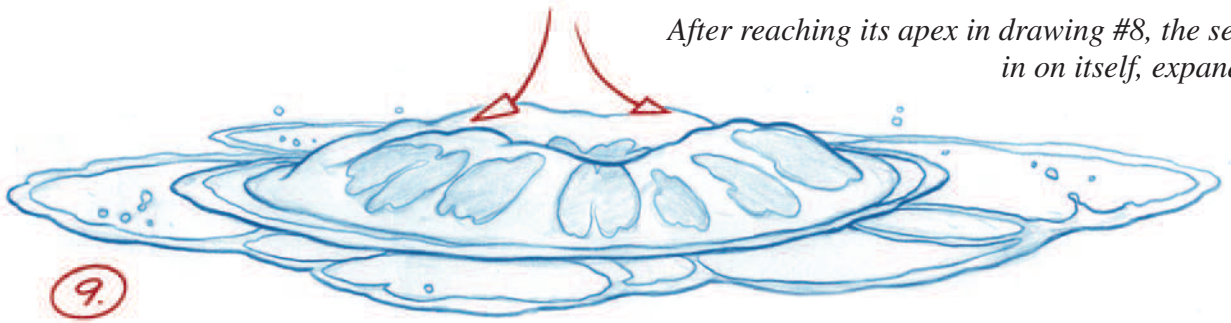
At this point we begin to see the "secondary splash" forming at the original point of impact.



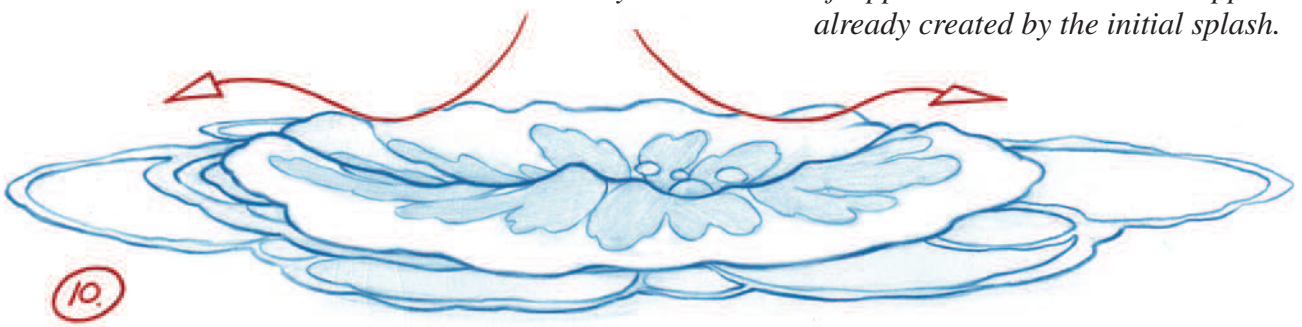
In this case, it is not so much a "splash" as a huge swelling bubble bursting back up to the surface, which is often the case when the object causing the splash is very heavy, or dense.



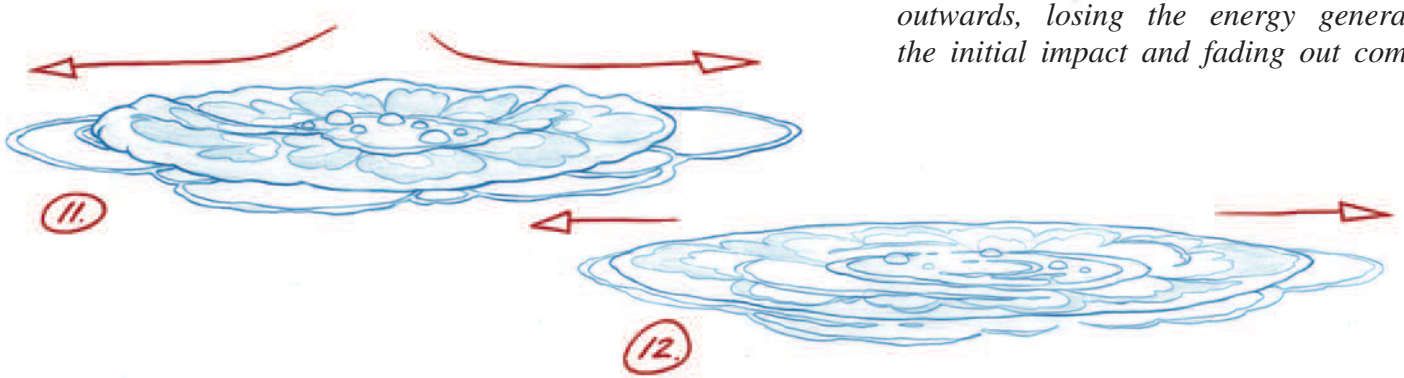
After reaching its apex in drawing #8, the secondary "swell" falls back in on itself, expanding outward as it resolves.



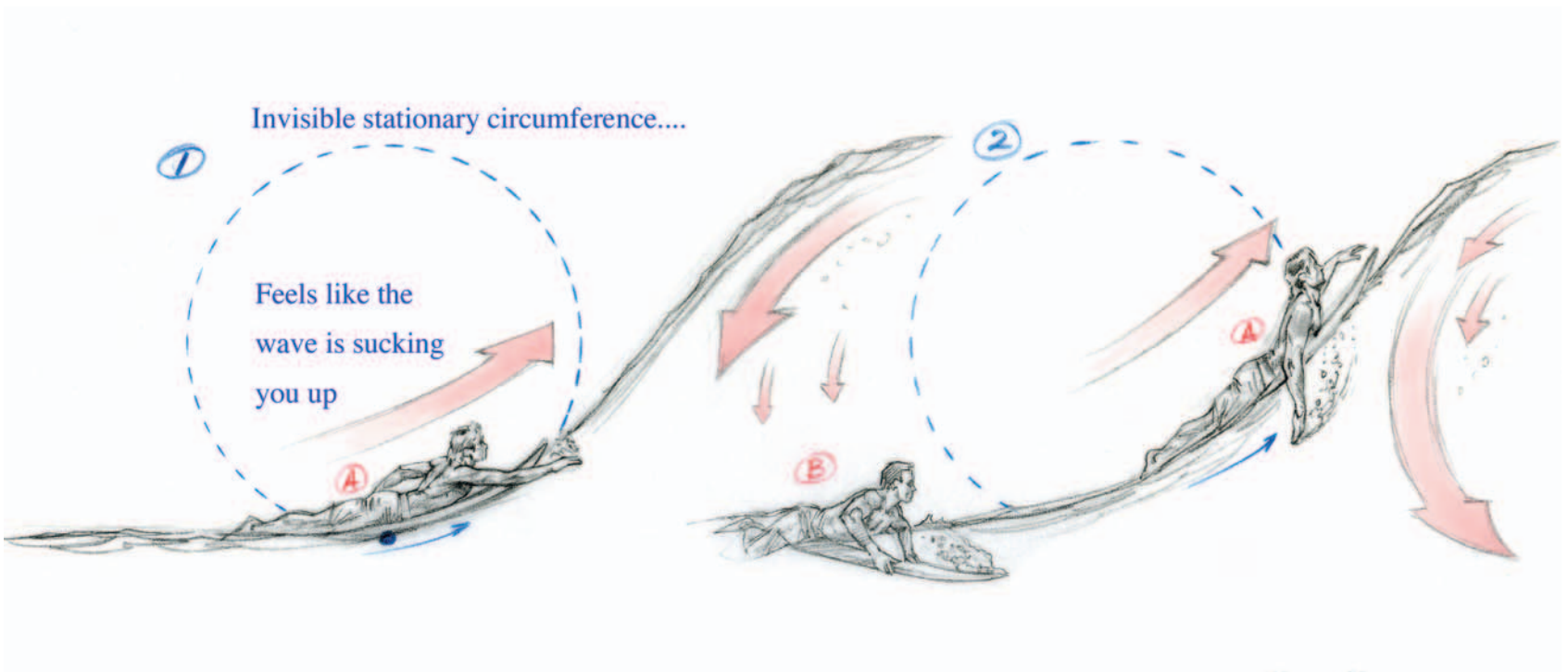
This creates yet another set of ripples which overtake the ripples already created by the initial splash.



Finally, the last set of ripples created by the secondary splash, flatten out as they expand outwards, losing the energy generated by the initial impact and fading out completely.

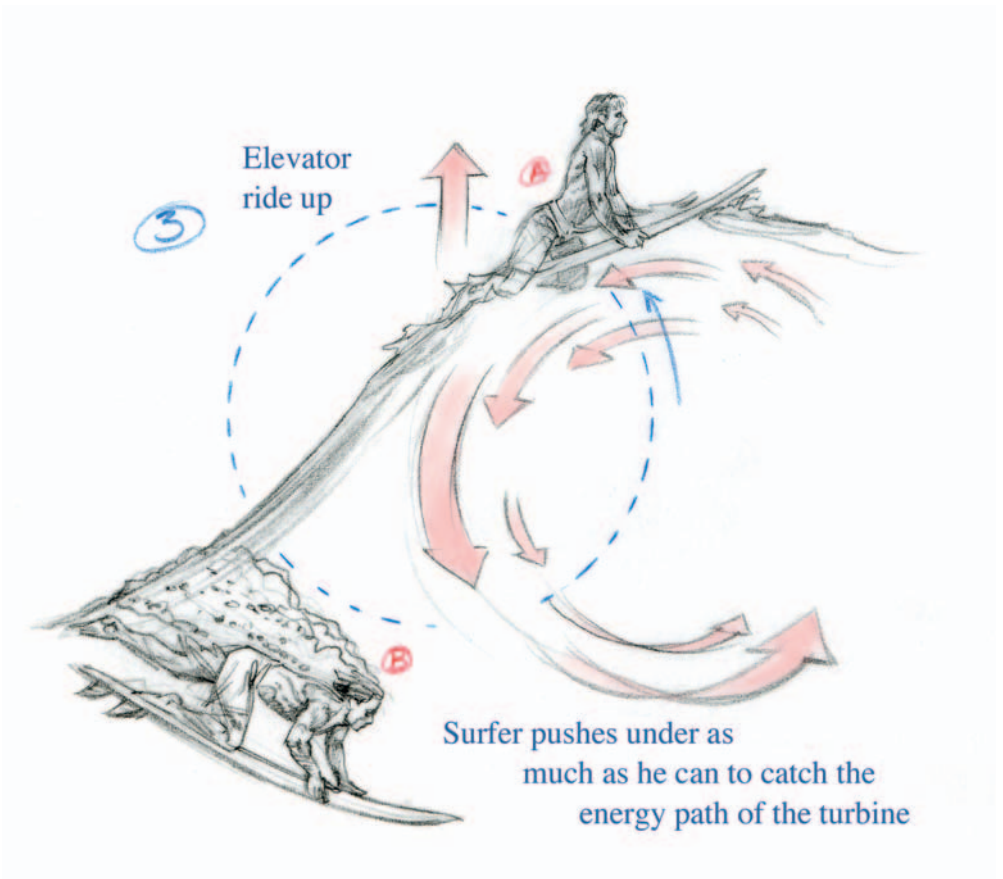


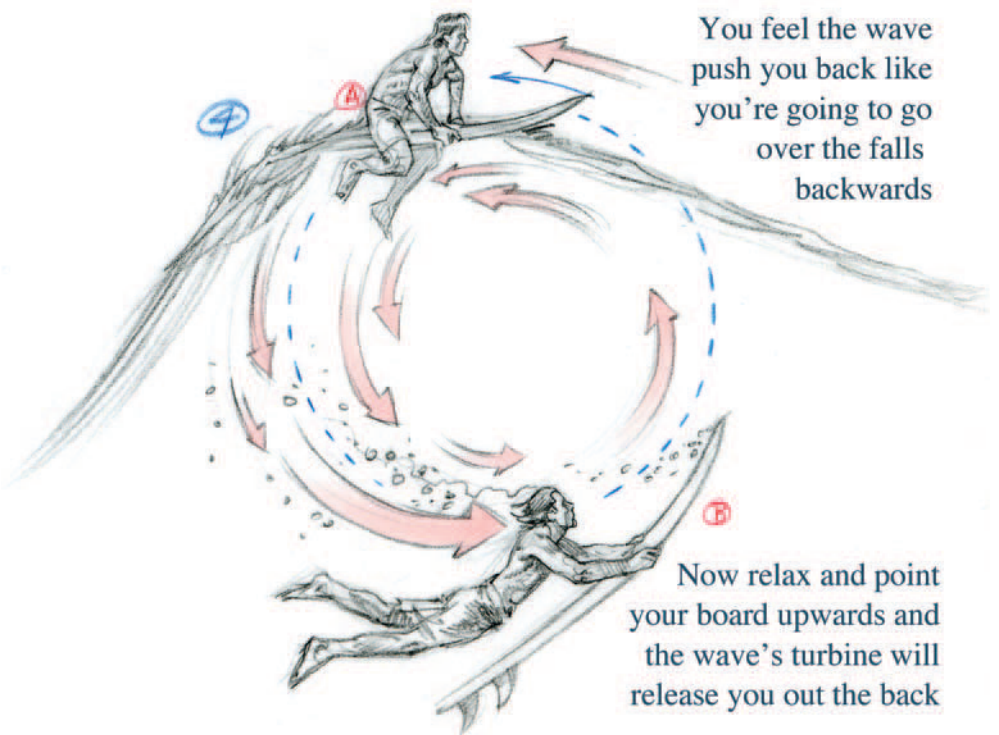
Wave action is another very commonly seen water effect, the basics of which are relatively easy to master once the underlying principles of wave motion are understood. There are some fun, easy tricks that can be used that mimic the principles of wave motion without the need to really get deeply into the actual scientific physics of wave movement. Understanding displacement, how a water surface always displaces itself as it moves from place to place is extremely important. Imagine tubular



Surfer A : Paddles over the top of the wave.
Surfer B : "Duck dives" under the wave and is lifted to surface by wave's turbine.

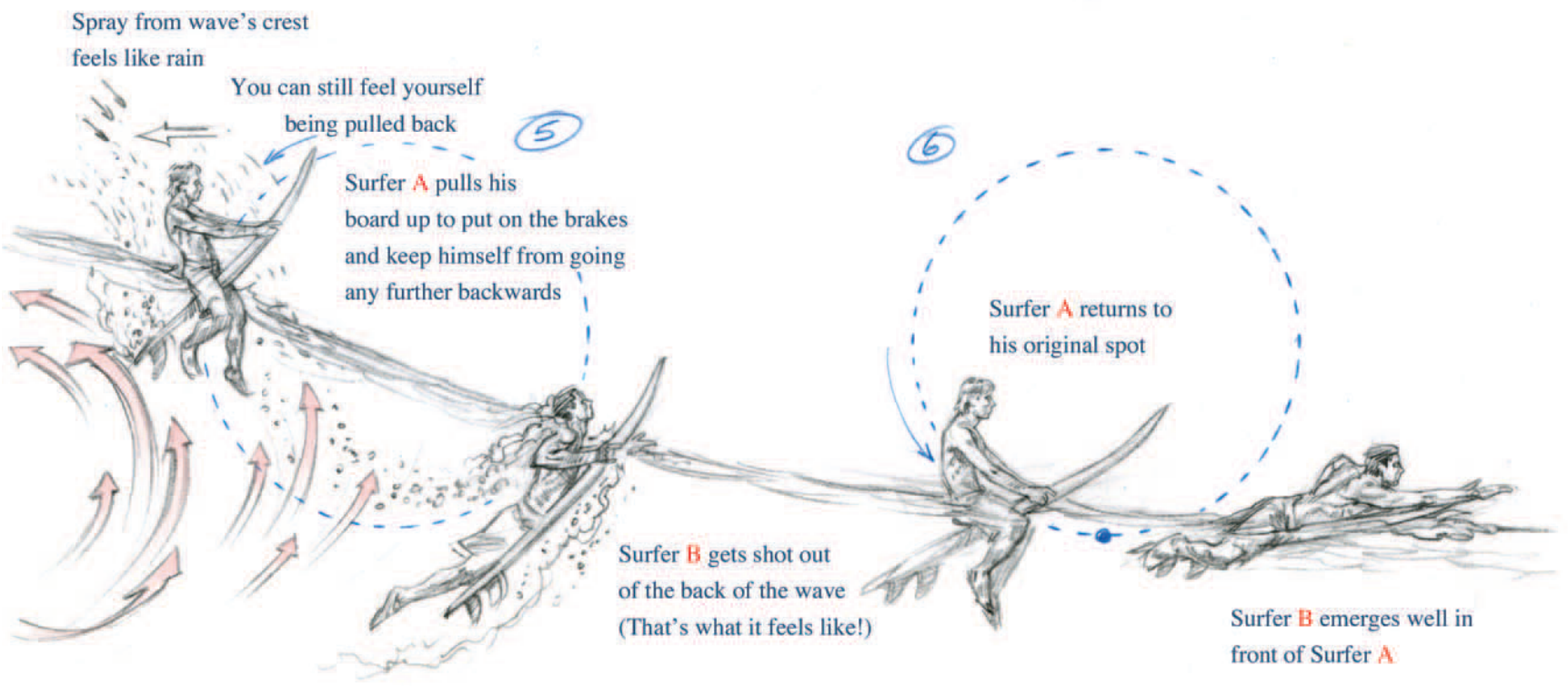
shapes of energy moving just under the surface of the water, which displace the surface textures as they roll underneath, allowing the details to fall back into their original place as they pass. The simplest way to illustrate this is the image of a tiny boat, bobbing up and down and to and fro as a wave passes under it, but winding up in the exact same place it was after the wave has passed. Whether a water surface is moved by





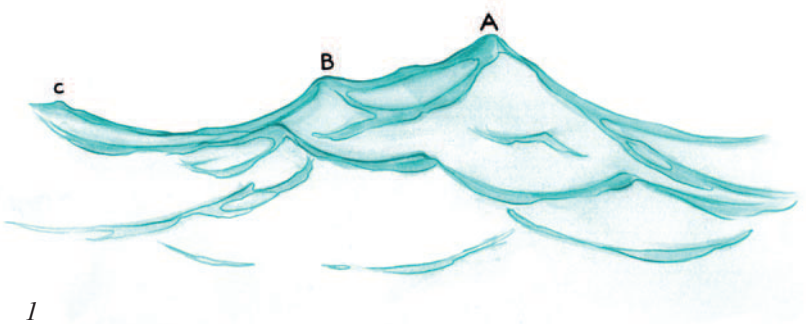
wind, the moon's gravity, a boat pushing through it, or an explosion that has caused its movement, waves all behave similarly once they get their initial directional movement. When attempting to animate more than one wave at a time, especially a water surface covered with waves, it is easy to get overwhelmed and find that the drawing is controlling us, rather than us controlling the drawing. Remember to keep your drawings as simple as possible, at least until the initial ruff animation is working

Note that Surfer A simply rolled around in a circle as the wave passed underneath him. He ends up in almost precisely the same spot he started in. If you follow his movement through this sequence of drawings, you see the classical rolling wave physics at work!

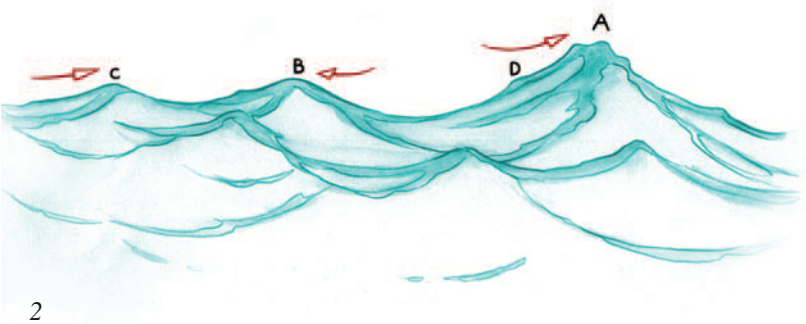


well. Then we can add details, and add the overlapping actions of different sized waves. It is extremely important with good effects design, to vary the size and shapes of our waves and their surface details—and their movement as well—to create believable, natural feeling water animation. Remember also, to always *push your drawings*. Always exaggerate, as animation needs that extra information to work well!

In this next wave sequence, we'll look at some of the tricks of the trade to consider when animating rolling waves, including overlapping our action and using counter action. These drawings are key poses about eight frames or one-third of a second apart. This entire sequence would last roughly 2.6 seconds, and it is a cycling wave action, with pose 8 hooking up and continuing through with pose 1. I have labeled specific wave



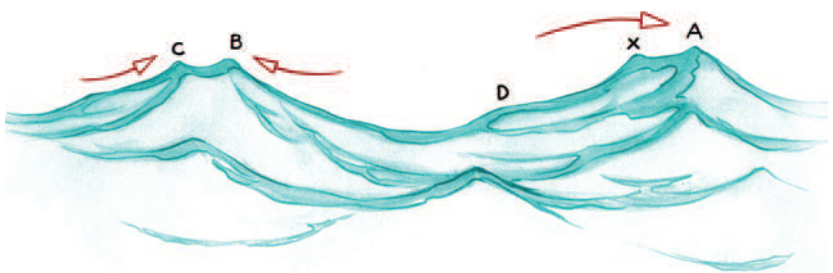
1



2

peaks with letters, as we often do in the business, so that an assistant animator can follow which wave is going where, especially when there are counter-acting waves. The main direction of these waves is from left to right. In pose 1, we see wave peak A figuring prominently. Imagine it is traveling from left to right, and wave peak B is sliding down the back of wave A in the opposite direction, from right to left. This action can be seen clearly in pose 2. Wave C has just entered screen left and is traveling toward the right, along with the general flow of this water surface. The larger waves might be big rolling swells that may have been created by a storm a thousand miles away, while the smaller counter-acting

waves could be a smaller, choppy type of wave caused by a local breeze. In pose 3, we see wave B and wave C converging on each other, moving in opposite directions and building up into a larger wave, as wave A rolls away to the right, with much smaller waves D and X rolling down its back much as wave B did. In pose 4, waves B and C have crossed over each other, with the much larger wave C pushing right through



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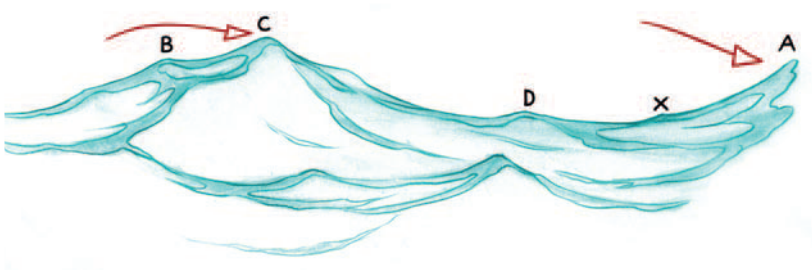
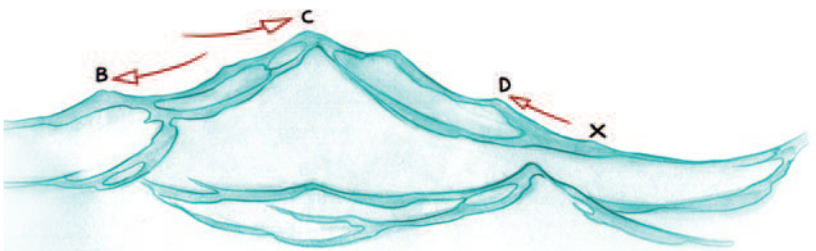


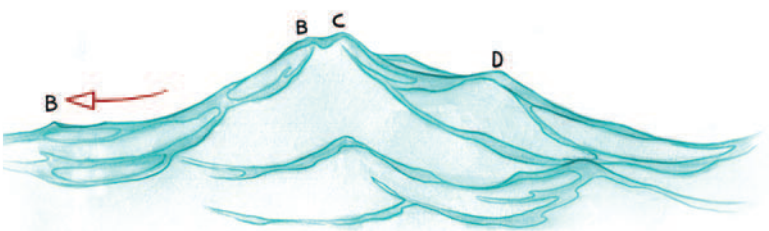
4

wave B, building momentum as it goes. The smaller waves D and X are still moving right to left against the larger waves, and wave A is rolling right out of frame right. It is important to keep in mind, if the camera and therefore the horizon line is fixed, our water will always displace itself evenly in frame. If we push it down on one side, it must rise on the other side. Water always displaces itself perfectly, as its molecules rush in to fill any space vacated by whatever energy is moving the water. It is always attempting to move back to its flat, even self, as it is held earthward by gravity. In pose 5, we see secondary wave B still

moving right to left, counter to the overall thrust of this water surface, while wave C has built up in size and continues left to right. All the smaller secondary waves, D and X and another unnamed ripple, are moving counter to the main wave, now moving toward the left, up the face of wave C as it rolls to the right. In pose 6, we see waves C and D converging, while wave X has been absorbed completely, and the



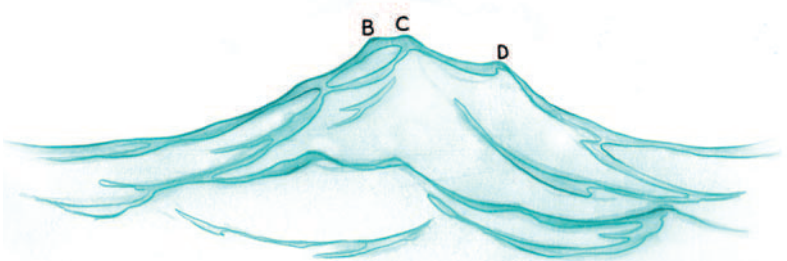
5



6

unnamed ripple gets pushed back and away from us as C and D converge. Behind wave peak C, I have reintroduced secondary wave B, as the original wave B continues of screen left. This is where this wave action begins to hook up with the original action in pose 1, so that it can be cycled over and over. In pose 7 we see wave C and D continue to converge and build up, approaching the size and weight of wave peak A from pose 1. The smaller secondary wave B is starting its way back down the backside of the larger more powerful wave.

In pose 8, secondary wave D is swallowed up by the strength and momentum of wave C, which I have also labeled (A) as this is where the cycle hooks up with pose 1. If you look back at pose 1 now and imagine it is the next pose, you can see how A continues its way left to right, as wave B slides down its back. This is a very stylized approach to animating wave action, with larger swells pushing against the smaller



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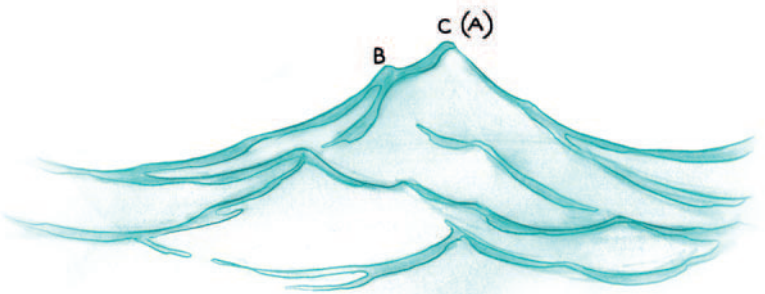
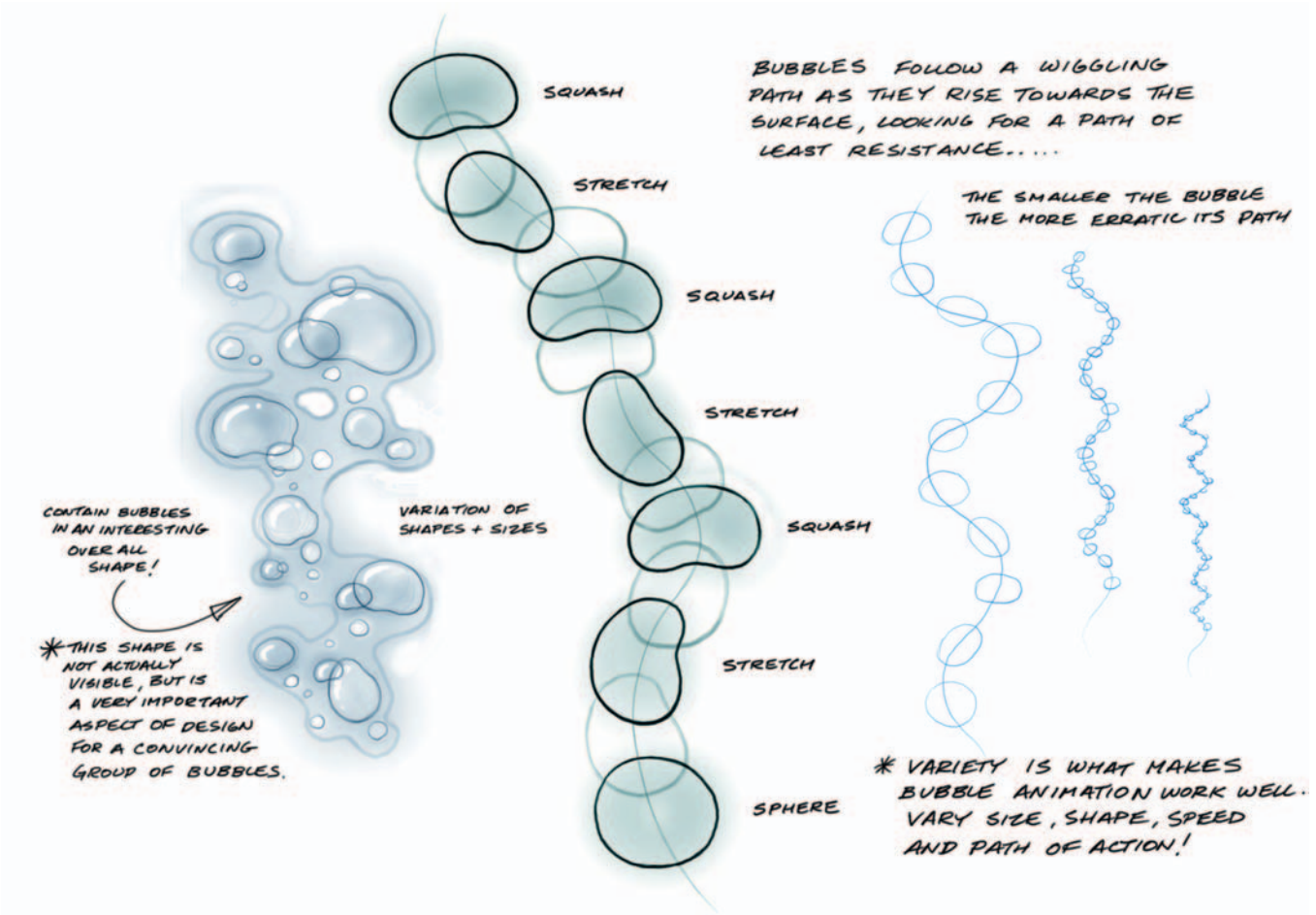


8

secondary chop moving in the opposite direction. The smaller waves in the foreground demonstrate the same type of wave action, and if you analyze these actions, there are some very useful techniques and tricks for successfully animating waves. Always keep overlapping counter-action and volume displacement in mind when you are animating a water surface. It's also very important to keep in mind that water is constantly morphing, and many wave shapes can and do disappear and appear mysteriously.



Bubbles are really fun, fascinating elements, little pockets of water pushing their way to the surface, looking for the path of least resistance, and just wobbling around madly in the process. They can take on an incredible variety of shapes, and they stretch and squash delightfully. When animating bubbles, remember to exaggerate the stretch and squash, and have fun with them!

Earlier on in this chapter we looked at how a splash looks, from a cut away side view. Here is a more detailed look at what happens underneath the water, as the object punches a pocket of air into the water. The initial air-filled hole gets pulled downward by the momentum of the object, and in the process it stretches into liquidly oblong shapes. Then as gravity continues to pull the object down, the air begins its wiggling journey back to the surface breaking up into smaller pieces in the process.